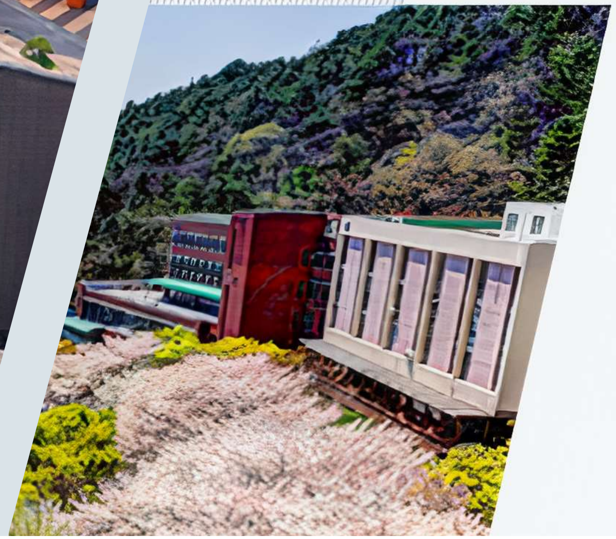
Dong-A Univ. (ISPL)

동아대학교
DONG-AUNIVERSITY

Backpropagation (역전파)

컴퓨터AI공학부
2024년 1학기 인공지능

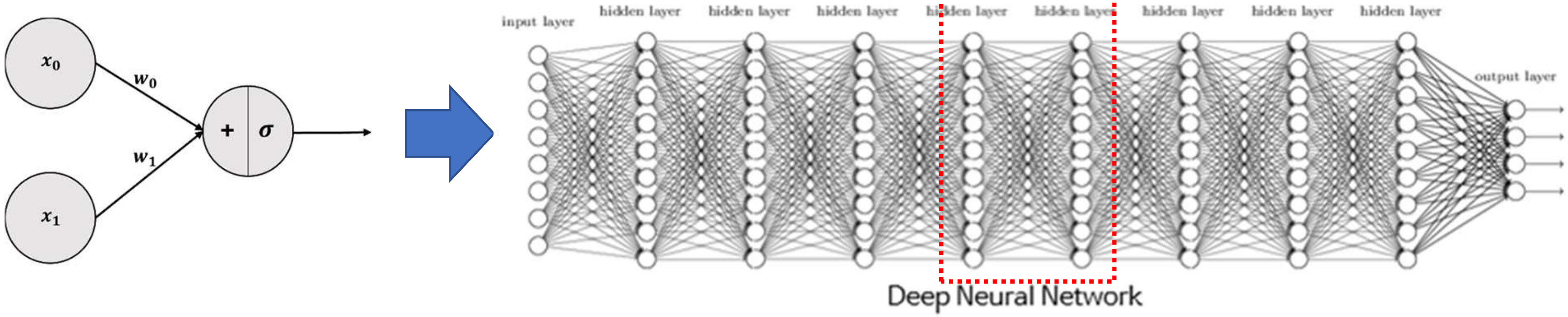


Review: MLP by Linear Algebra

- **SLP → MLP (One of the Deep Neural Network: DNN)**

  → **행렬에 대한 Forward propagation**

$\boldsymbol{x} \in \boldsymbol{R}^{\boldsymbol{m}}$ $\boldsymbol{q} \in \boldsymbol{R}^{\boldsymbol{n}}$



$$\boldsymbol{W} \in \boldsymbol{R}^{\boldsymbol{m}\times\boldsymbol{n}} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}, \quad \boldsymbol{W}^{\boldsymbol{T}} \in \boldsymbol{R}^{\boldsymbol{n}\times\boldsymbol{m}} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}, \quad \boldsymbol{x} \in \boldsymbol{R}^{\boldsymbol{m}} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}, \quad \boldsymbol{b} \in \boldsymbol{R}^{\boldsymbol{n}} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

$$\boldsymbol{q} = \boldsymbol{\sigma}(\boldsymbol{W}^{\boldsymbol{T}} \cdot \boldsymbol{x} + \boldsymbol{b}) = \boldsymbol{\sigma}\left\{\begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}\right\} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in \boldsymbol{R}^{\boldsymbol{n}}$$

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Introduction

- **Backpropagation(역전파)**
- 모험가 이야기: 어떤 모험가는 여행을 하며 가장 깊고 낮은 골짜기를 찾아가려 한다.
  - ✓ 조건 1: 지도를 보지 않을 것
  - ✓ 조건 2: 눈가리개를 쓰는 것
  - ➢ **어떻게 모험가는 가장 깊고 낮을 골짜기를 찾아 갈 수 있을까?**

GD Method (역전파를 위한 최적화: Optimization for Backpropagation)

Gradient Descent Method (GD Method / 경사하강법)

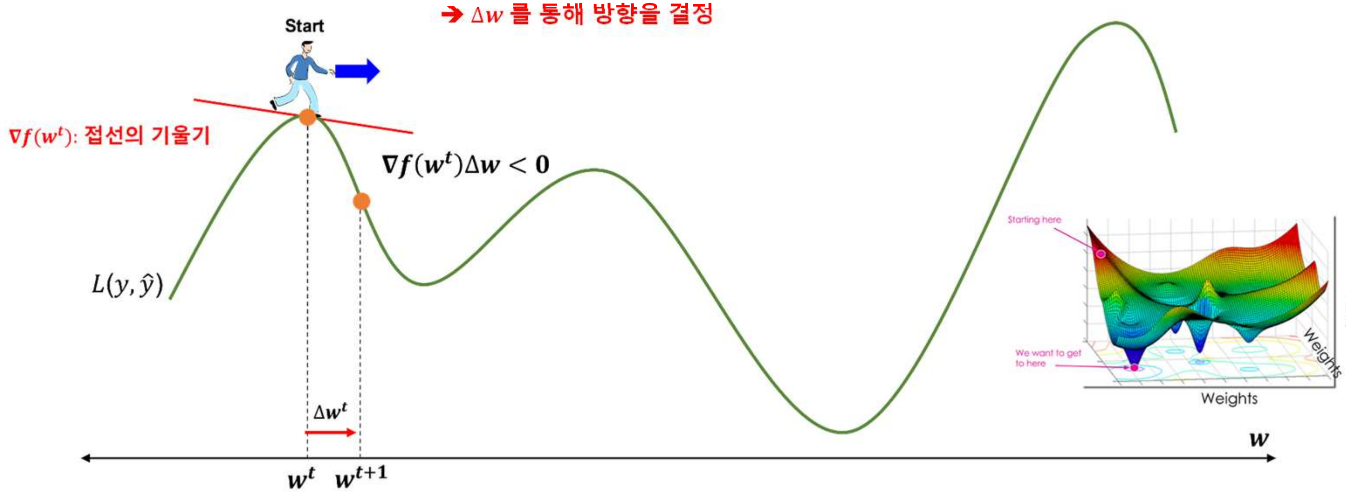
학습률(learning rate, step size)

$$\boldsymbol{w}^{t+1} \leftarrow \boldsymbol{w}^{t} - \mu \nabla f(\boldsymbol{w}^{t}), \quad \textit{where } \nabla f(\boldsymbol{w}^{t})\Delta \boldsymbol{w} < 0$$

탐색 방향

하강법에서 접선의 기울기와 w 는 항상 다른 부호를 가짐

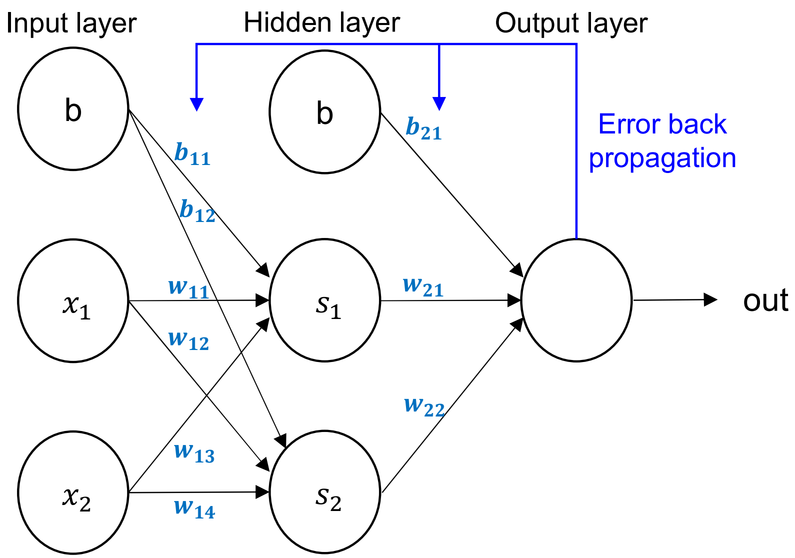
➔ Δw 를 통해 방향을 결정



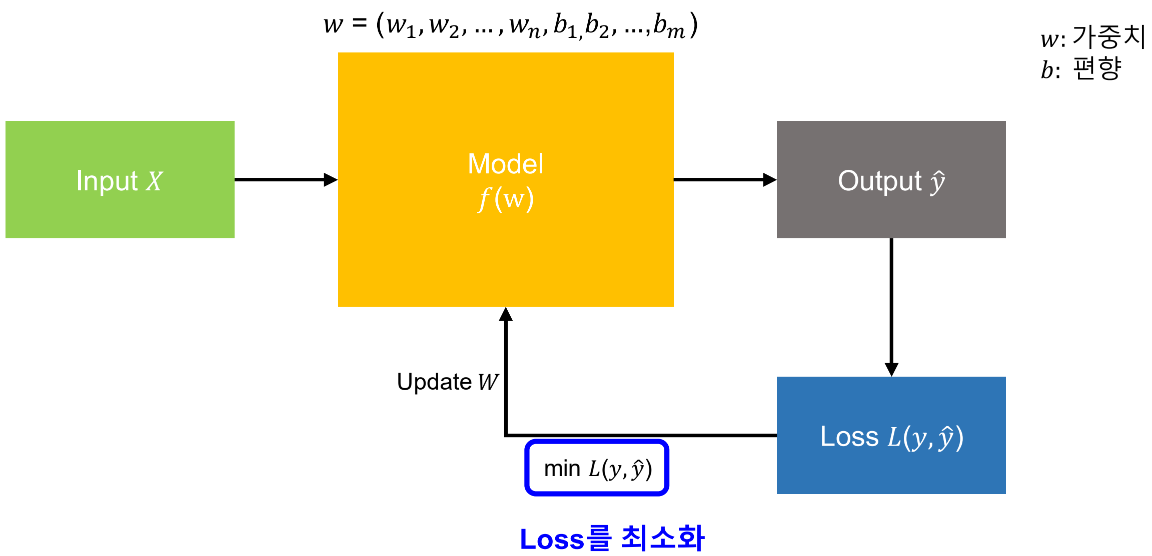
Contents

Introduction

- Backpropagation: 계산 결과와 정답의 오차를 구해서 오차에 관여하는 노드 값들의 가중치와 편향을 수정하는데, **오차가 작아지는 방향으로 반복해서 수정**하는 기법



<Multi-Layer Perceptron>



Introduction

- Backpropagation: 계산 결과와 정답의 오차를 구해서 오차에 관여하는 노드 값들의 가중치와 편향을 수정하는데,
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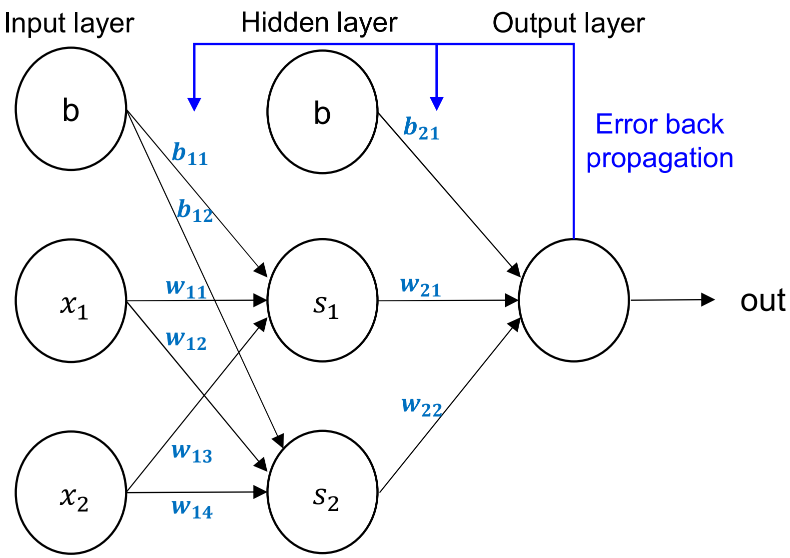

<Multi-Layer Perceptron>

- ✓ Parameter (w, b) 업데이트: per Mini-batch or Epoch (Iteration)
- ✓ Iteration 수가 늘어나면 정확성을 증가됨 ➔ 시간多
- ✓ Iteration 수가 줄어들면 정확성을 떨어짐 ➔ 시간小

Introduction

- 모험가 이야기: 어떤 모험가는 여행을 하며 가장 깊고 낮은 골짜기를 찾아가려 한다.
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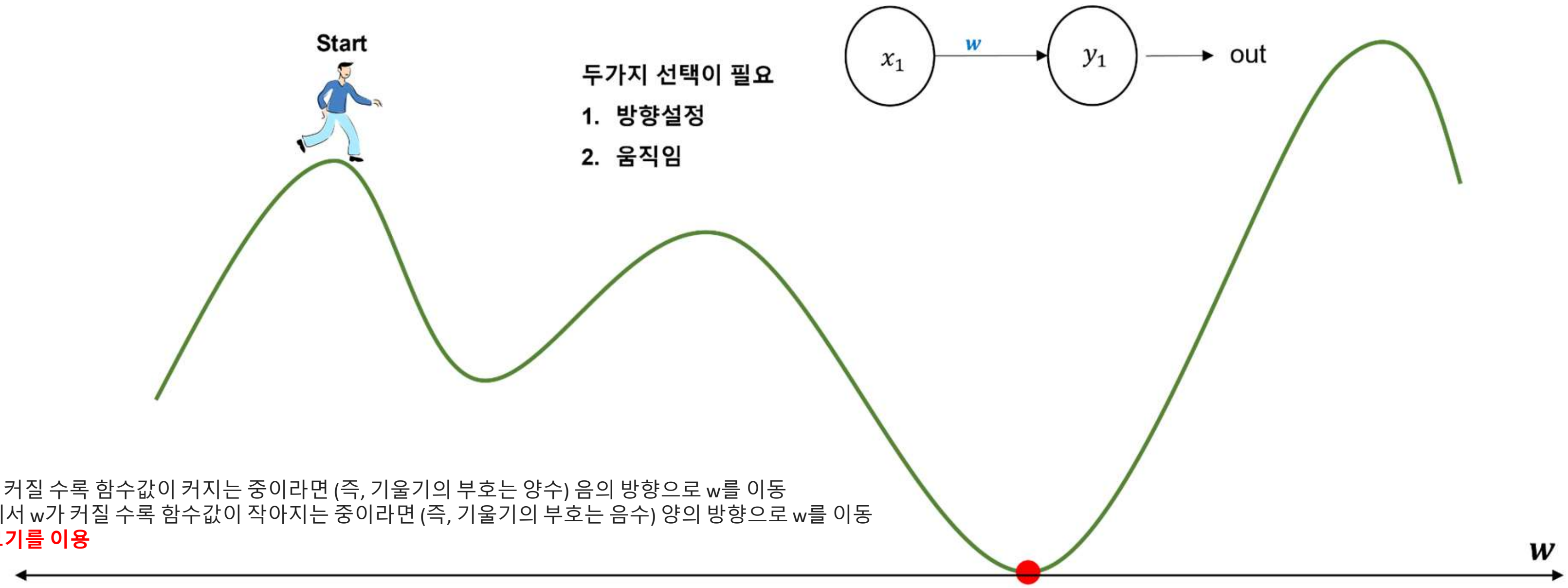

- 특정 포인트 w에서 w가 커질 수록 함수값이 커지는 중이라면 (즉, 기울기의 부호는 양수) 음의 방향으로 w를 이동
- 반대로 특정 포인트 w에서 w가 커질 수록 함수값이 작아지는 중이라면 (즉, 기울기의 부호는 음수) 양의 방향으로 w를 이동
- 이동거리 → **Gradient 크기를 이용**

Introduction

- 모험가 이야기: 어떤 모험가는 여행을 하며 <u>가장 깊고 낮은 골짜기</u>를 찾아가려 한다.
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두가지 선택이 필요

1. **방향설정**
2. **움직임**

1. **전진/후진**
2. **보폭**

Introduction

- 모험가 이야기: 어떤 모험가는 여행을 하며 가장 깊고 낮은 골짜기를 찾아가려 한다.
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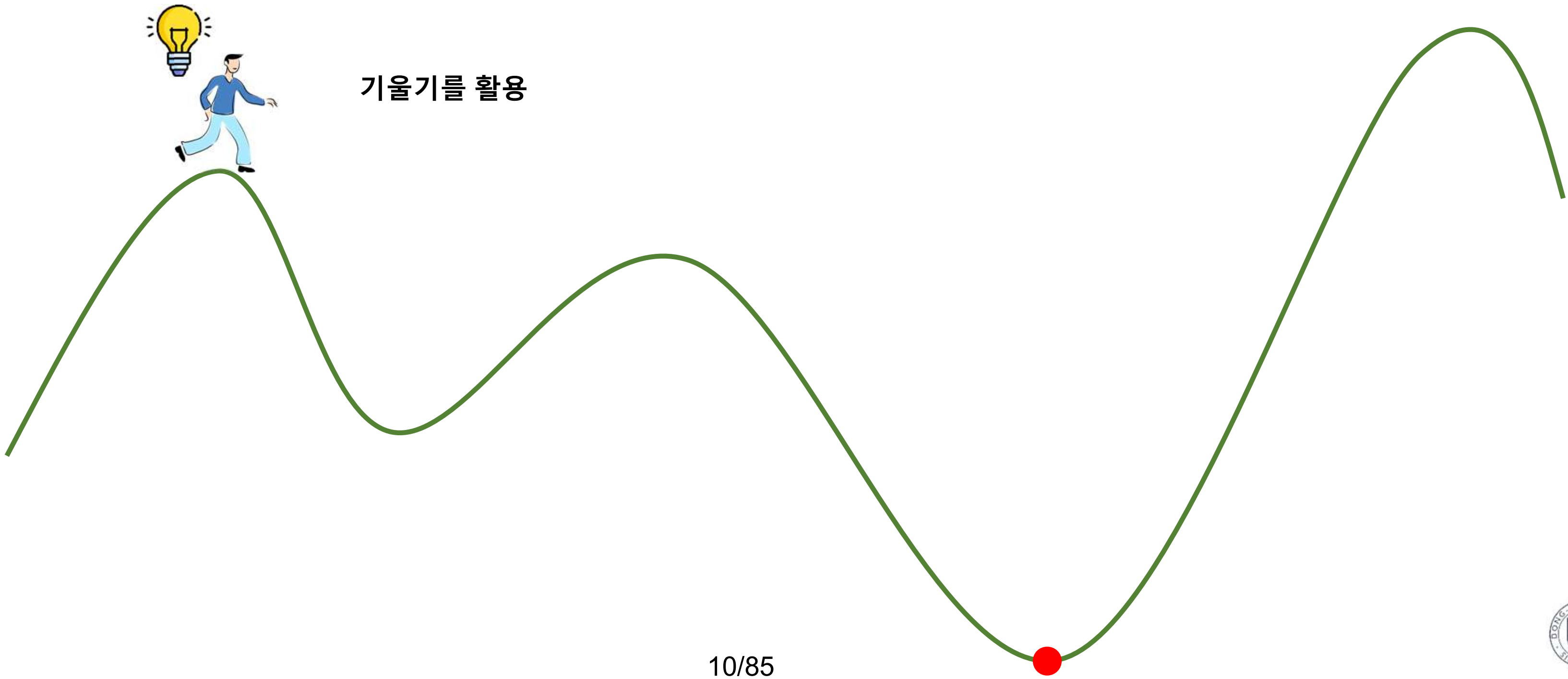

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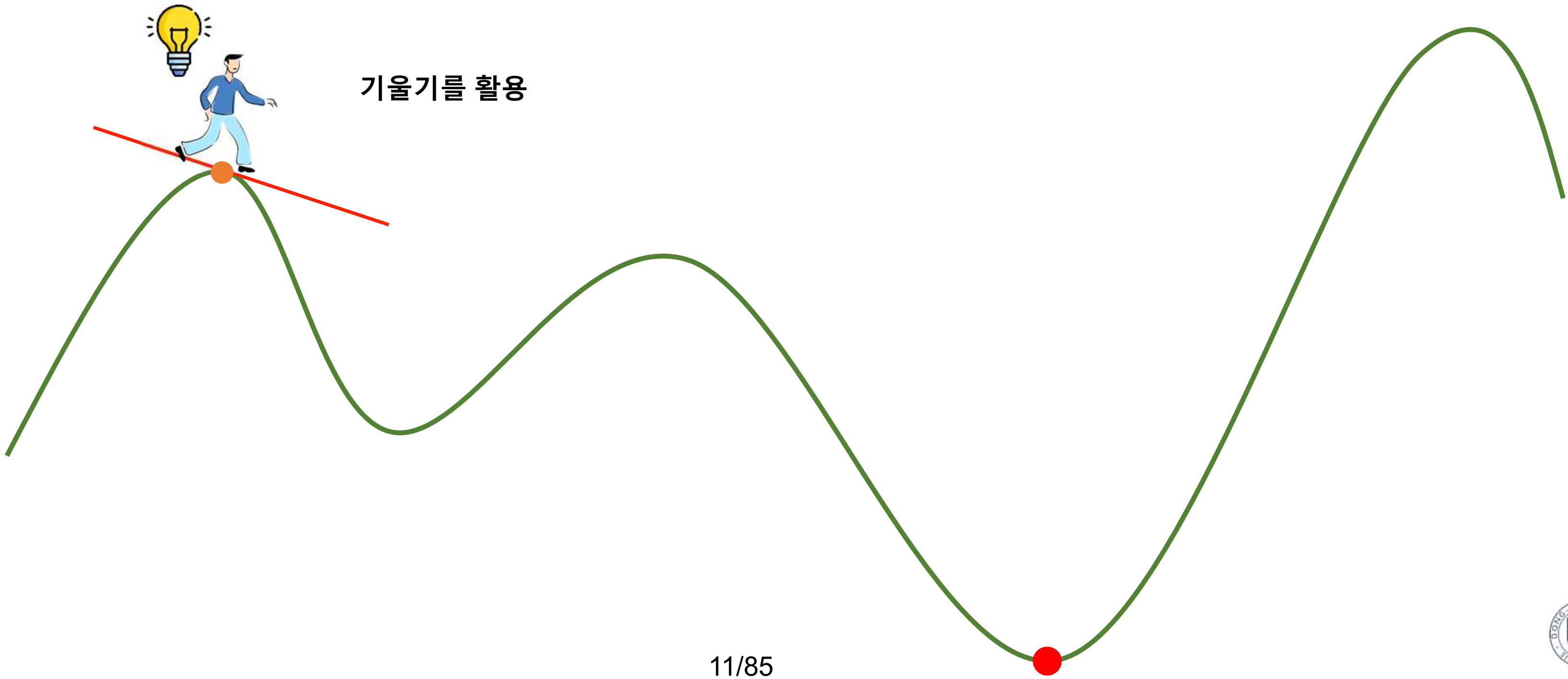
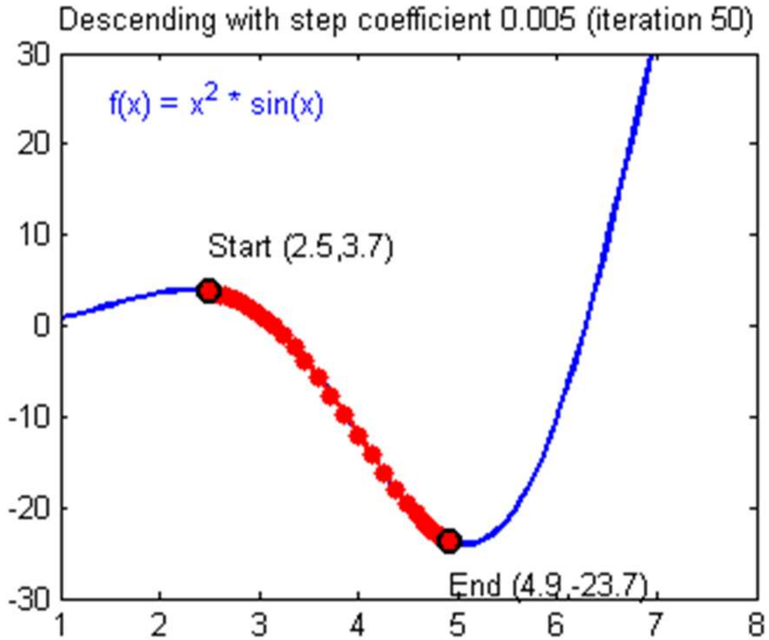
  - ➢ **어떻게 모험가는 가장 깊고 낮을 골짜기를 찾아 갈 수 있을까? → Optimization (최적화)**



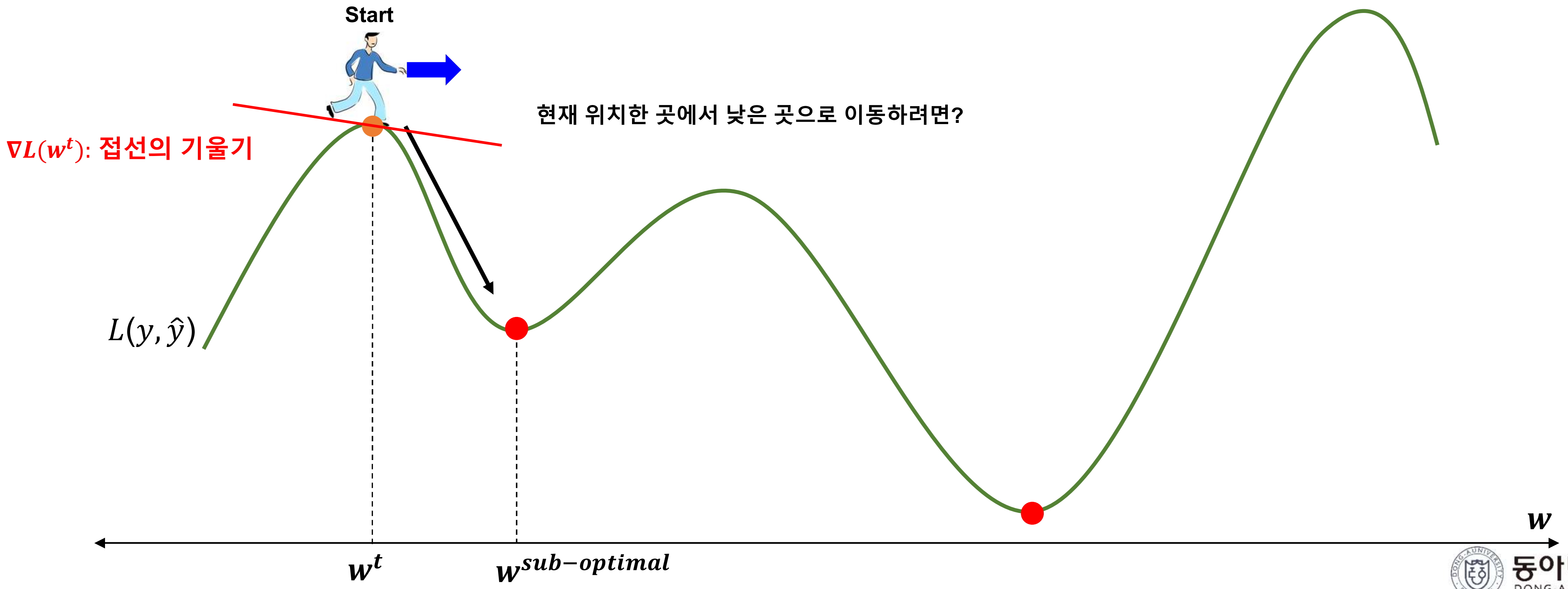


이동거리에 사용할 값을 gradient의 크기와 비례하는 factor를 이용하면 현재 xx의 값이 극소값에서 멀 때는 많이 이동하고, 극소값에 가까워졌을 때는 조금씩 이동할 수 있게 된다.

Contents

Descent Method(하강법)

- 주어진 어떤 지점에서부터 오차가 더 작은 곳으로 이동하려는 방법



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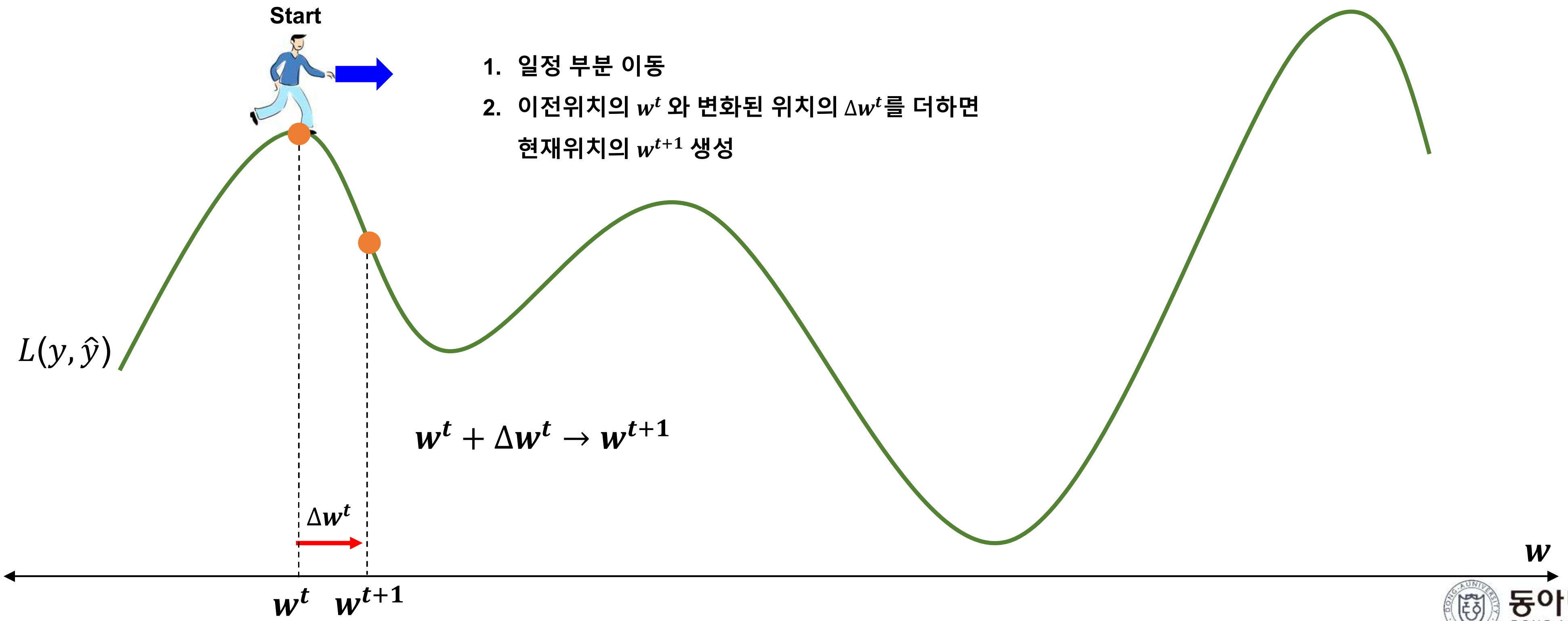

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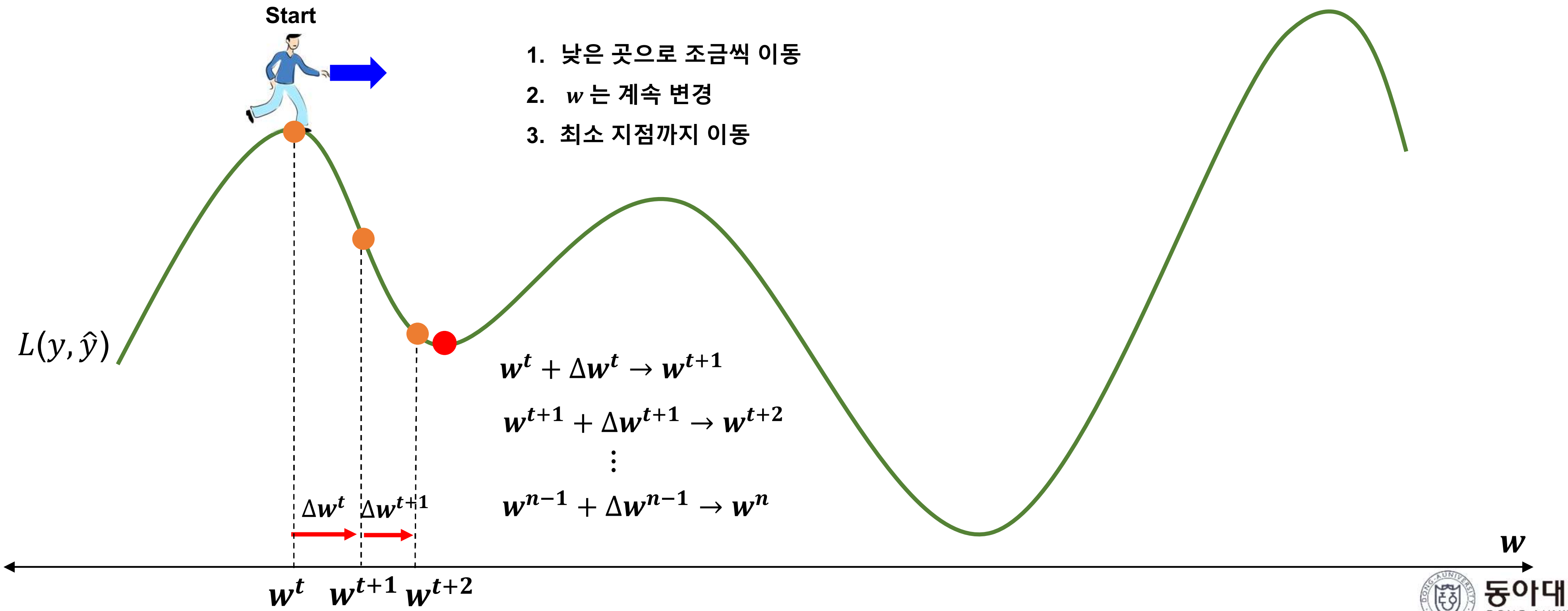

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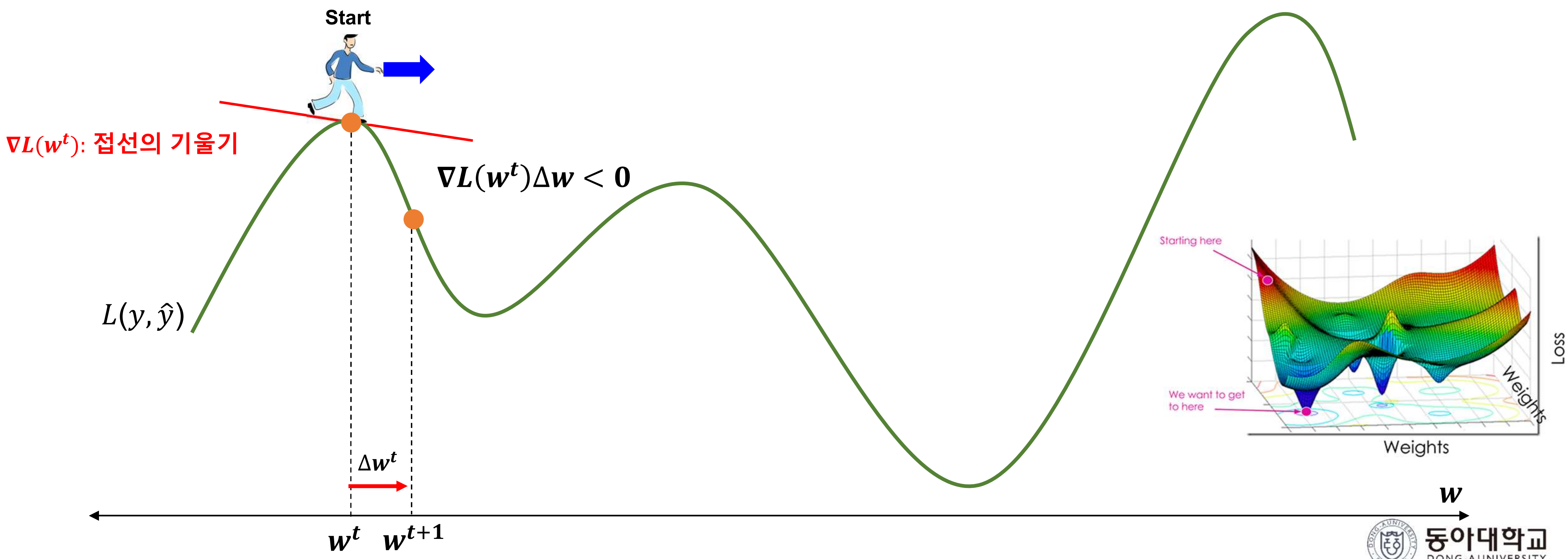
1. 하강법에서 접선의 기울기와 w 는 항상 다른 부호를 가짐 ➔ Δw 를 통해 방향을 결정



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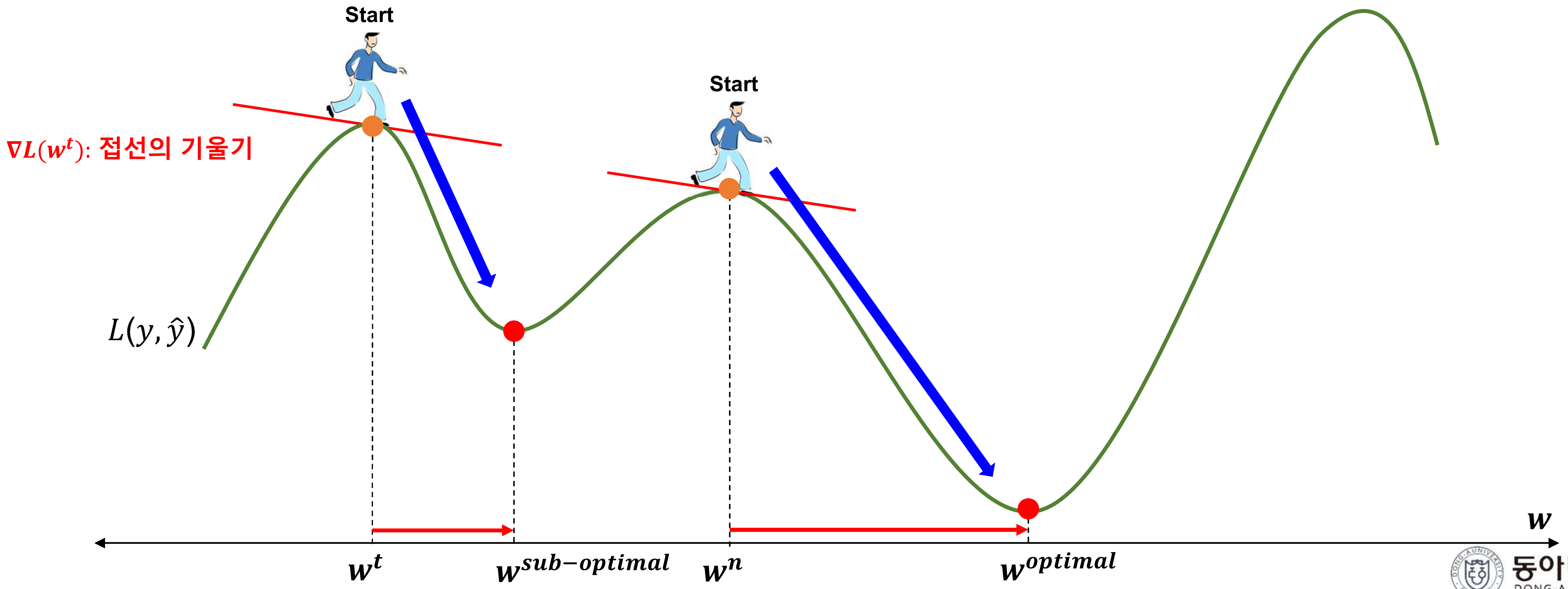
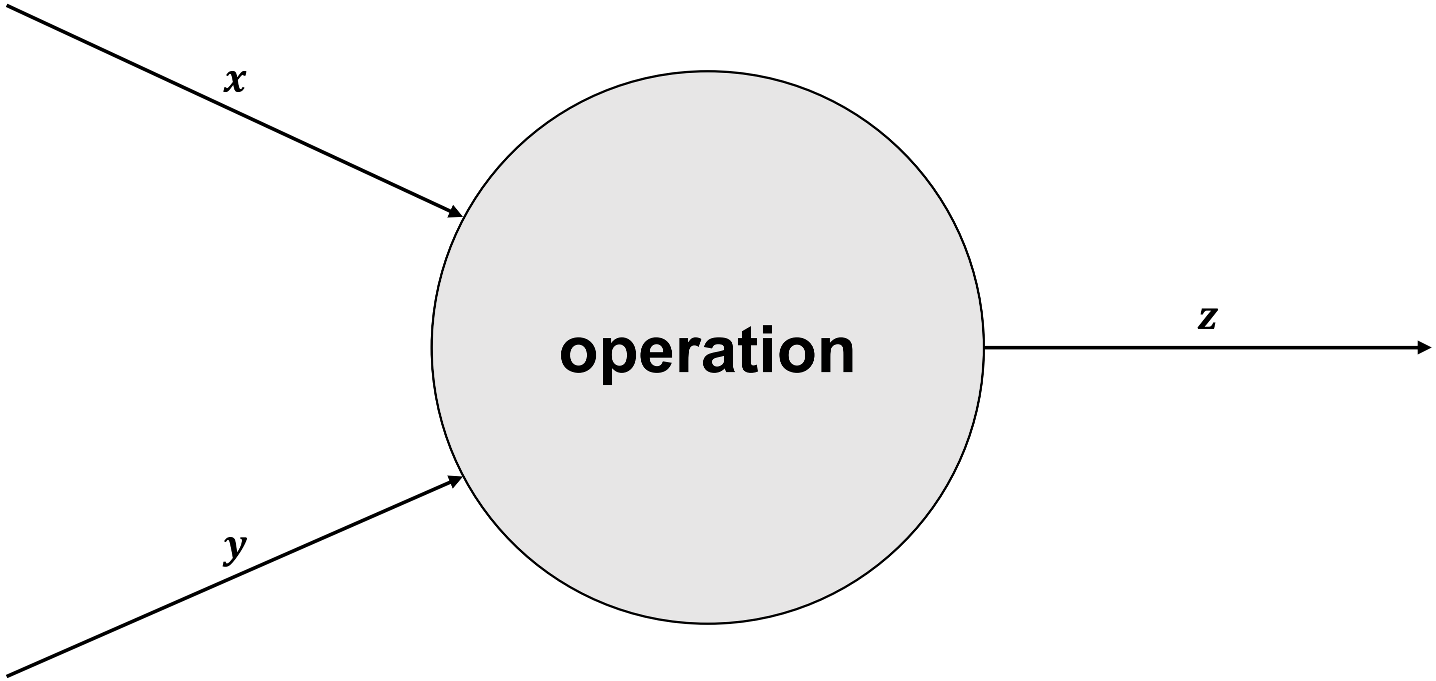
2. 하강법은 초기 위치에 따라 도착지점이 변경 → Weight Initialization



Backpropagation 동작원리

- Chain rule



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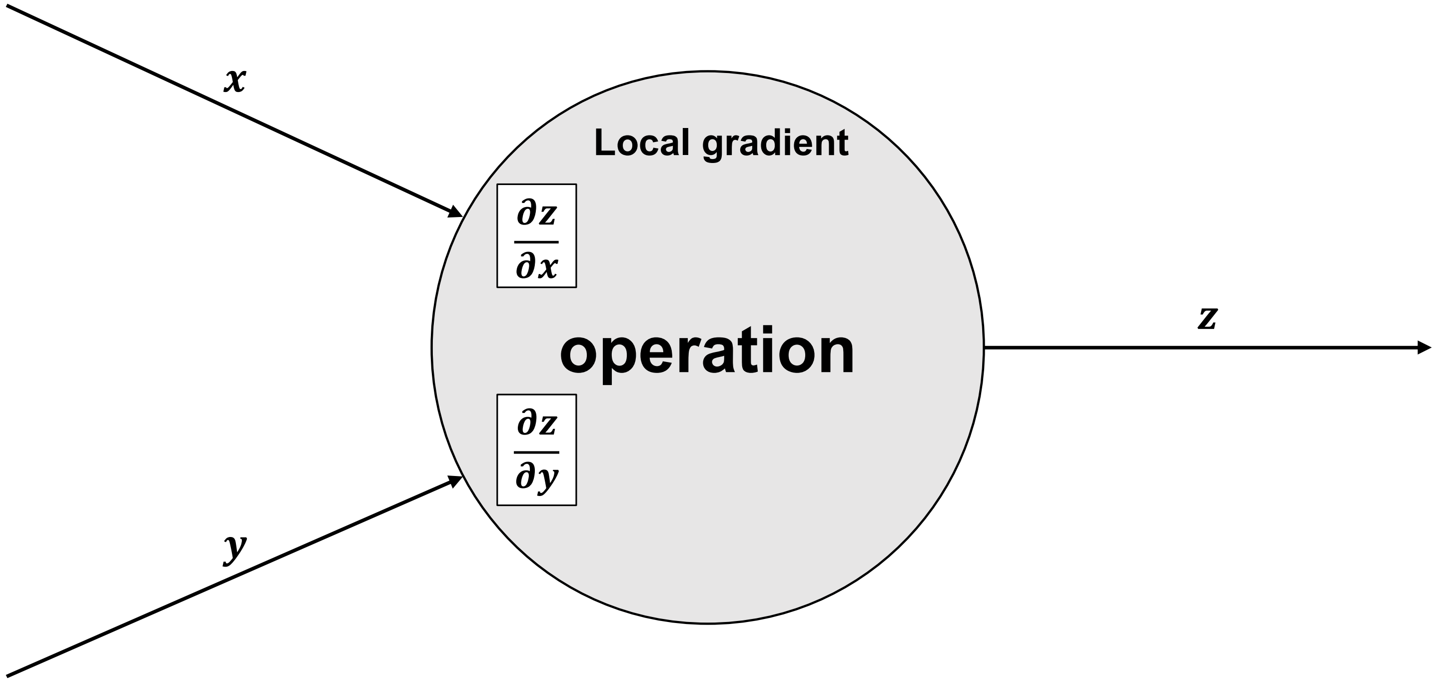

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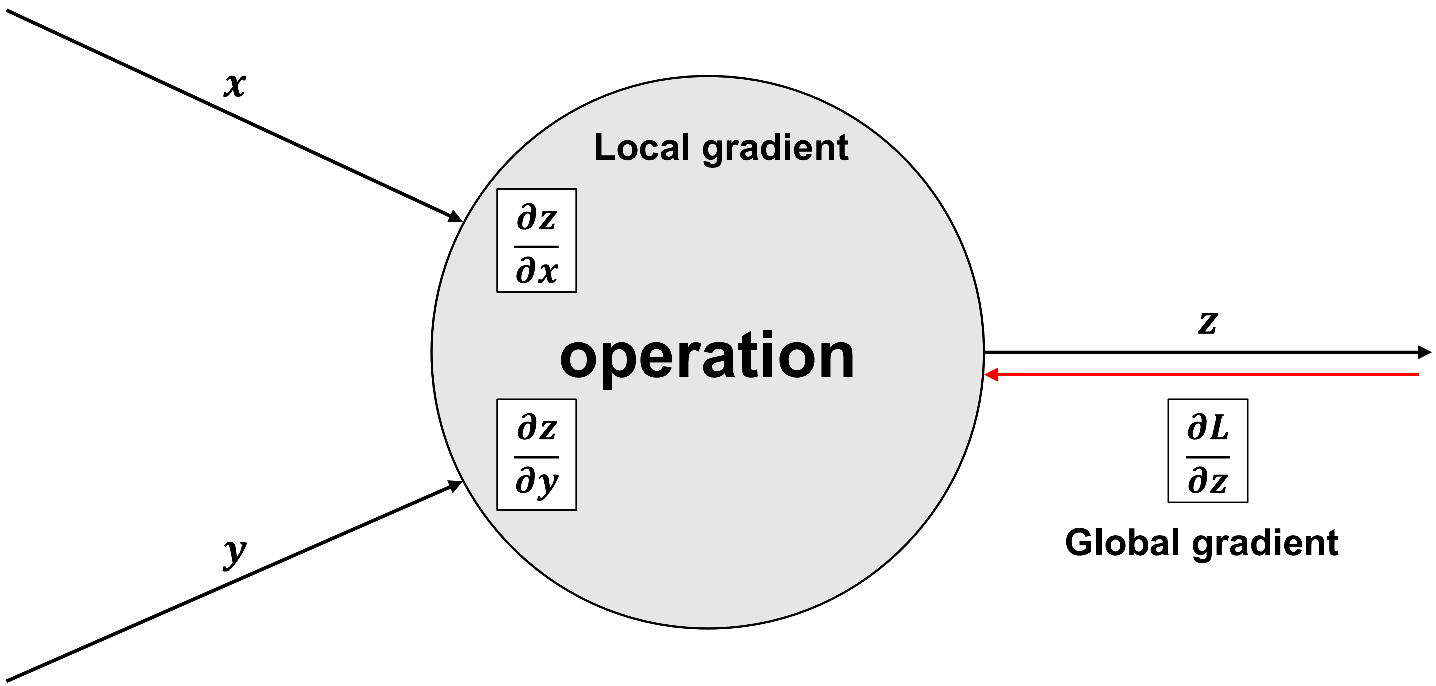

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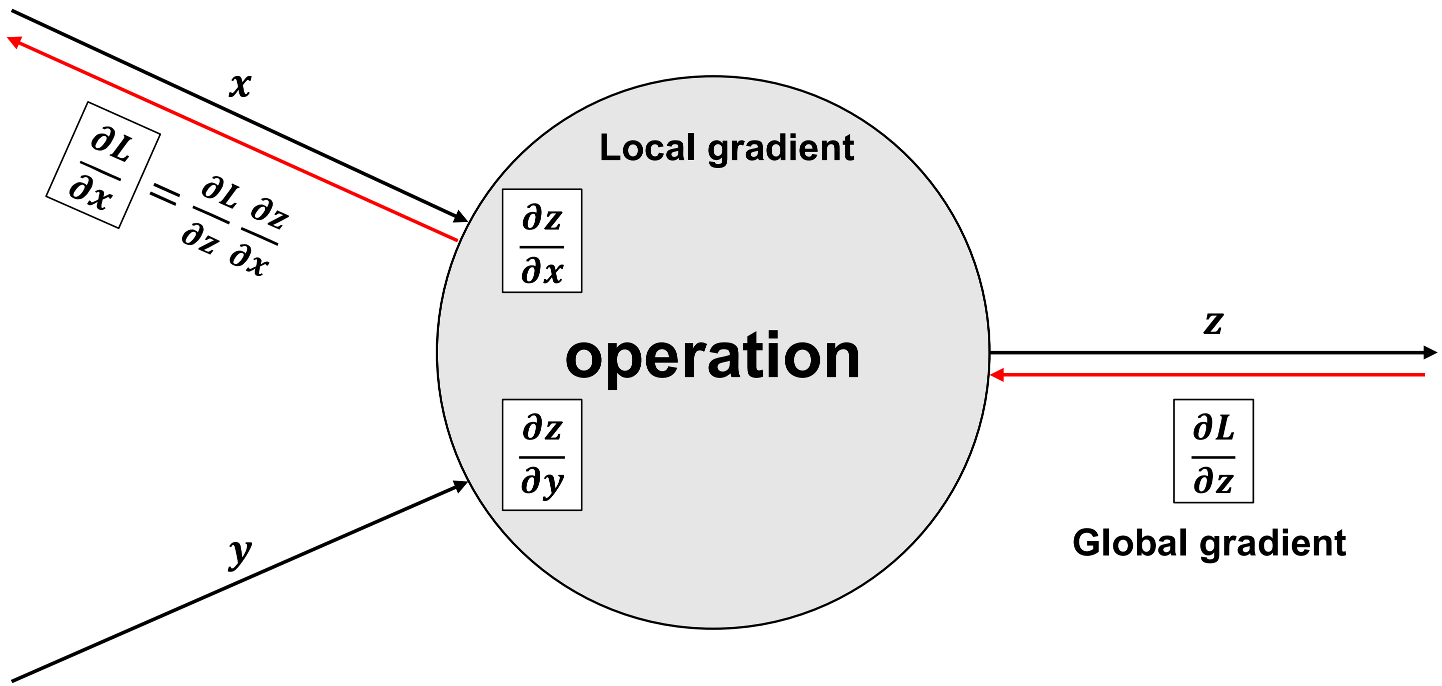

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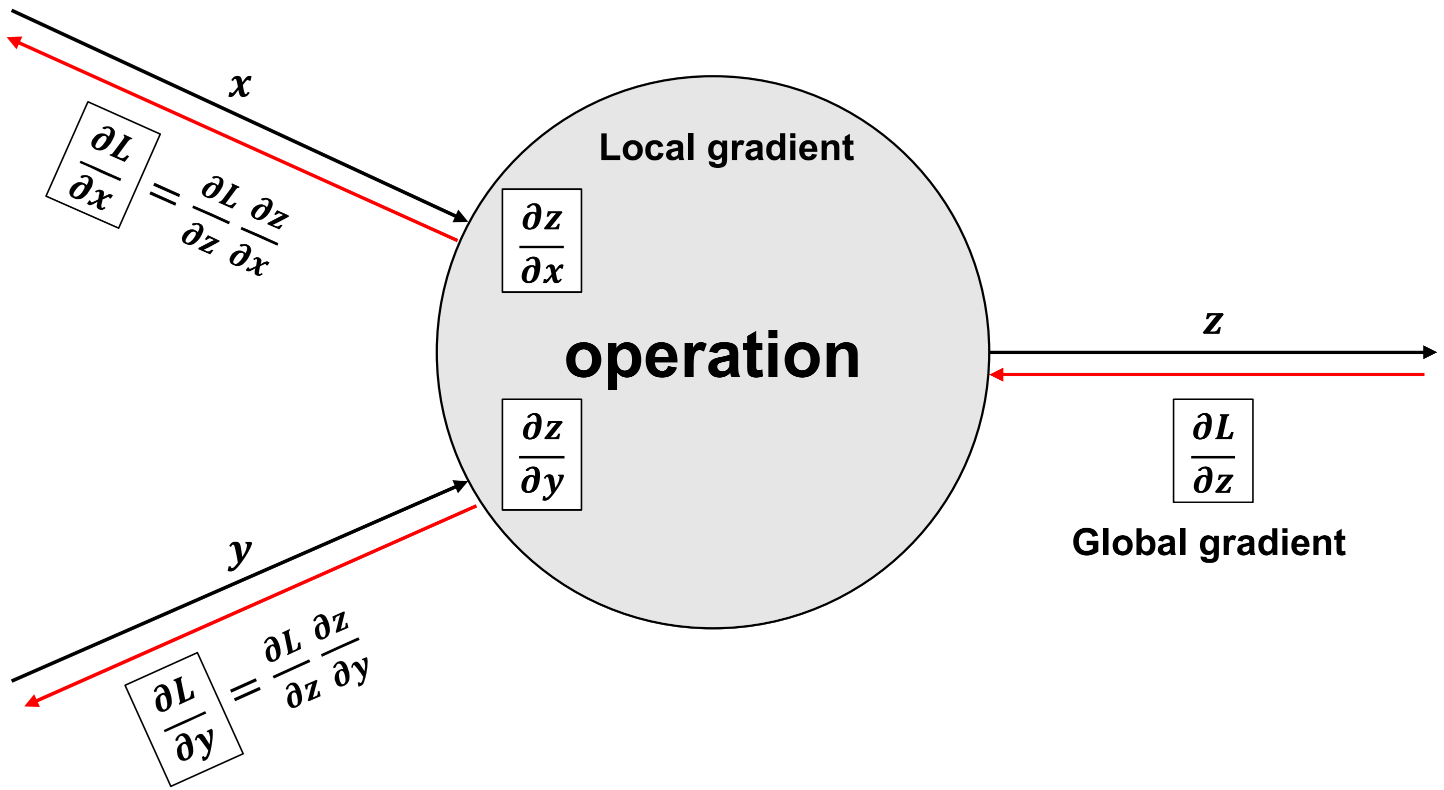

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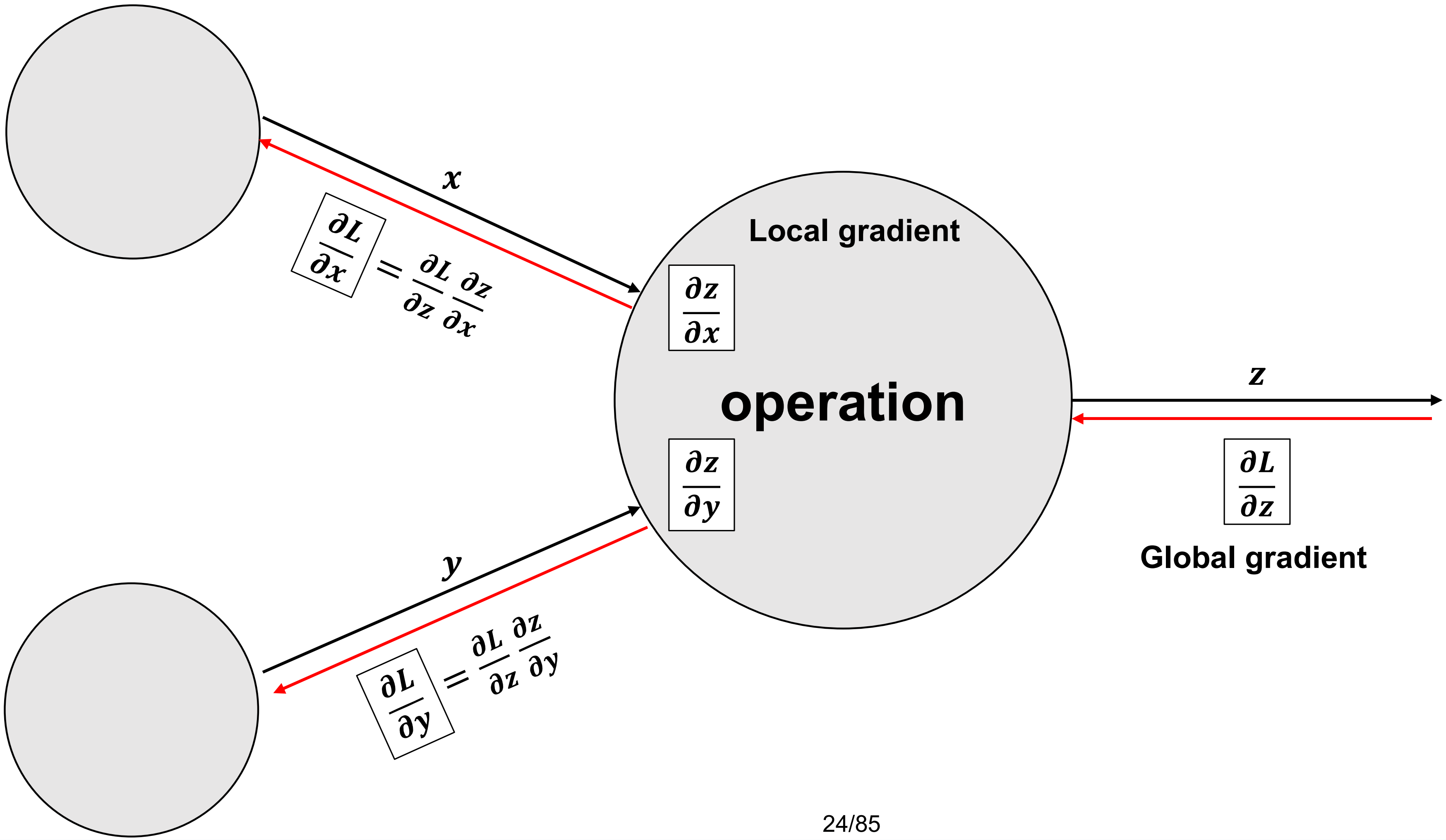


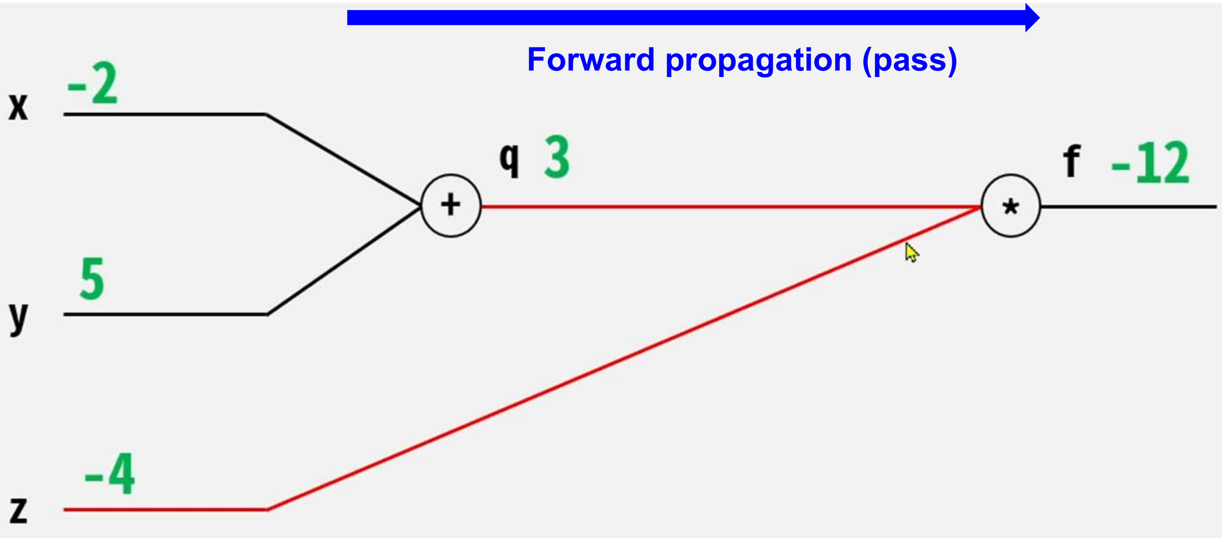
Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

$$q = x + y$$

$$f = qz$$

x=-2, y=5, z=-4



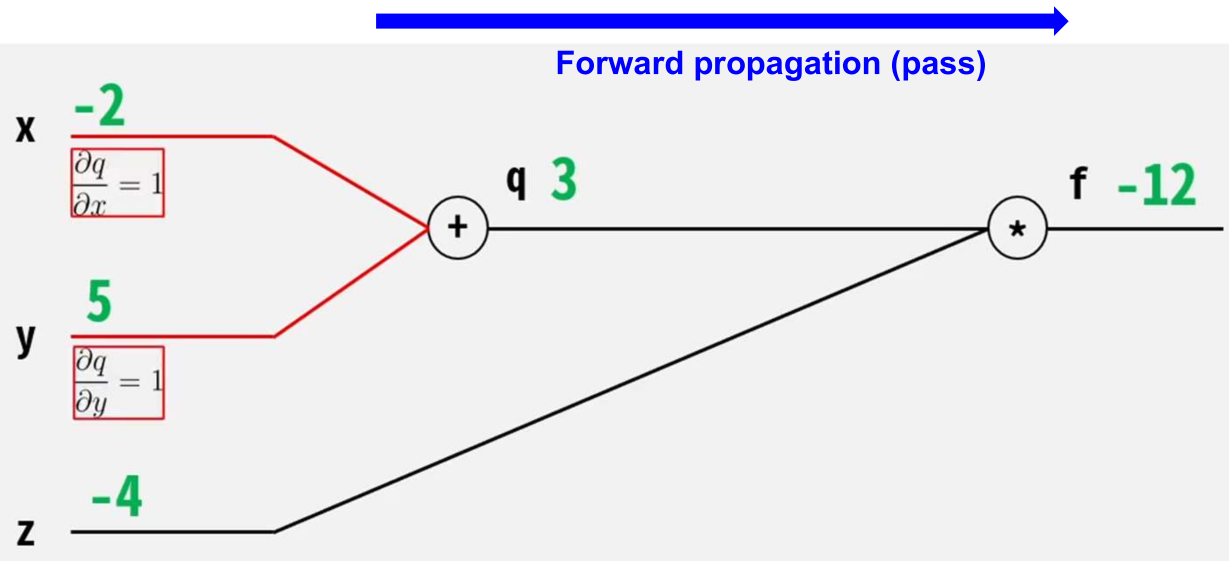
Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

x=-2, y=5, z=-4

$$q = x + y$$

$$\frac{\partial q}{\partial x} = 1 \quad \frac{\partial q}{\partial y} = 1$$



Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

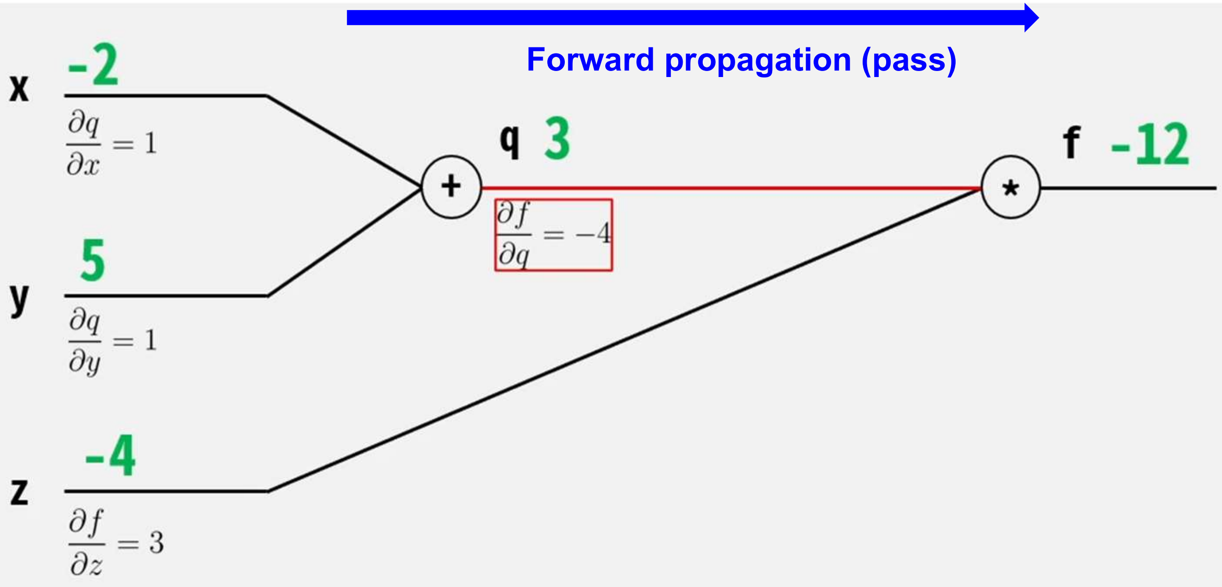
x=-2, y=5, z=-4

$$q = x + y$$

$$f = qz$$

$$\boxed{\frac{\partial f}{\partial q} = z = -4}$$

$$\frac{\partial f}{\partial z} = q = (x + y) = (-2 + 5) = 3$$



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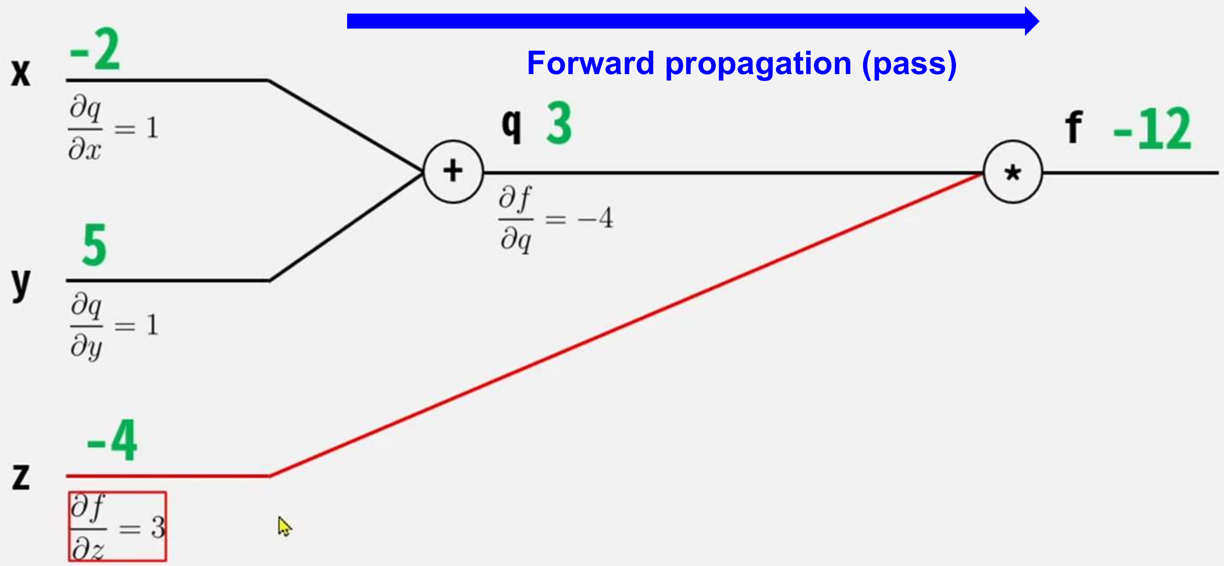


Backpropagation 동작원리

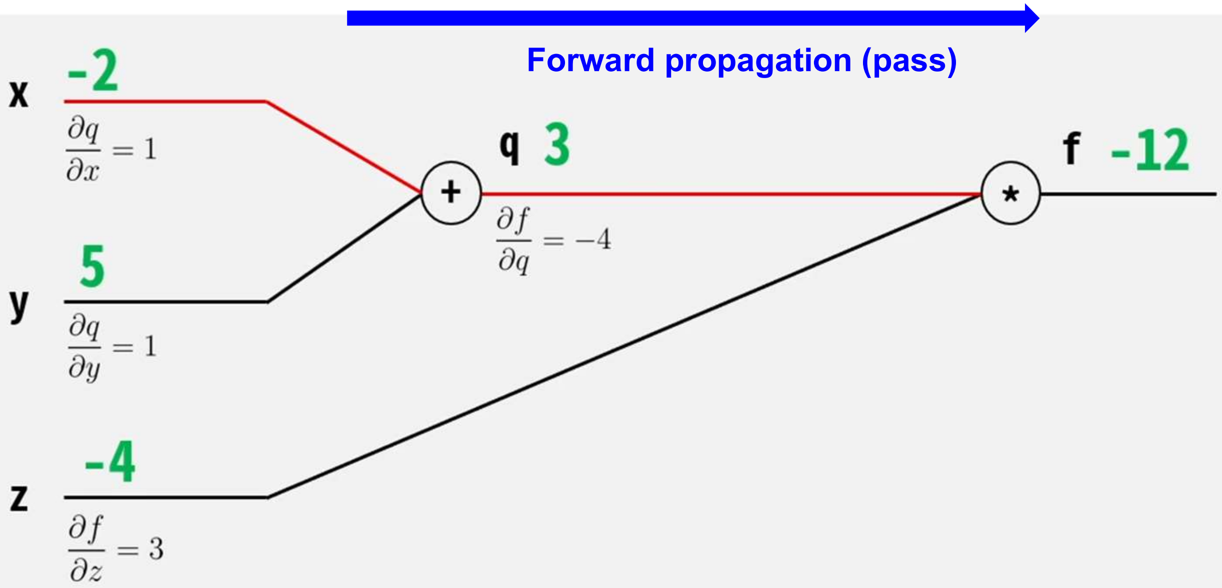
$$f(x, y, z) = (x + y)z$$

x=-2, y=5, z=-4

$$q = x + y$$

$$f = qz$$

$$\boxed{\frac{\partial f}{\partial x} = ?}$$



Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

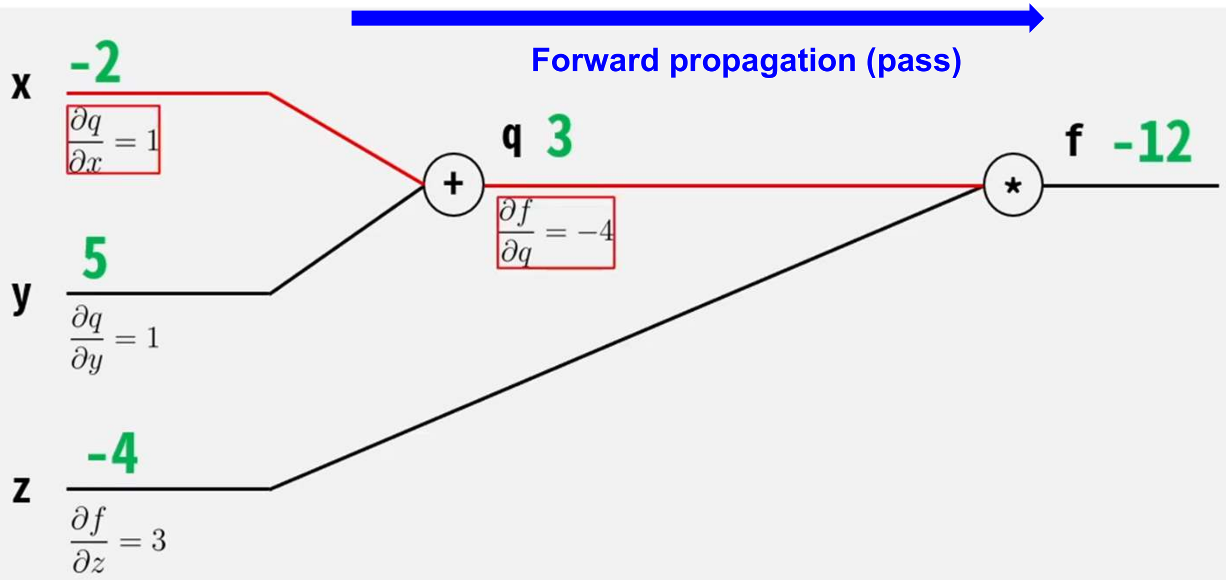
x=-2, y=5, z=-4

$$q = x + y$$

$$f = qz$$

Chain Rule

$$\boxed{\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial x}}$$



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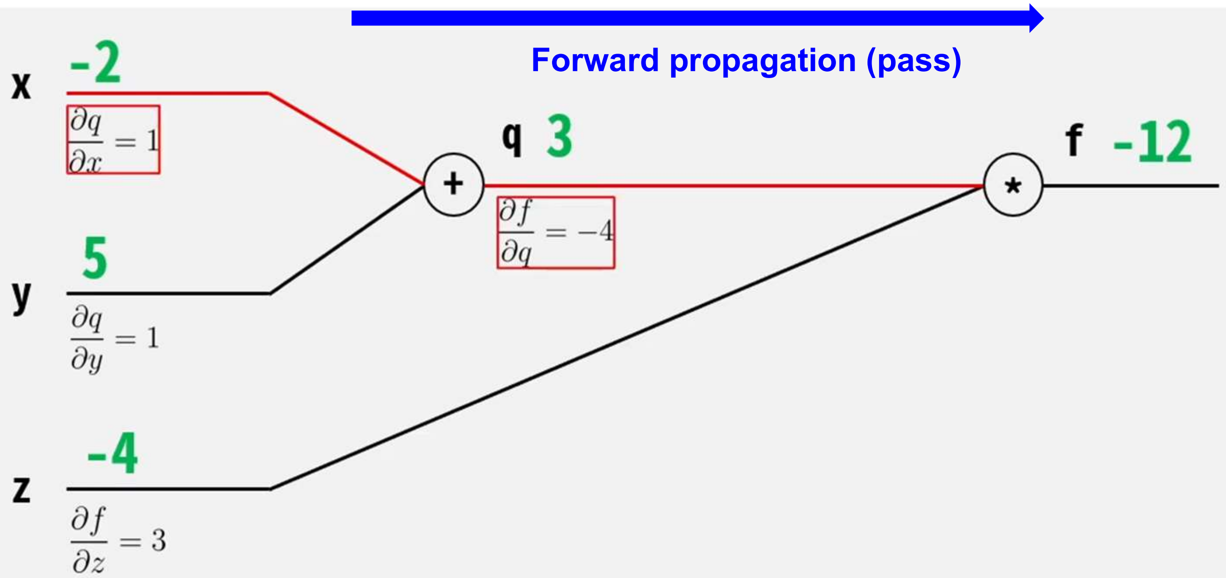
Chain Rule

$$\boxed{\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial x} = (-4)(1) = -4}$$



잠깐!

Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

x=-2, y=5, z=-4

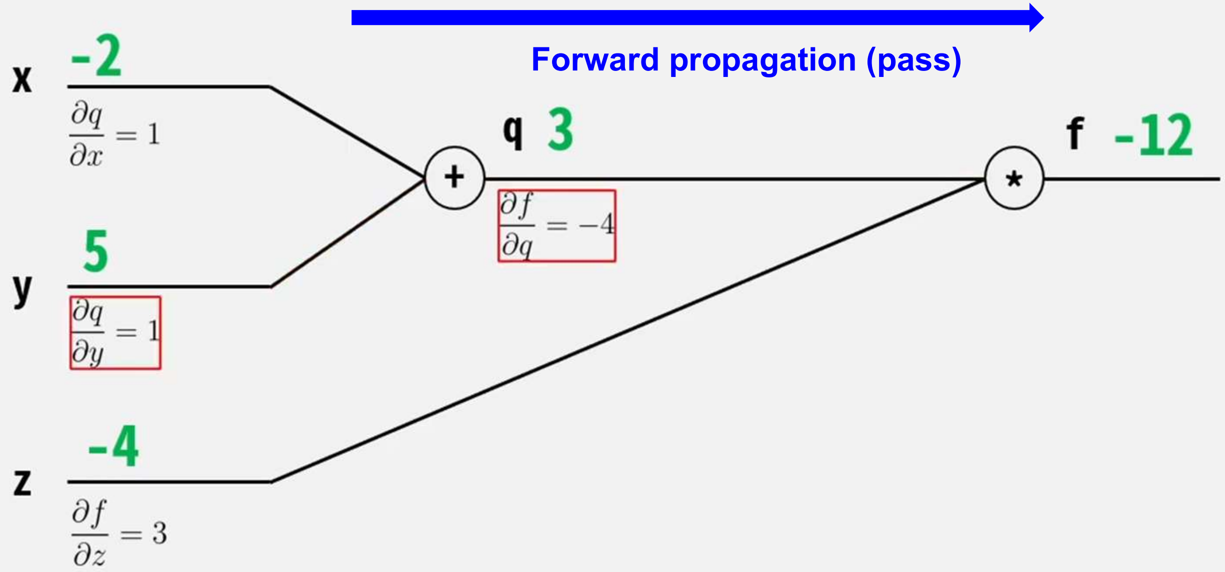
$$q = x + y$$

$$f = qz$$

Chain Rule

$$\frac{\partial f}{\partial x} = -4$$

$$\boxed{\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial y} = (-4)(1) = -4}$$



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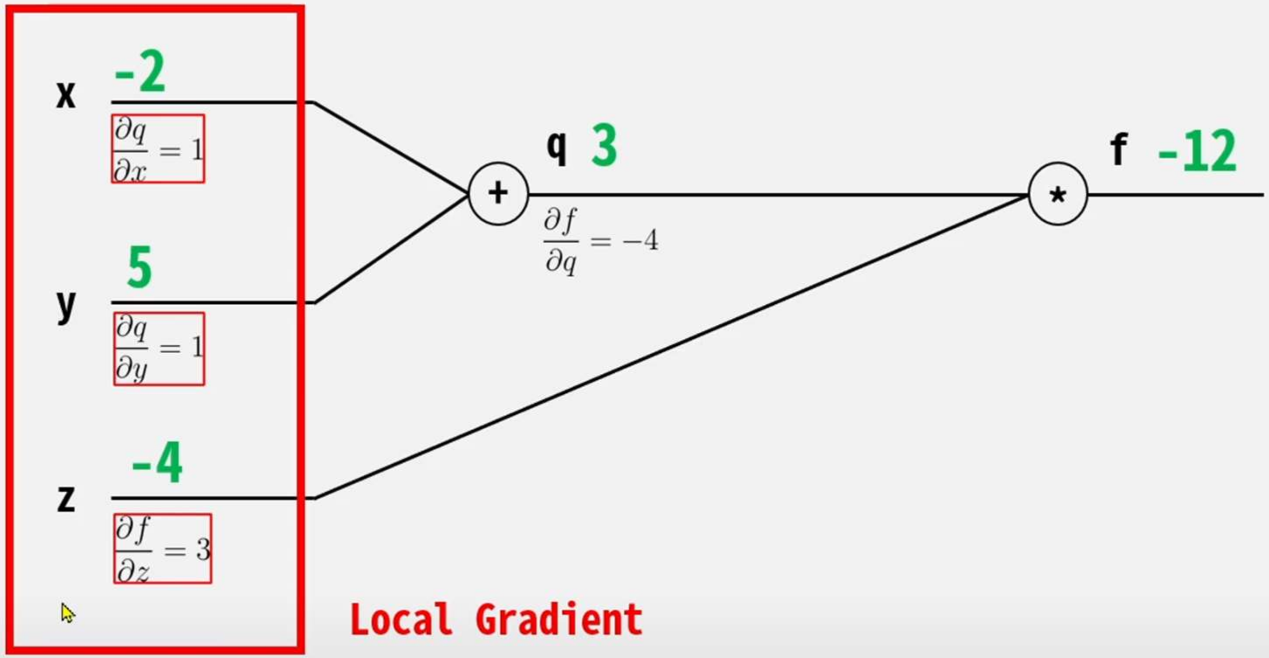

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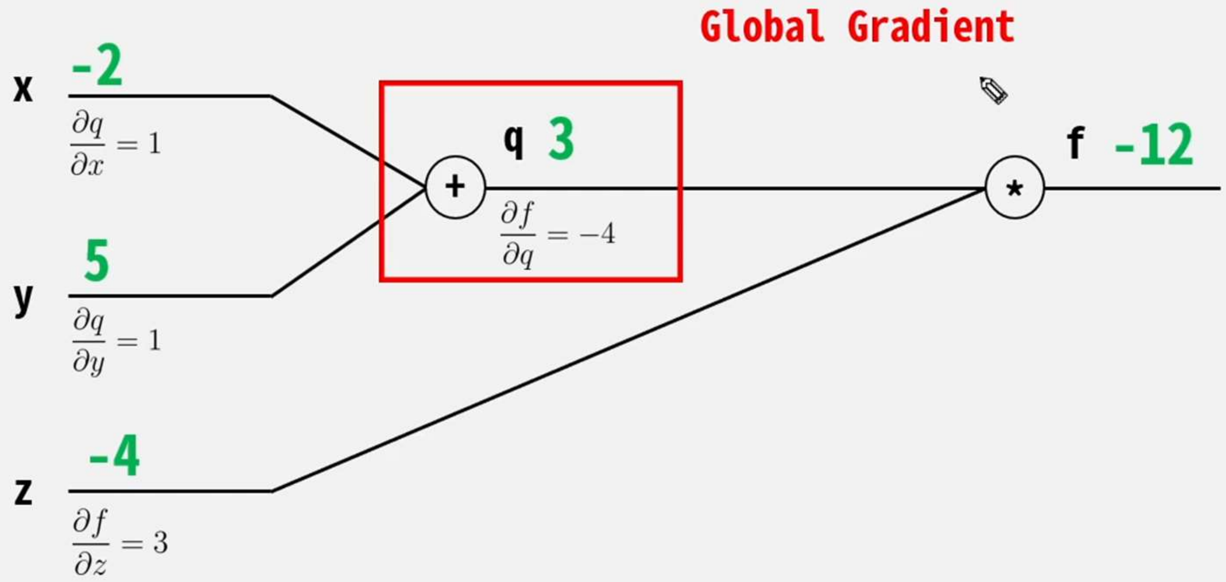

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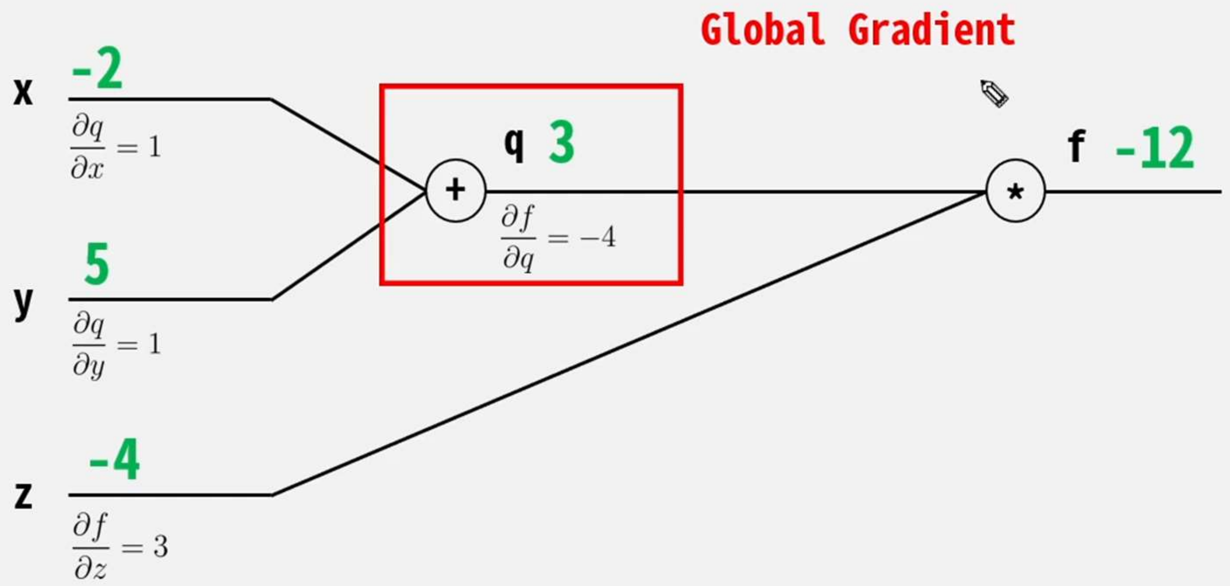


Backpropagation 동작원리

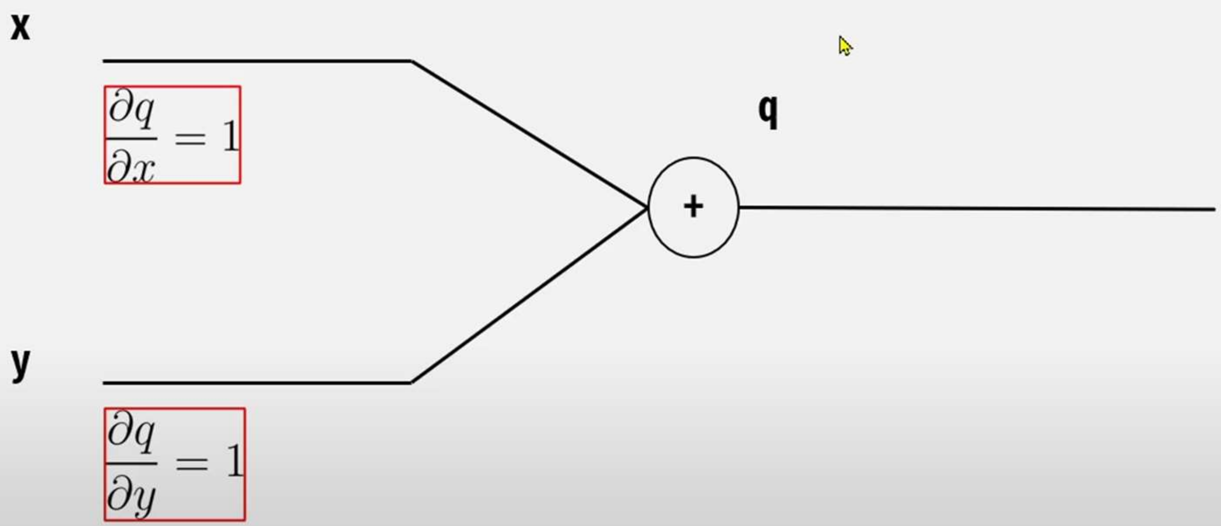
$$q = x + y$$

$$\frac{\partial q}{\partial x} = 1 \qquad \frac{\partial q}{\partial y} = 1$$

$$f = qz$$

$$\frac{\partial f}{\partial q} = z \qquad \frac{\partial f}{\partial z} = q$$

Forward Pass 시 우리는 Local Gradient를 미리 구하여 저장할 수 있다!



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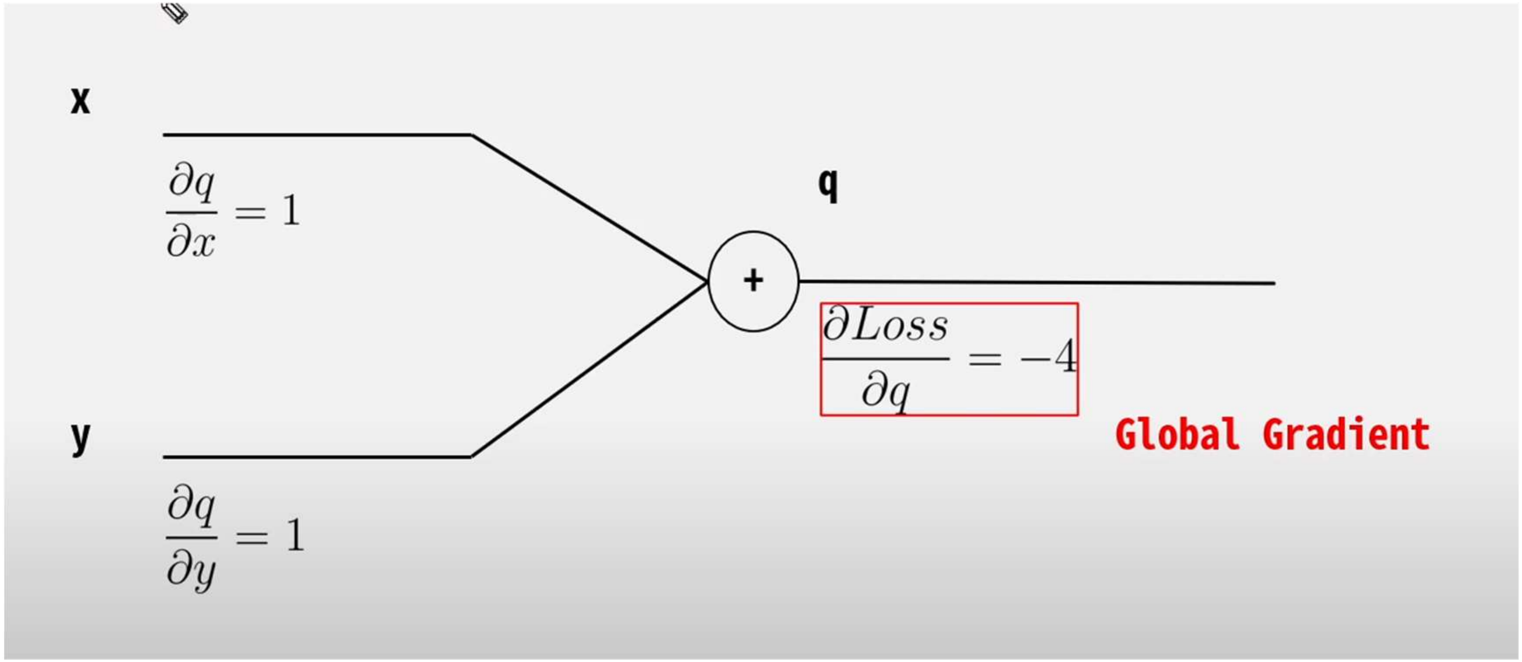


Backpropagation 동작원리

Chain Rule 을 활용하여
Local Gradient * Global Gradient를 곱하여 계산한다.

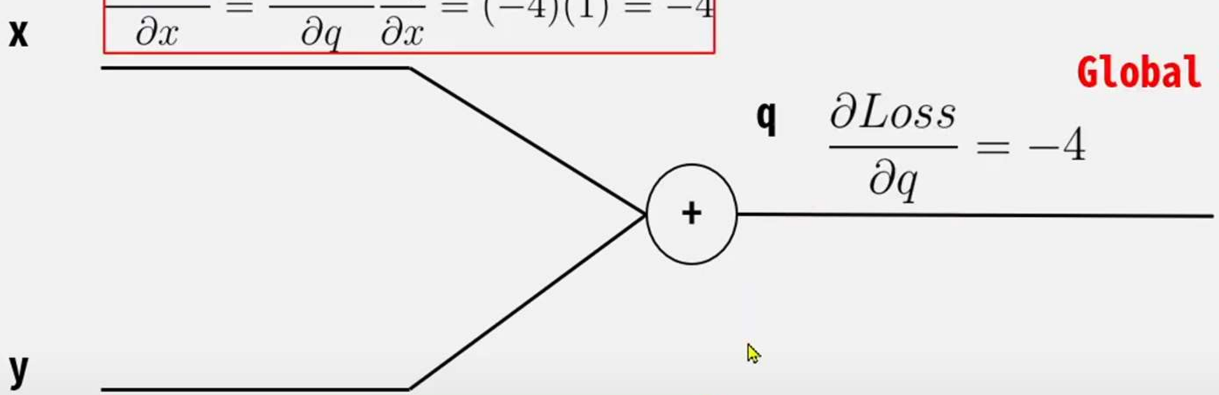
$$\frac{\partial q}{\partial x} = 1$$

Local Gradient

$$\boxed{\frac{\partial Loss}{\partial x} = \frac{\partial Loss}{\partial q}\frac{\partial q}{\partial x} = (-4)(1) = -4}$$

Global Gradient

$$\frac{\partial Loss}{\partial q} = -4$$



$$\frac{\partial q}{\partial y} = 1$$

Local Gradient

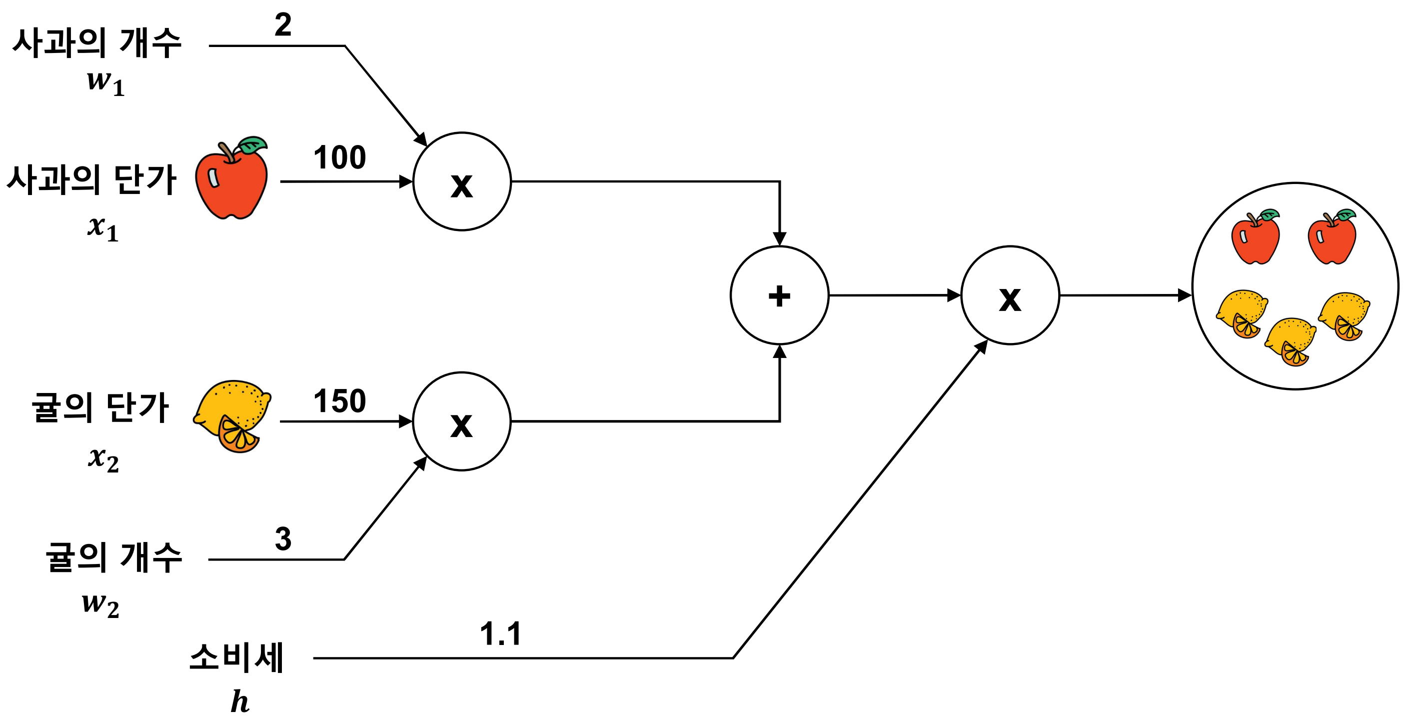
$$\boxed{\frac{\partial Loss}{\partial y} = \frac{\partial Loss}{\partial q}\frac{\partial q}{\partial y} = (-4)(1) = -4}$$

Backpropagation 동작원리

- 아무리 깊고 복잡한 층으로 구성되어 있다 하더라도 Chain Rule을 활용하여 미분 값을 얻어낼 수 있다.
- Forward Pass시 Local Gradient를 미리 계산하여 저장해 둔다.
- 저장해둔 Local Gradient와 Global Gradient를 Backward Pass시 곱하여 최종 미분 값을 얻는다.

Backpropagation 동작원리

- 예제1: 슈퍼에서 사과를 2개, 귤을 3개 구매 시 지불금액은?
  - ✓ 사과는 1개 100원, 귤은 1개 150원
  - ✓ 소비세 10%



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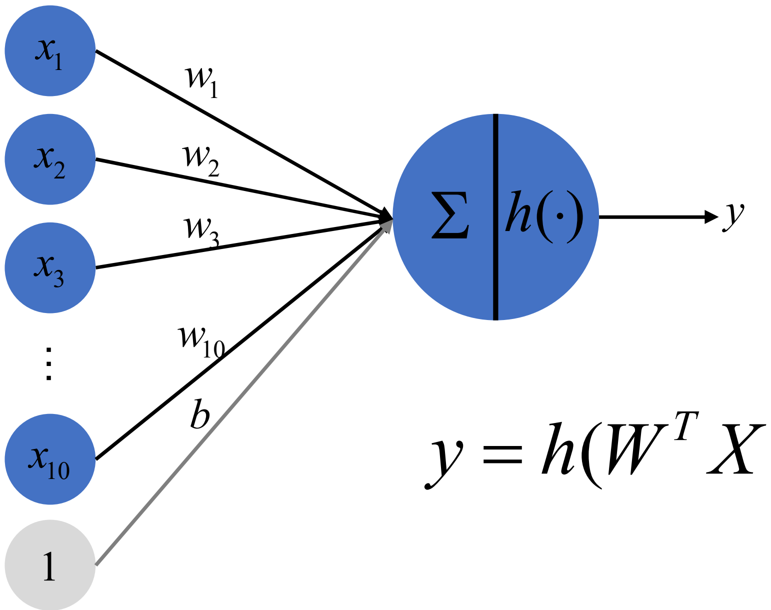
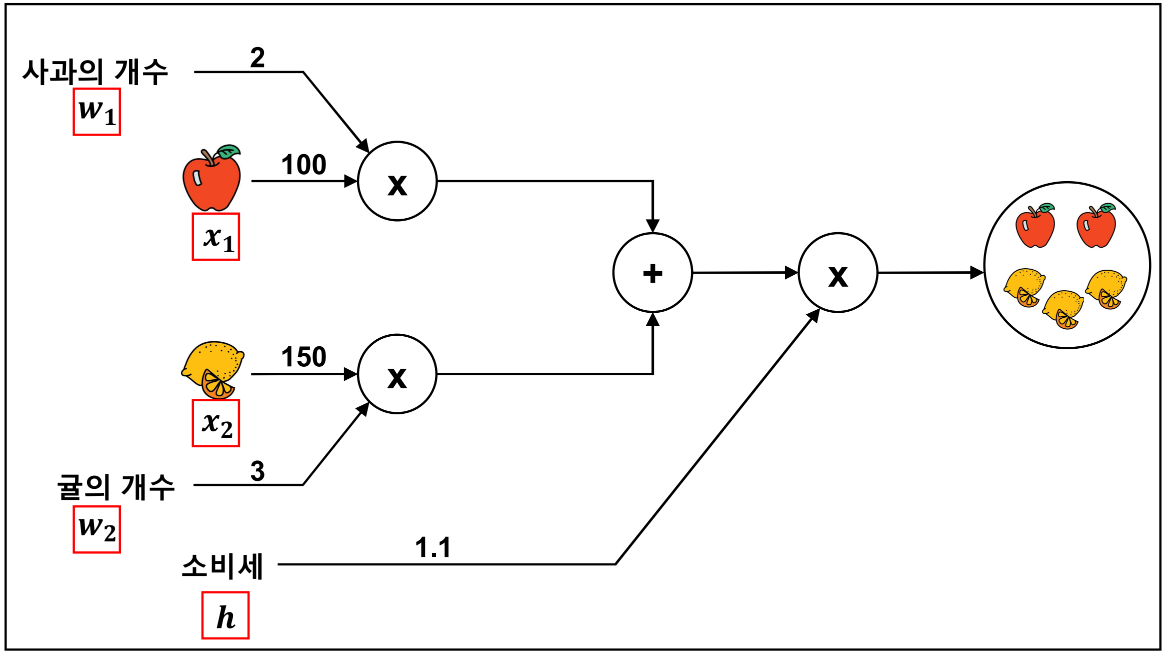


$$y = h(W^T X + b)$$



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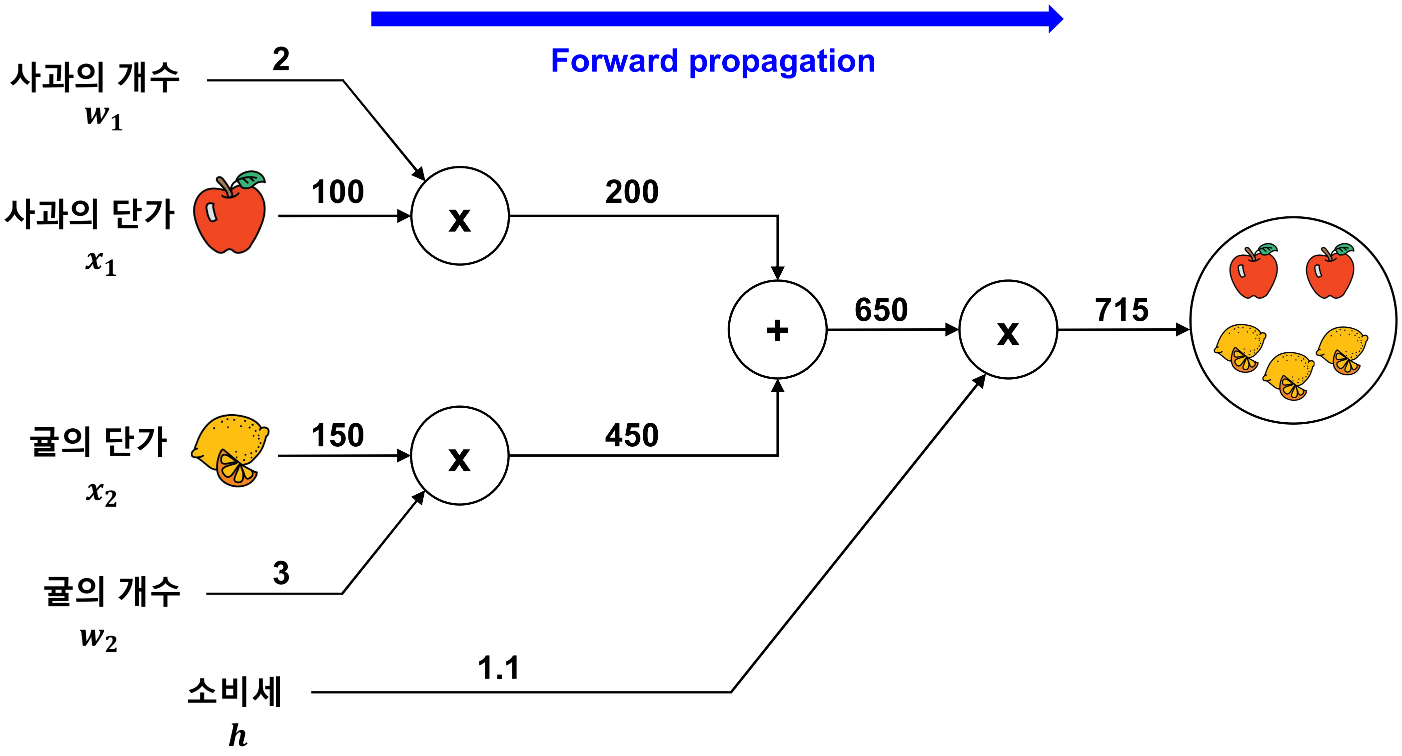

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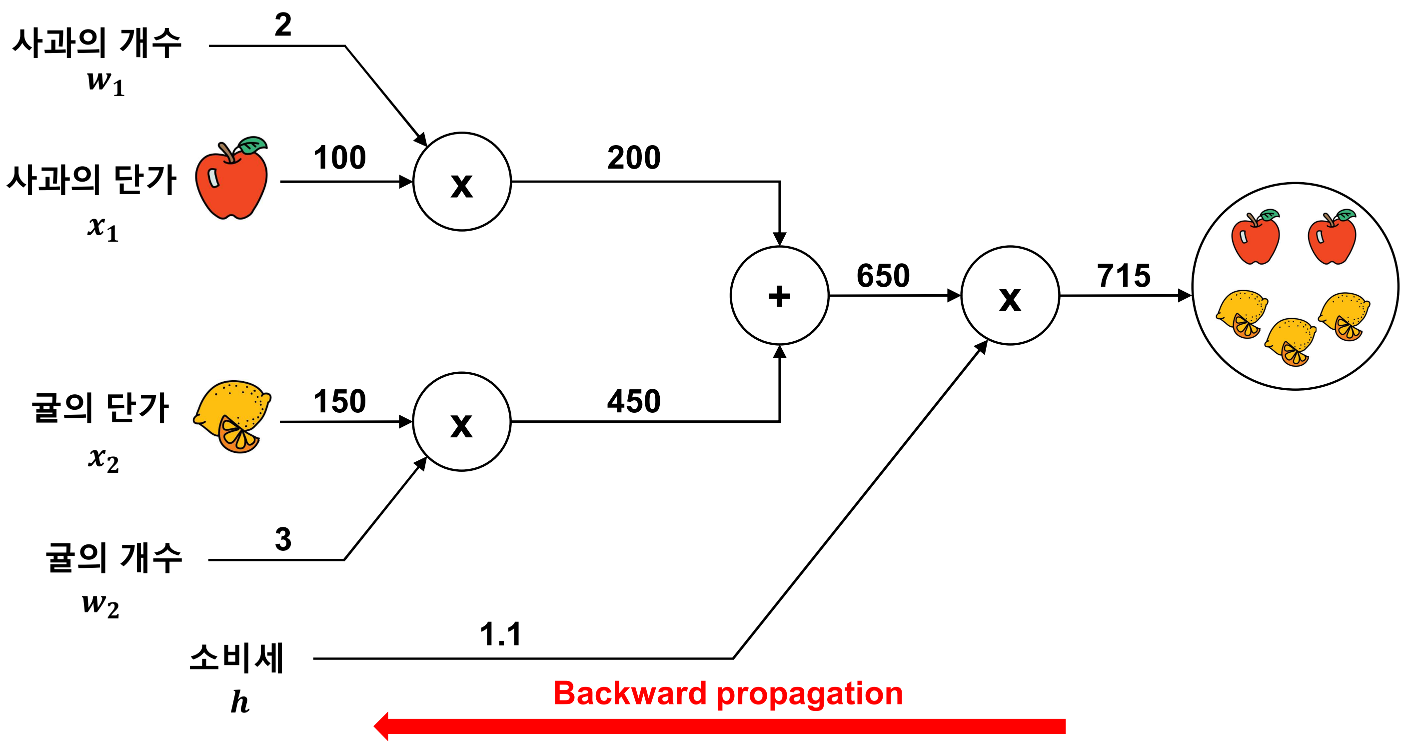

Backpropagation 동작원리

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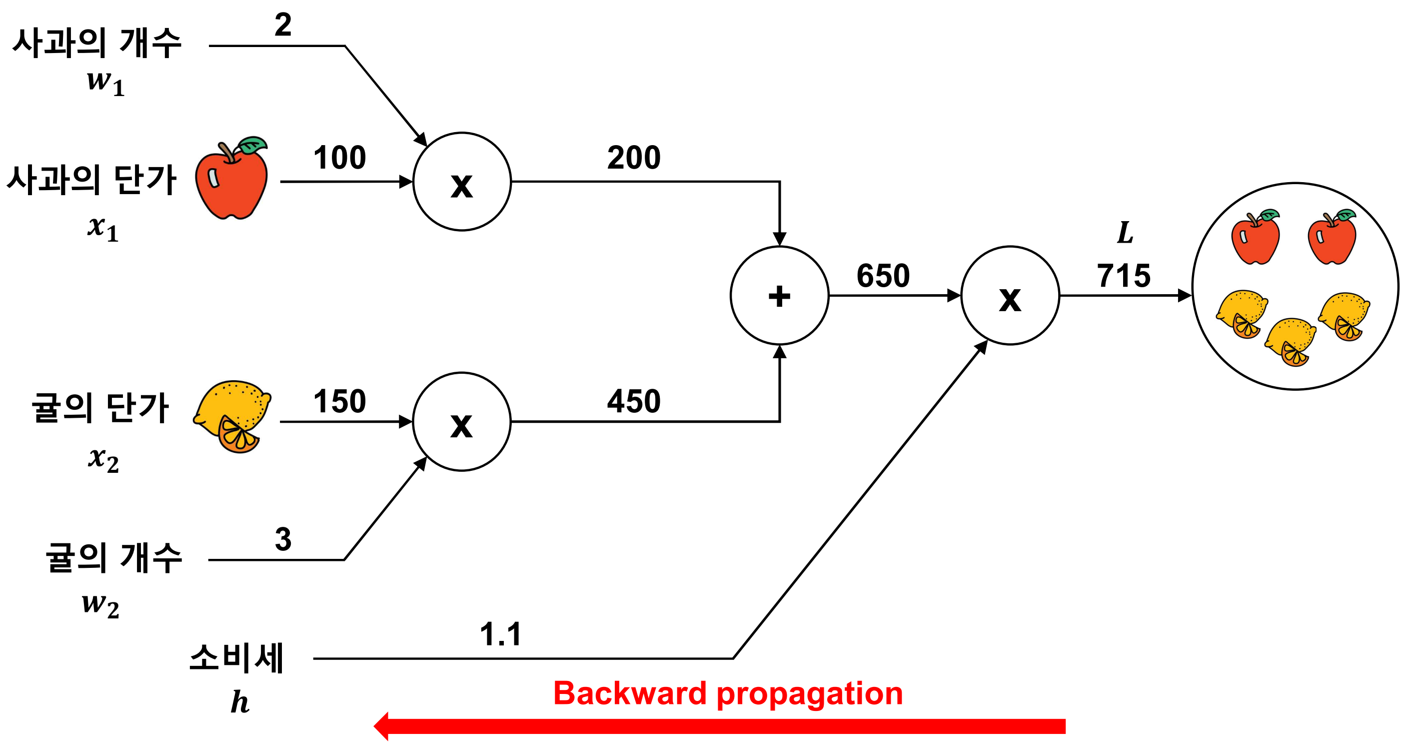

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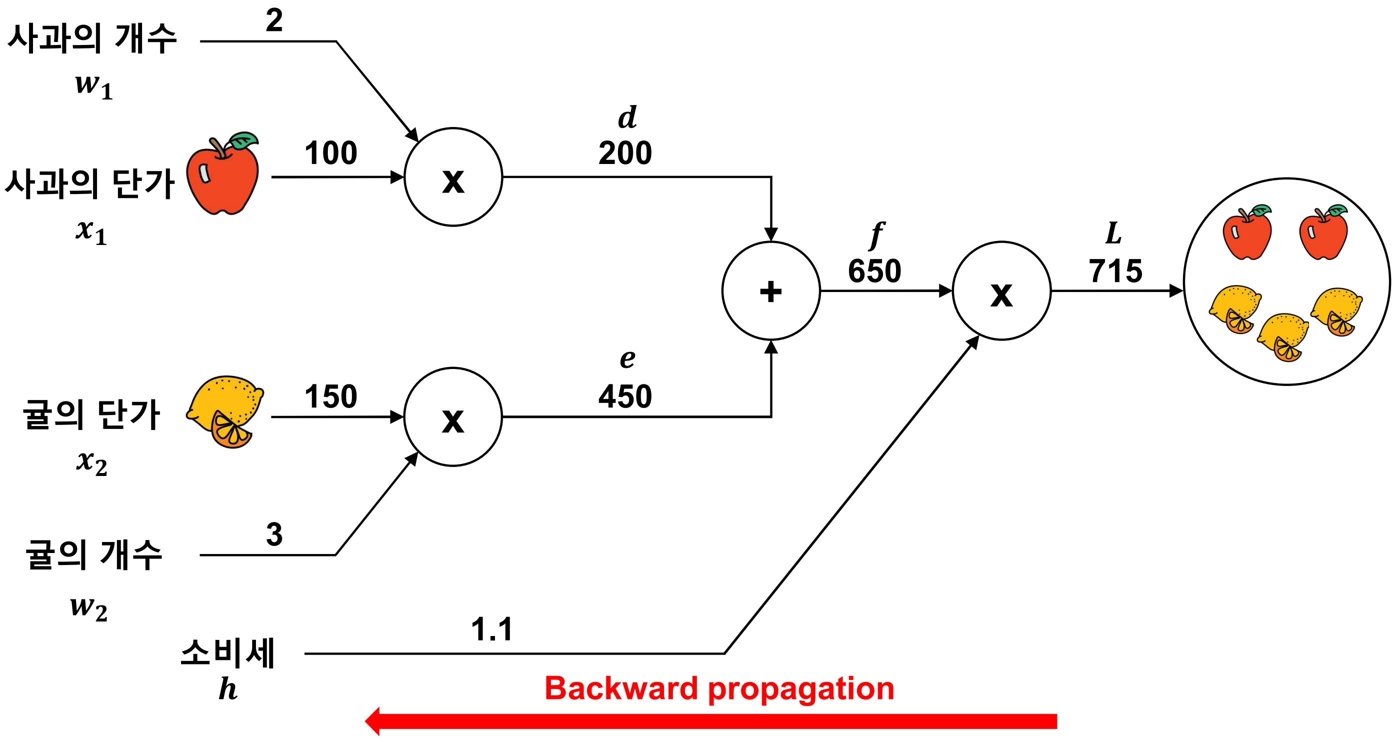

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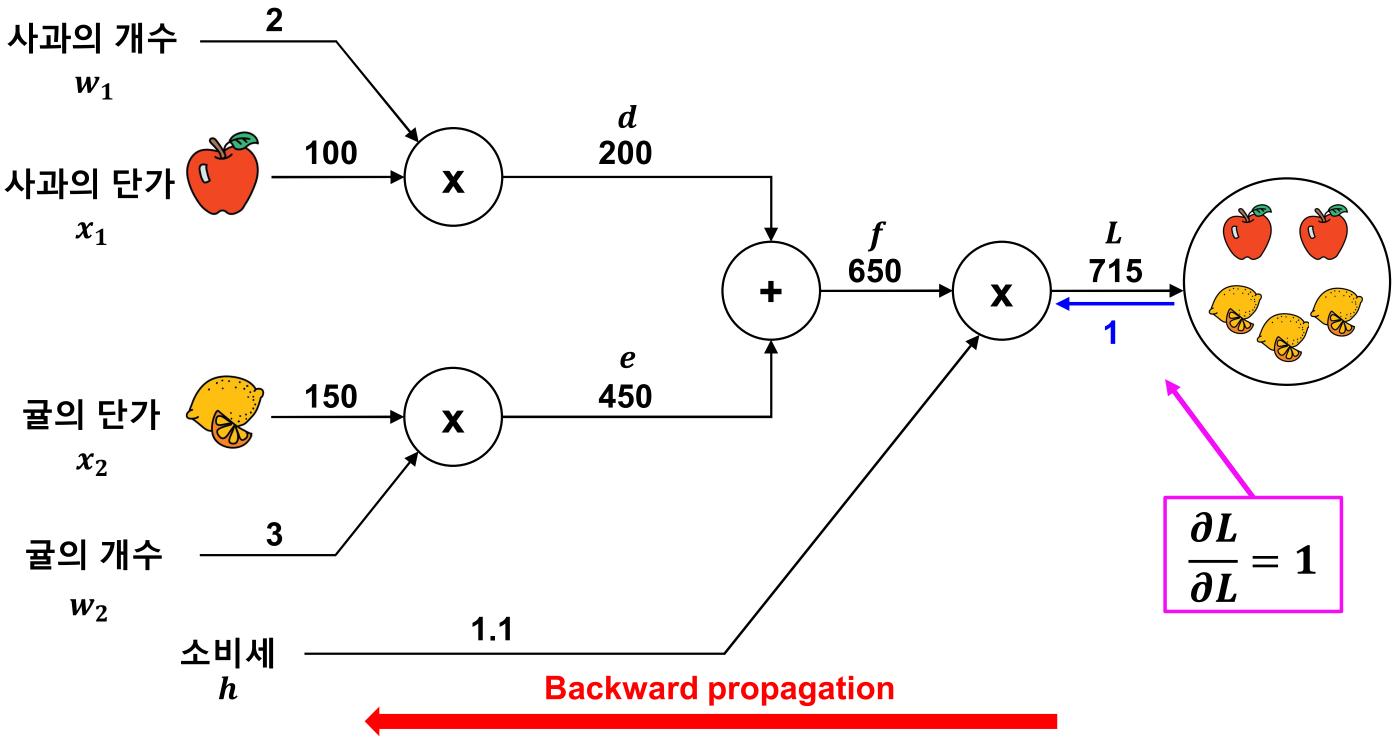

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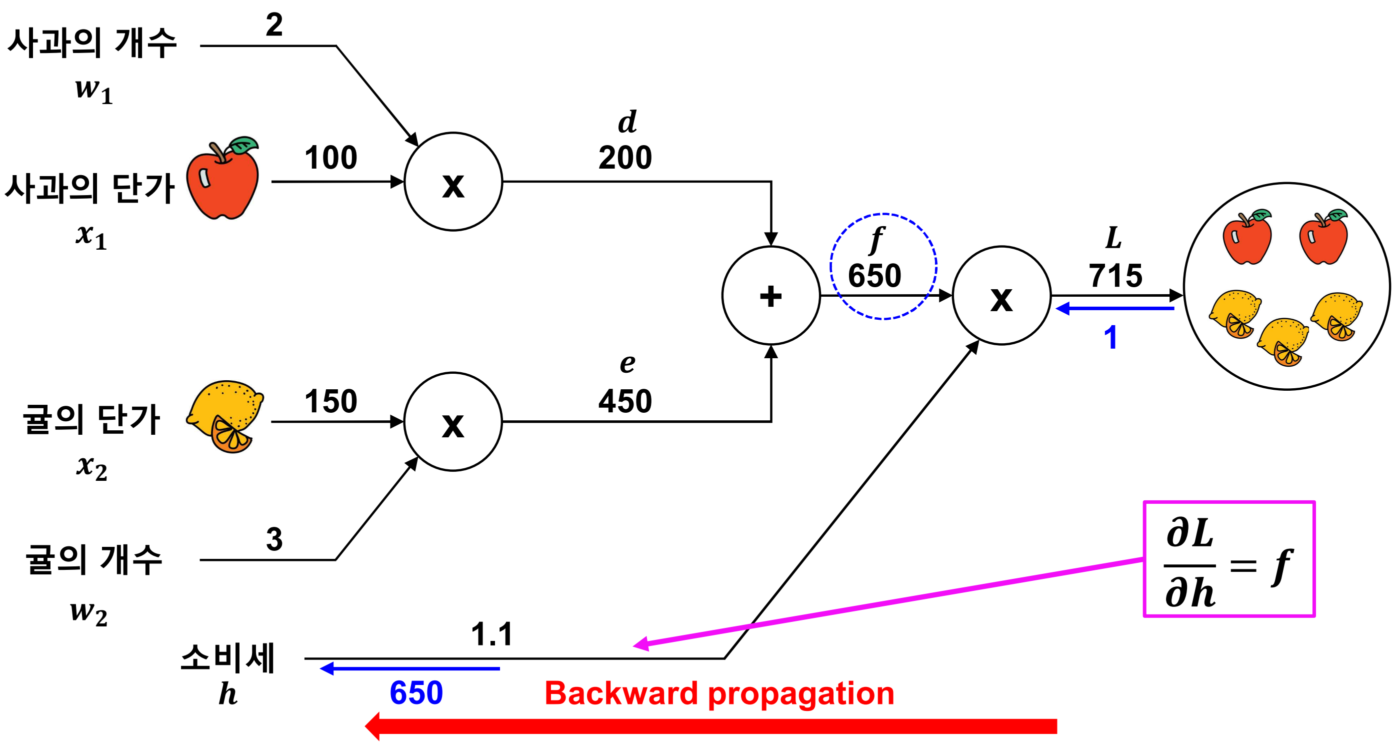

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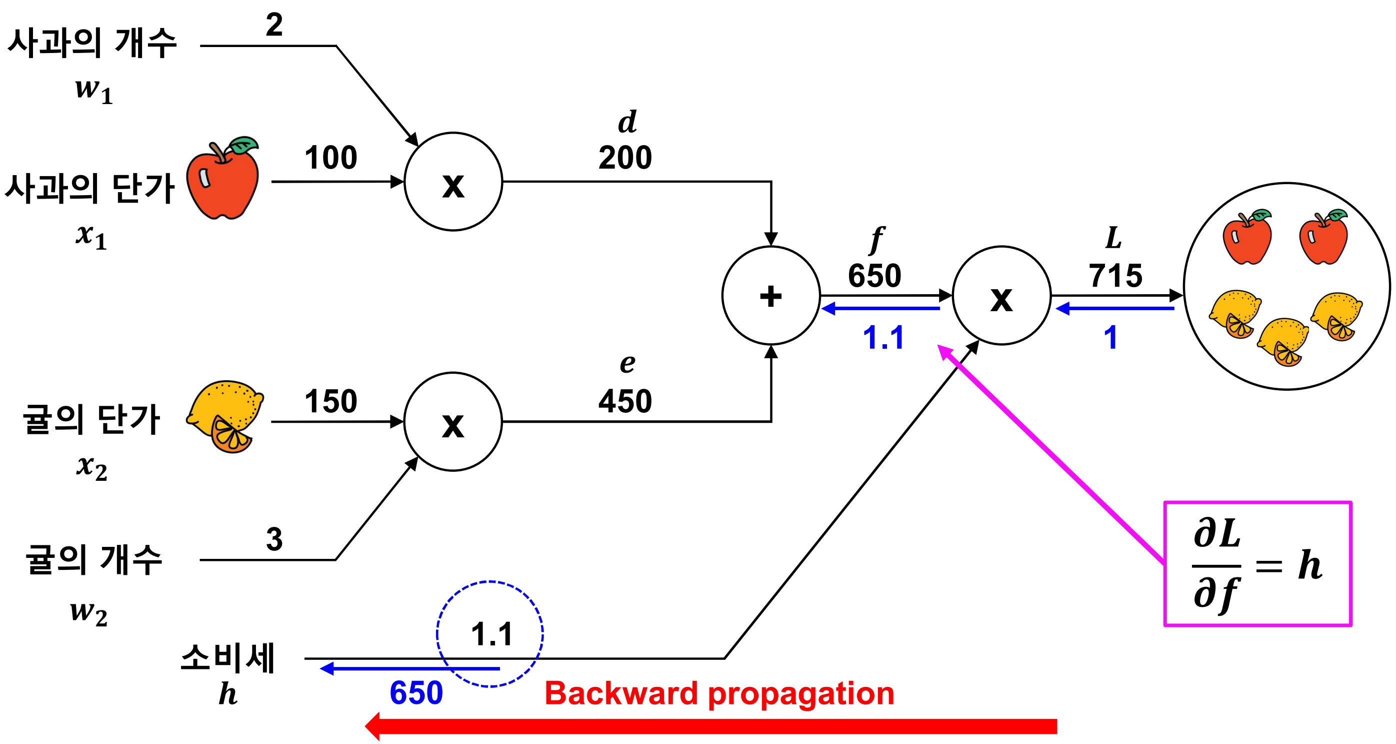

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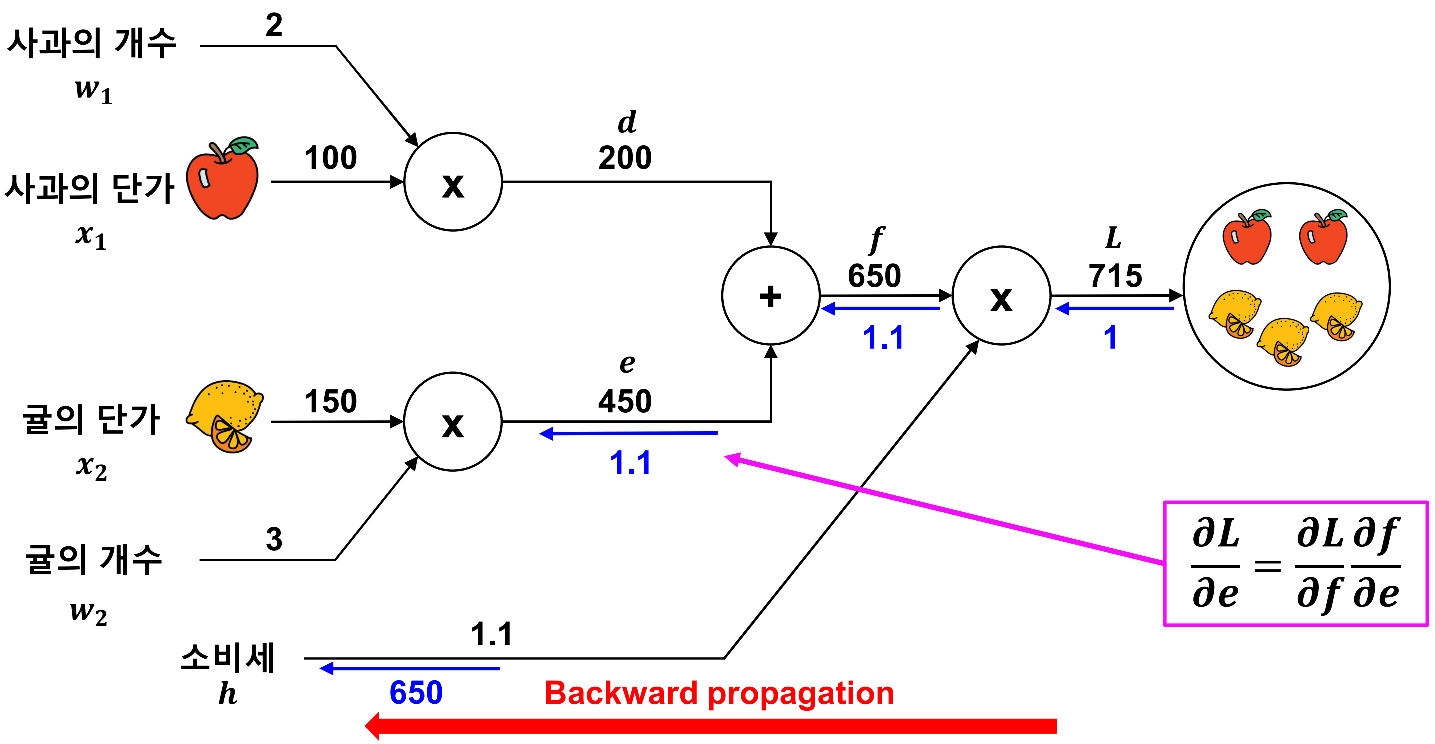

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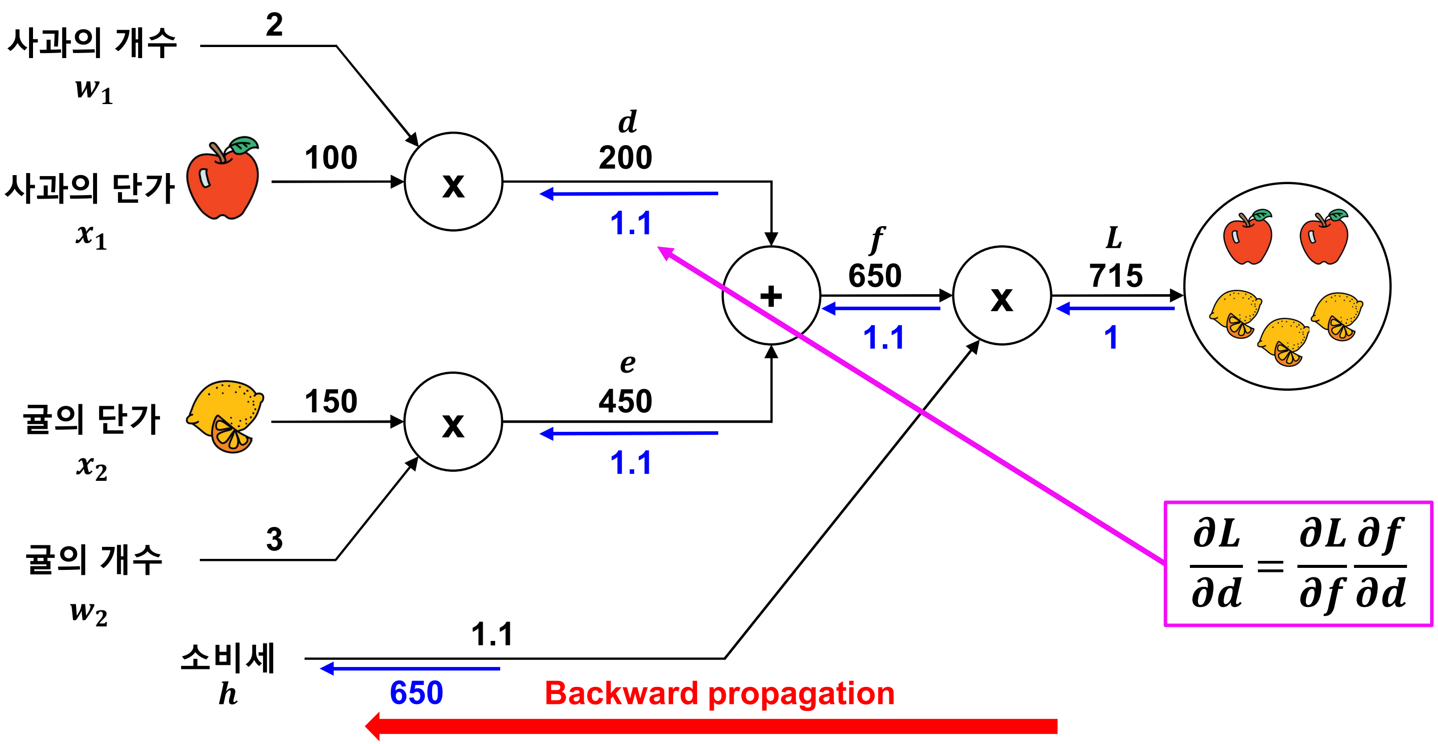

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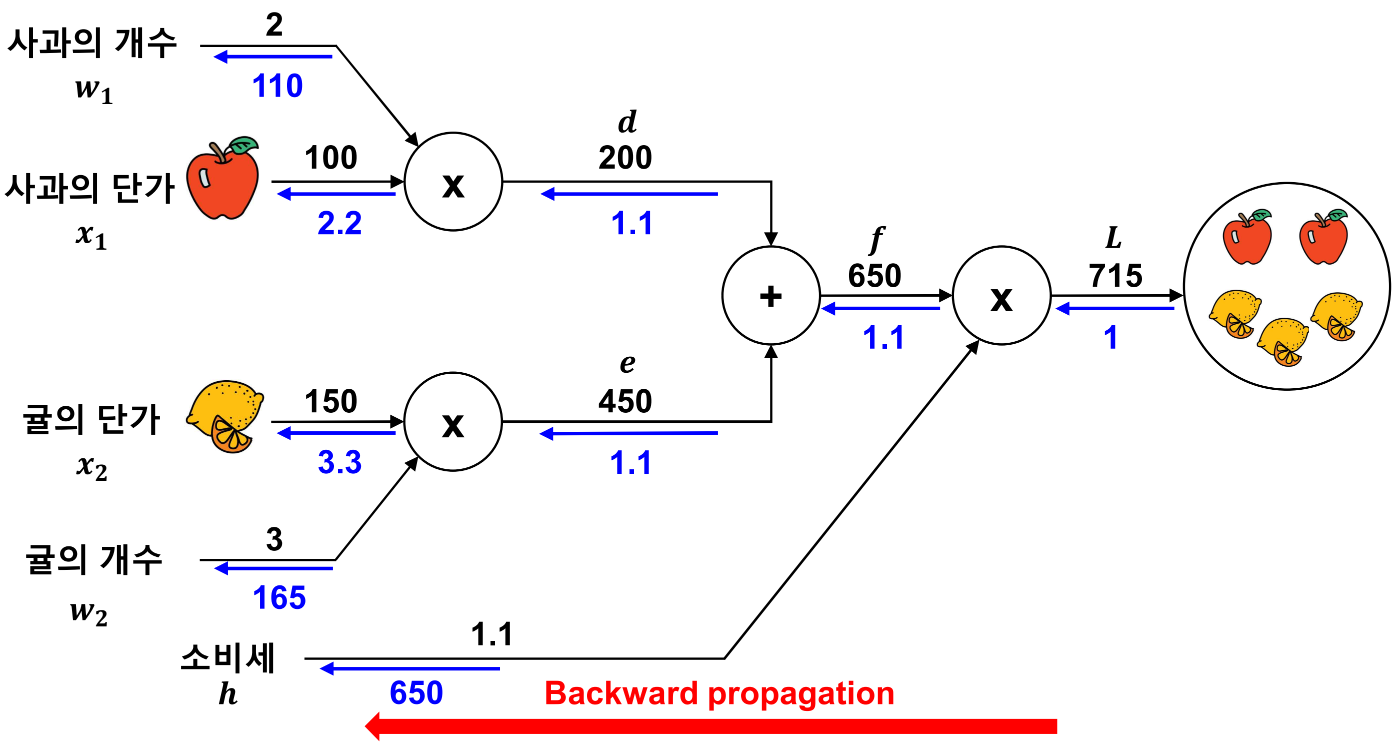

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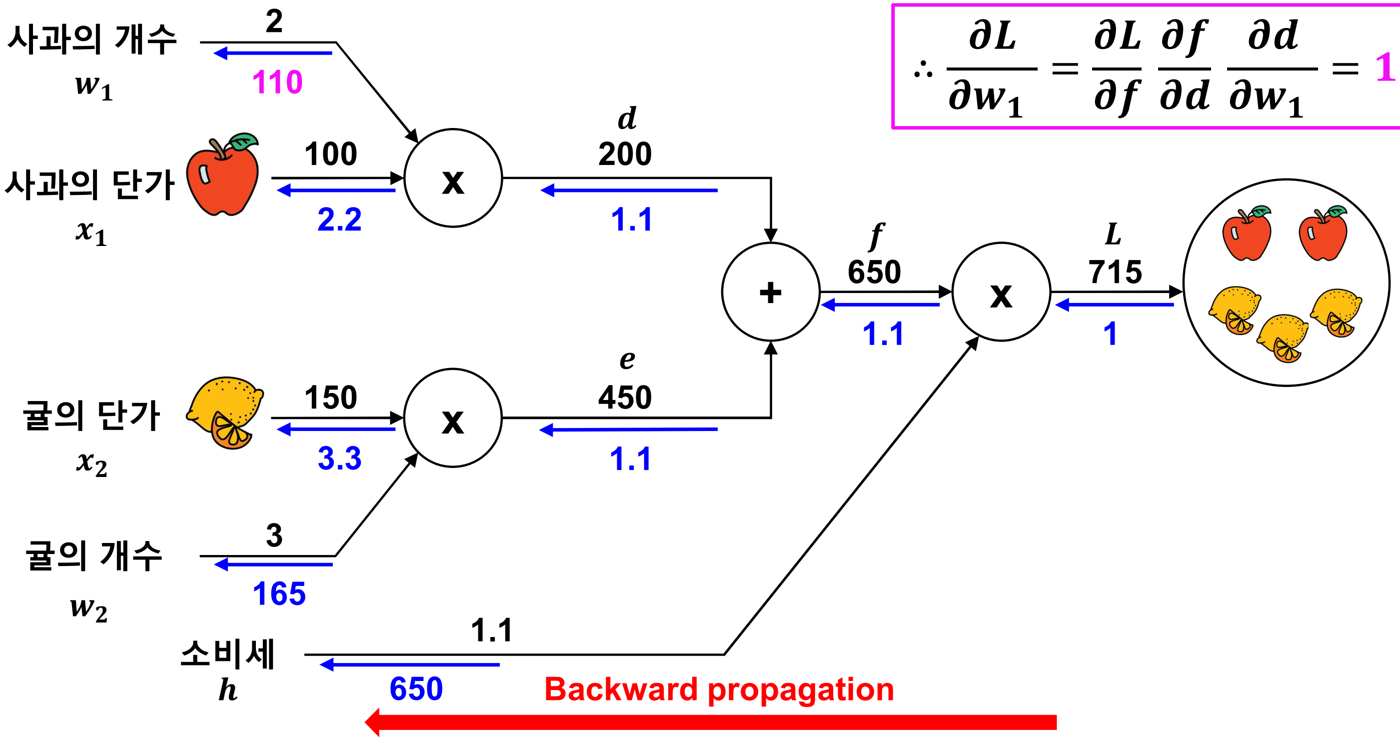
Chain rule

$$\therefore \frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial f}\frac{\partial f}{\partial d}\frac{\partial d}{\partial w_1} = 110$$



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Backpropagation 동작원리

- Weight Update (a: 사과의 개수)

Chain rule

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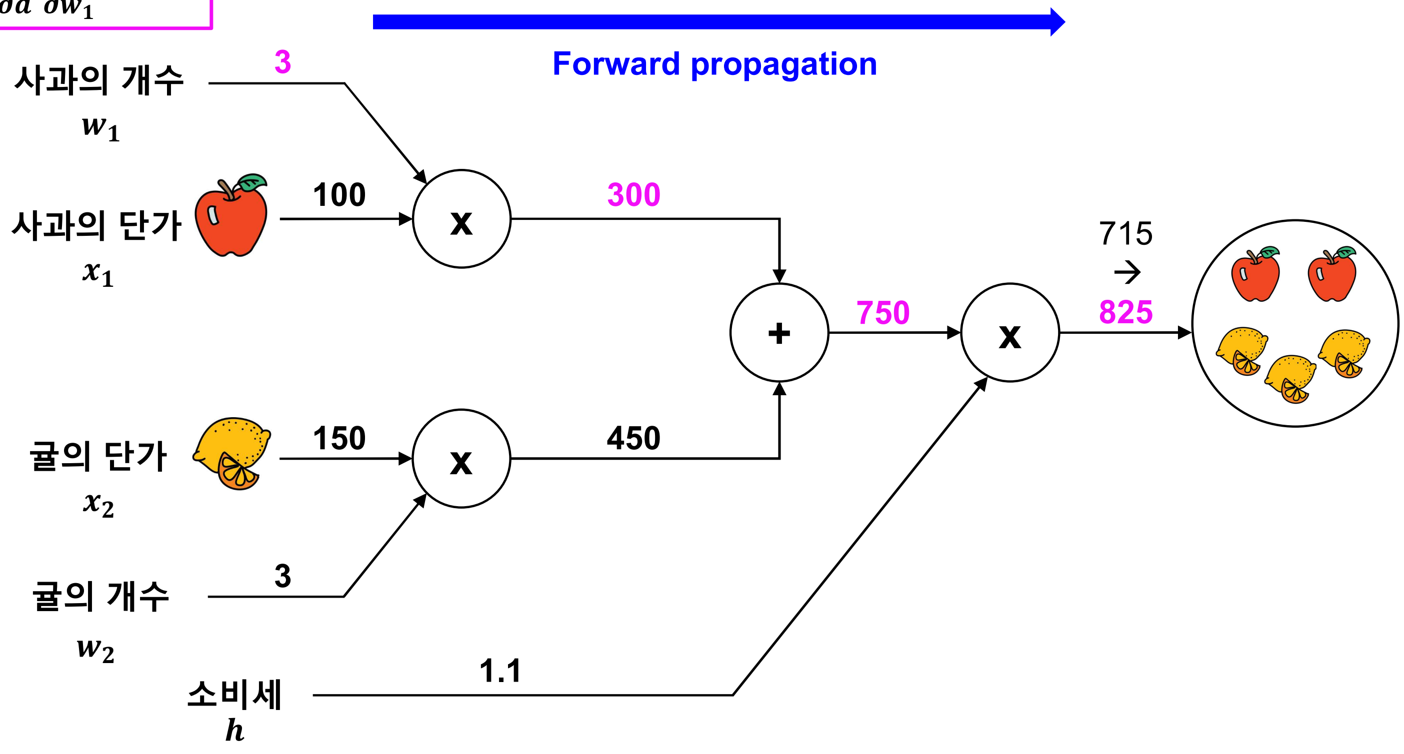
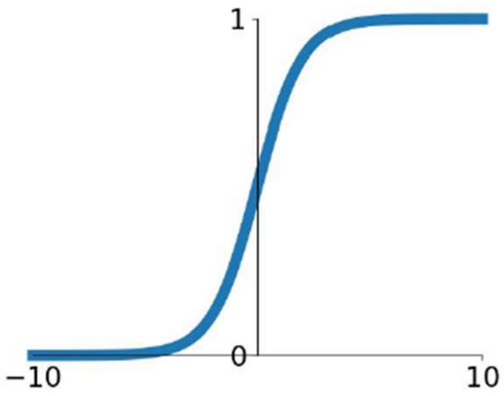


Backpropagation 동작원리

- Sigmoid 계층의 backpropagation



$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\begin{aligned}
\frac{d}{dx} sigmoid(x) &= \frac{d}{dx}(1+e^{-x})^{-1} \\
&= (-1)\frac{1}{(1+e^{-x})^2}\frac{d}{dx}(1+e^{-x}) \\
&= (-1)\frac{1}{(1+e^{-x})^2}(0+e^{-x})\frac{d}{dx}(-x) \\
&= (-1)\frac{1}{(1+e^{-x})^2}e^{-x}(-1) \\
&= \frac{e^{-x}}{(1+e^{-x})^2} \\
&= \frac{1+e^{-x}-1}{(1+e^{-x})^2} \\
&= \frac{(1+e^{-x})}{(1+e^{-x})^2} - \frac{1}{(1+e^{-x})^2} \\
&= \frac{1}{1+e^{-x}} - \frac{1}{(1+e^{-x})^2} \\
&= \frac{1}{1+e^{-x}}(1 - \frac{1}{1+e^{-x}}) \\
&= sigmoid(x)(1 - sigmoid(x))
\end{aligned}$$

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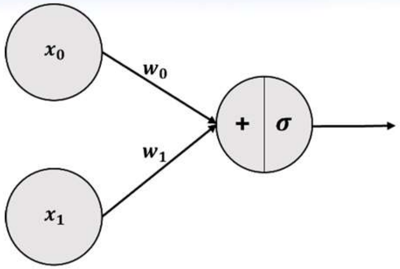

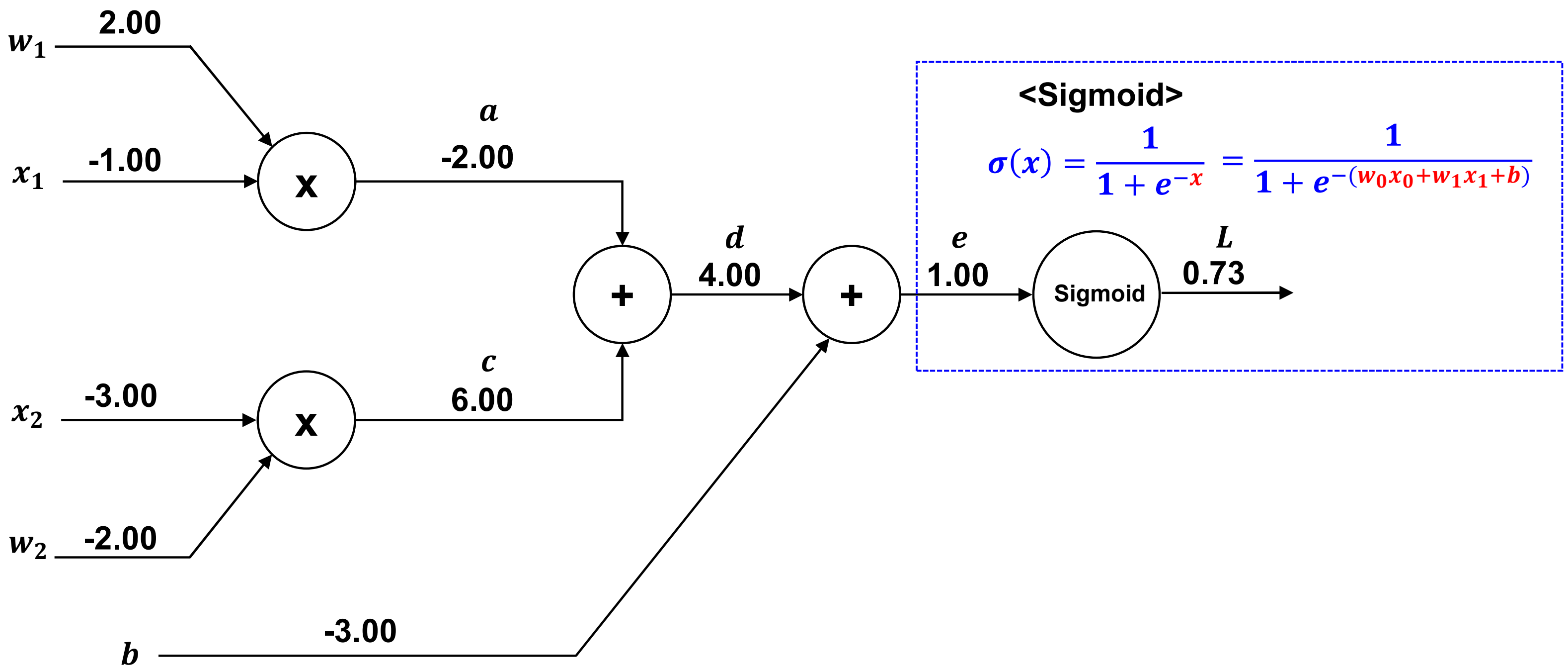

$$\sigma(x) = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-(w_0x_0+w_1x_1+b)}}$$

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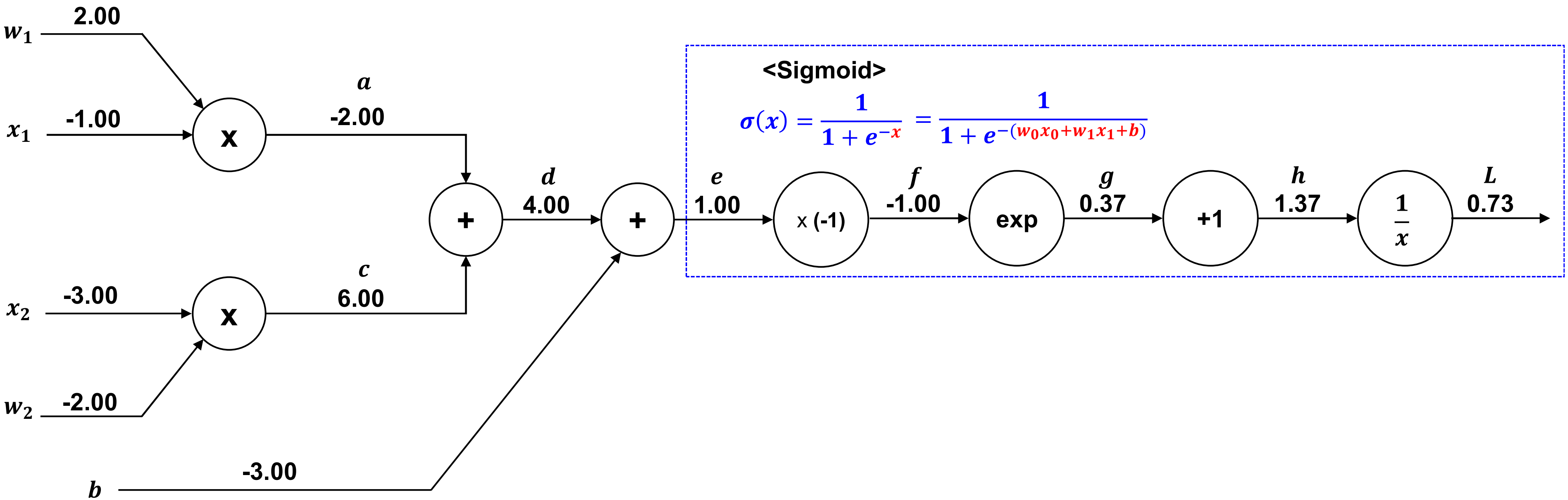

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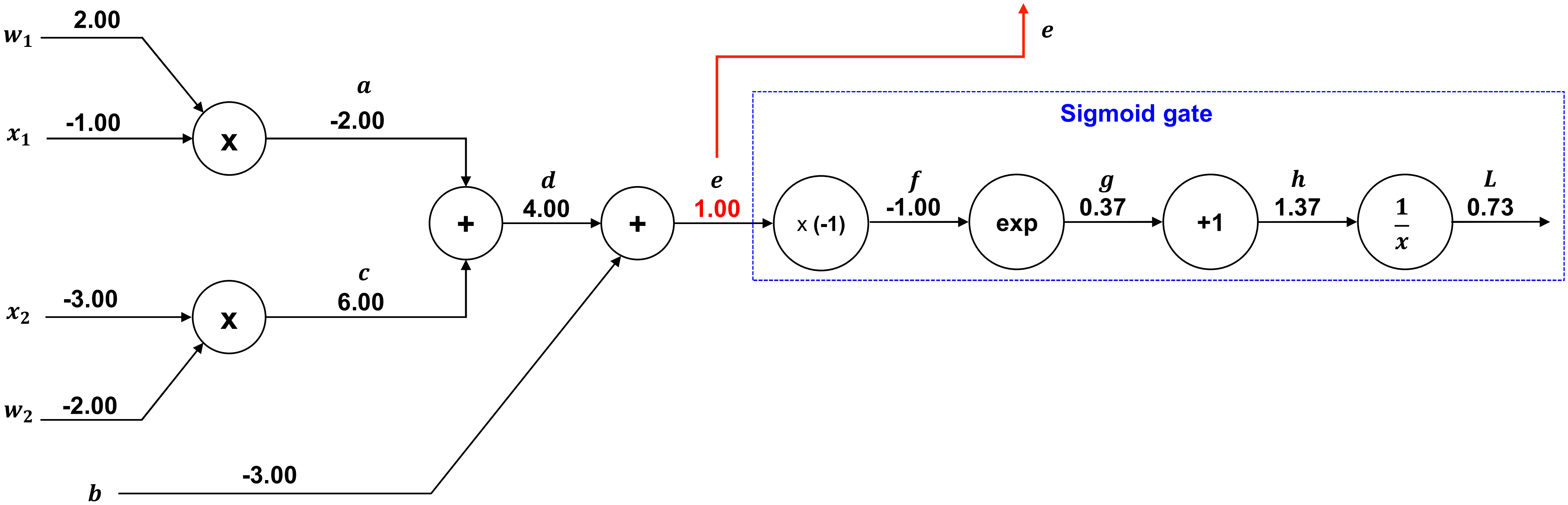

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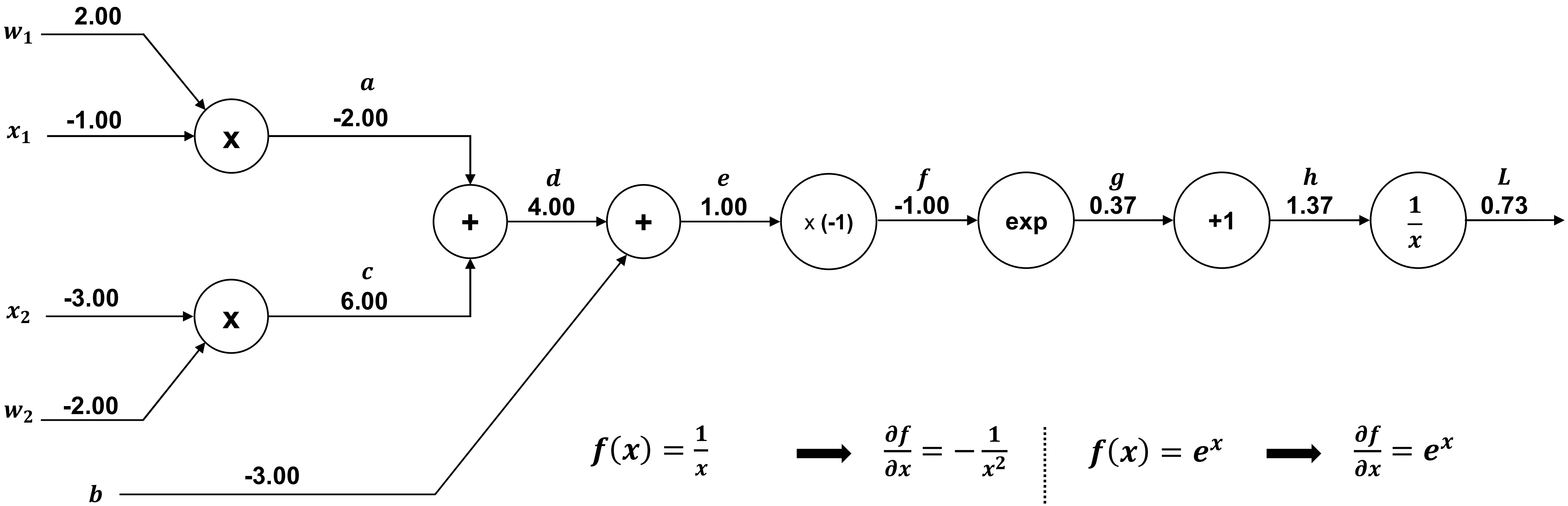

$$f(x) = \frac{1}{x} \rightarrow \frac{\partial f}{\partial x} = -\frac{1}{x^2}$$

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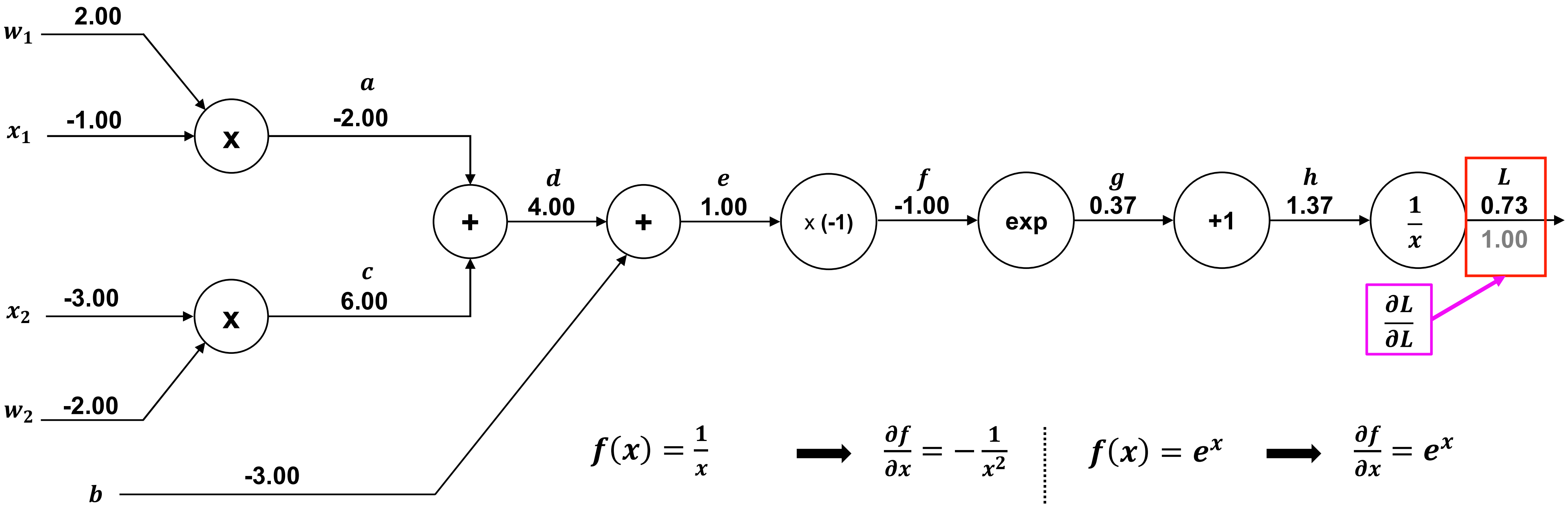

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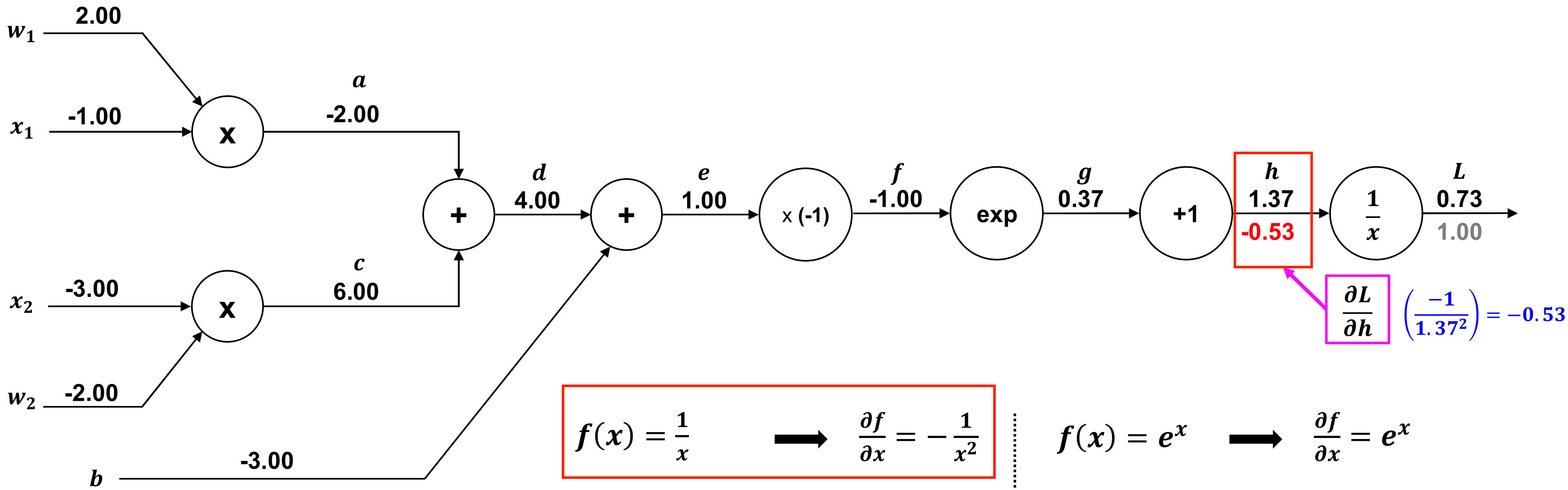

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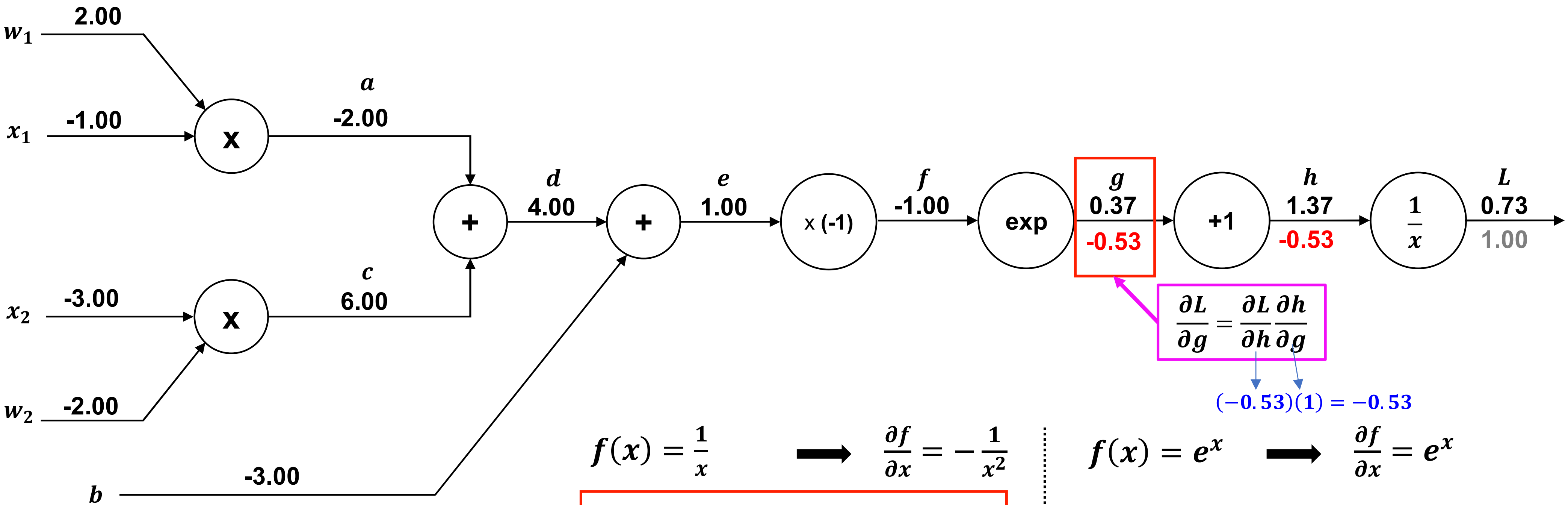

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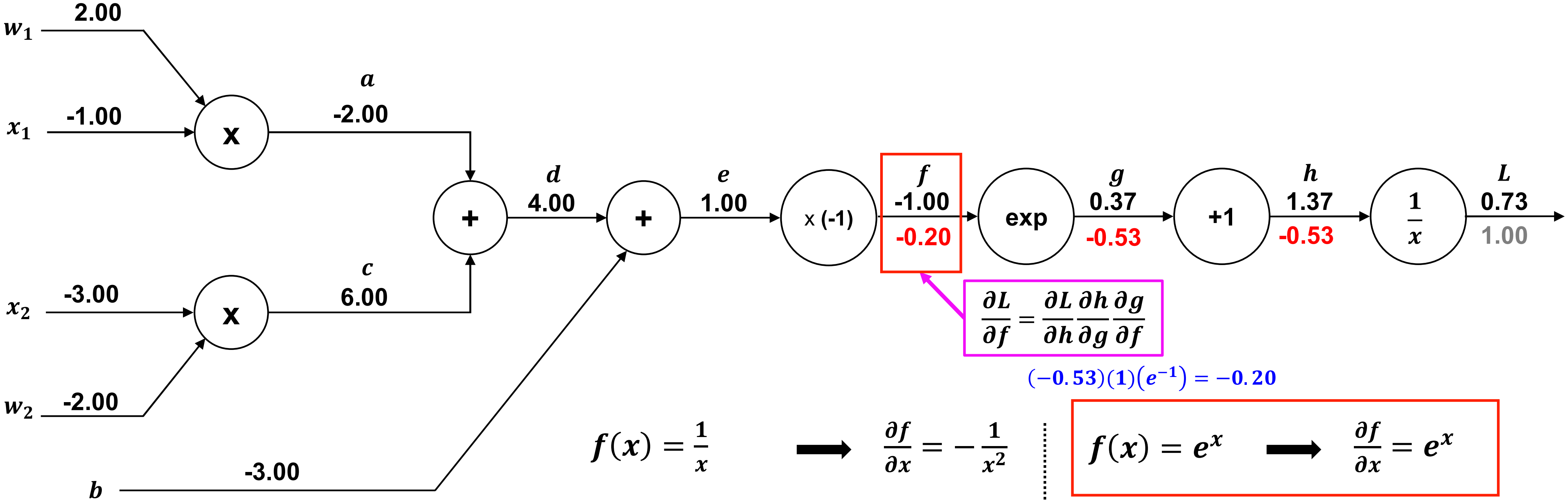

$$\frac{\partial L}{\partial f} = \frac{\partial L}{\partial h}\frac{\partial h}{\partial g}\frac{\partial g}{\partial f}$$

$$(-0.53)(1)(e^{-1}) = -0.20$$

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$$f(x) = x + c \longrightarrow \frac{\partial f}{\partial x} = 1$$

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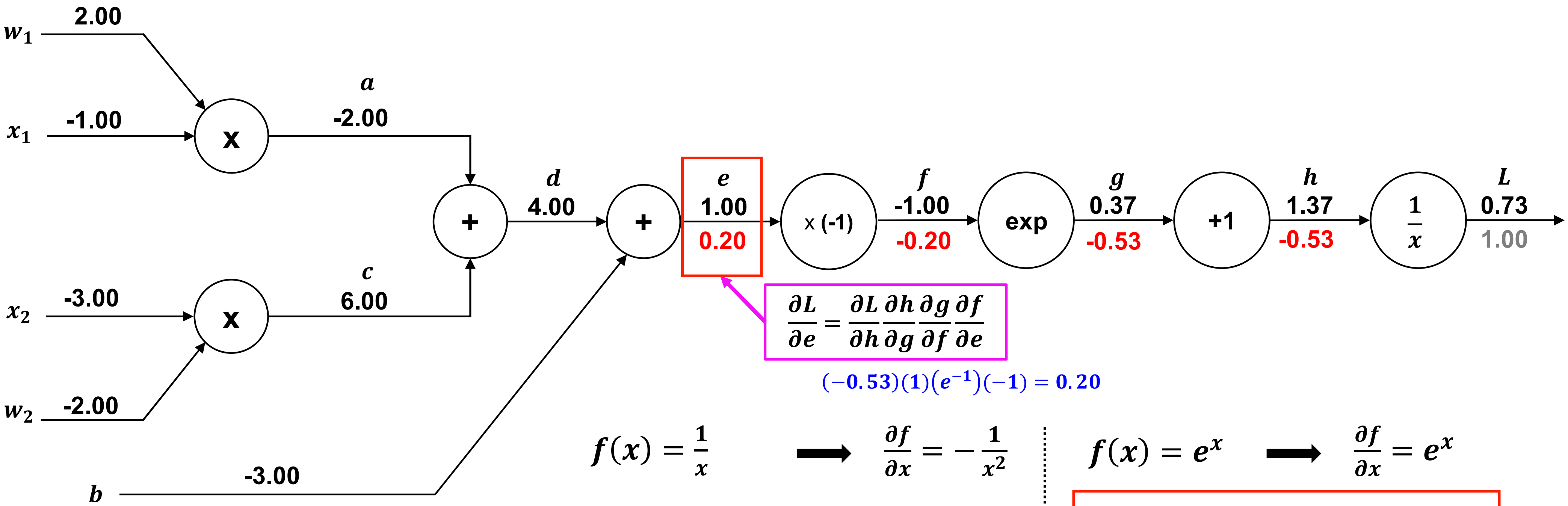
Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

<Sigmoid>

$$\sigma(x) = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-(w_0x_0+w_1x_1+b)}}$$



$$\frac{\partial L}{\partial e} = \frac{\partial L}{\partial h}\frac{\partial h}{\partial g}\frac{\partial g}{\partial f}\frac{\partial f}{\partial e}$$

$$(-0.53)(1)(e^{-1})(-1) = 0.20$$

$$f(x) = \frac{1}{x} \Rightarrow \frac{\partial f}{\partial x} = -\frac{1}{x^2}$$

$$f(x) = x + c \Rightarrow \frac{\partial f}{\partial x} = 1$$

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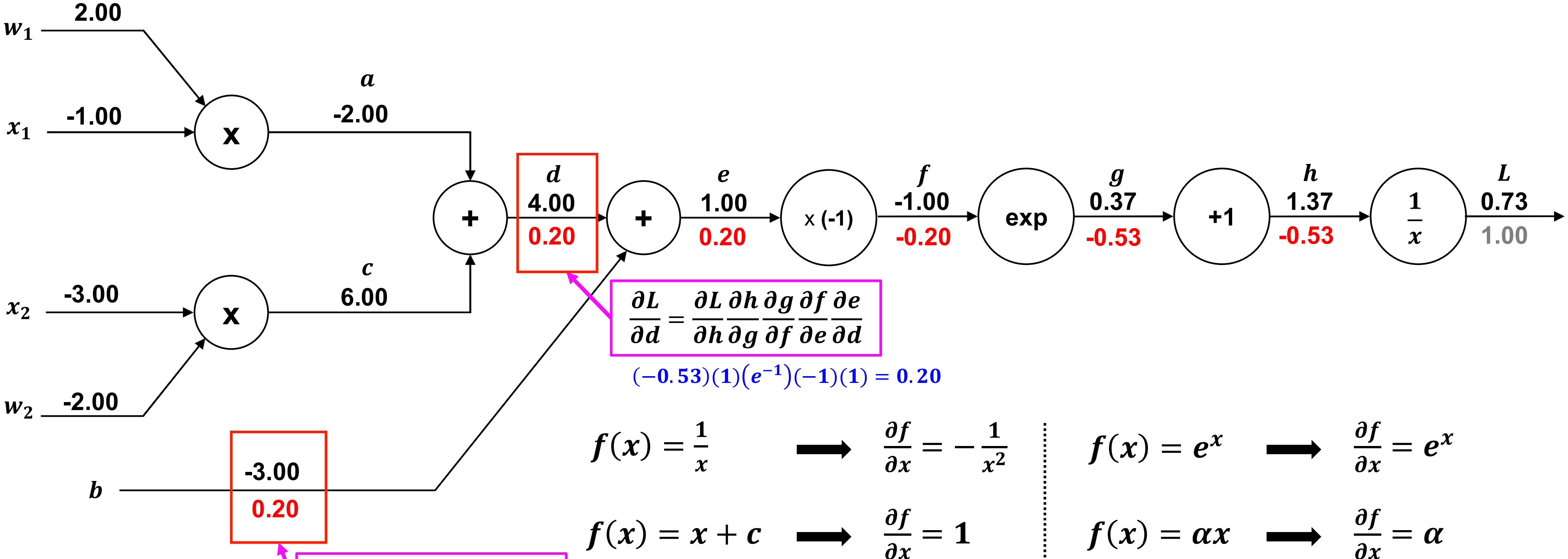

$$\frac{\partial L}{\partial d} = \frac{\partial L}{\partial h}\frac{\partial h}{\partial g}\frac{\partial g}{\partial f}\frac{\partial f}{\partial e}\frac{\partial e}{\partial d}$$

$$(-0.53)(1)\left(e^{-1}\right)(-1)(1) = 0.20$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial h}\frac{\partial h}{\partial g}\frac{\partial g}{\partial f}\frac{\partial f}{\partial e}\frac{\partial e}{\partial b}$$

$$(-0.53)(1)\left(e^{-1}\right)(-1)(1) = 0.20$$

$$f(x) = \frac{1}{x} \Rightarrow \frac{\partial f}{\partial x} = -\frac{1}{x^2}$$

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Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

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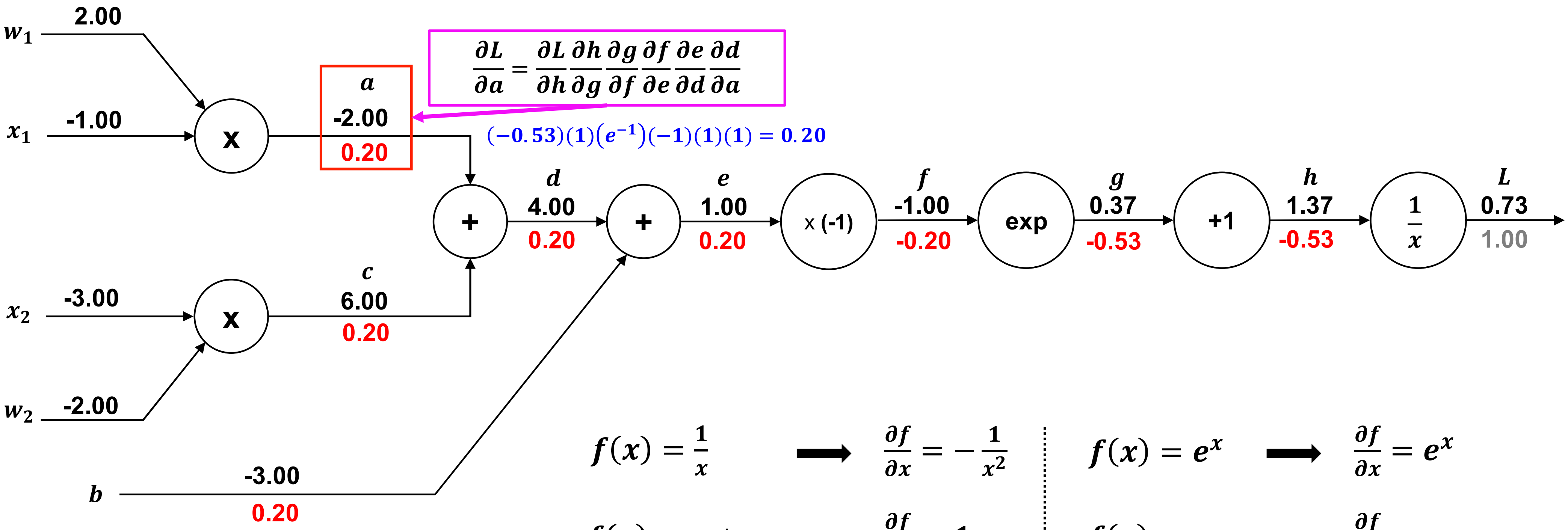

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Backpropagation 동작원리

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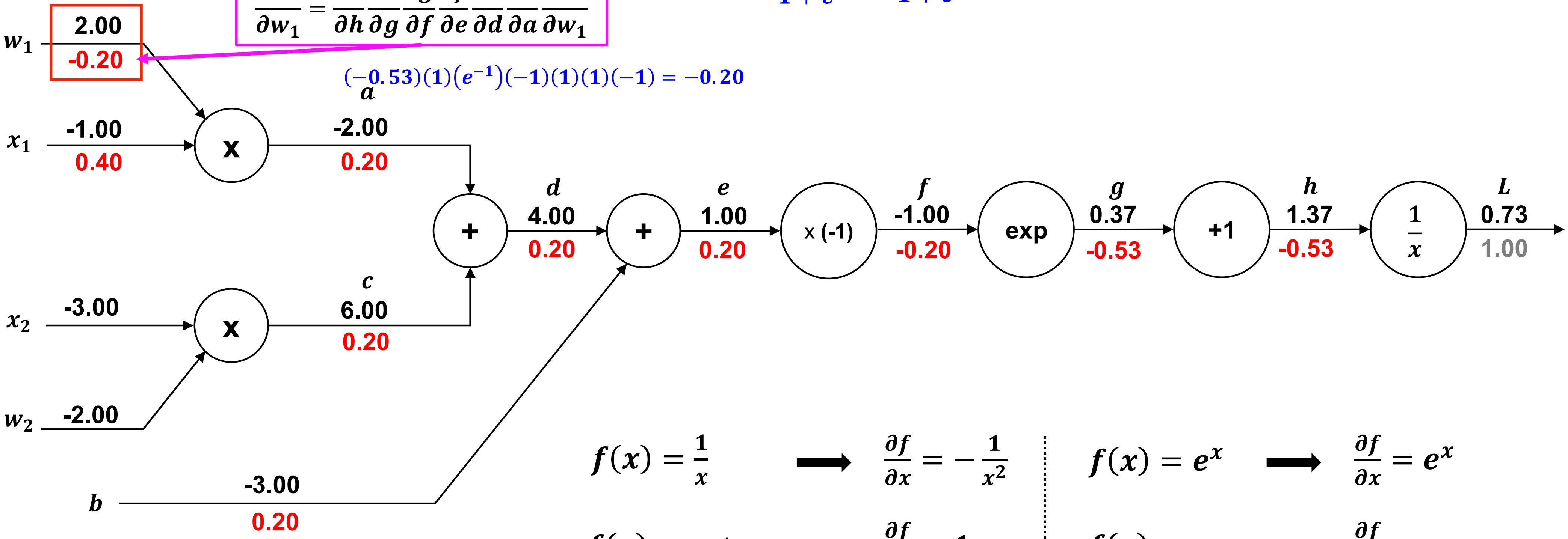
$$\sigma(x) = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-(w_0x_0+w_1x_1+b)}}$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial h}\frac{\partial h}{\partial g}\frac{\partial g}{\partial f}\frac{\partial f}{\partial e}\frac{\partial e}{\partial d}\frac{\partial d}{\partial a}\frac{\partial a}{\partial w_1}$$

$$(-0.53)(1)(e^{-1})(-1)(1)(1)(-1) = -0.20$$



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Backpropagation 동작원리

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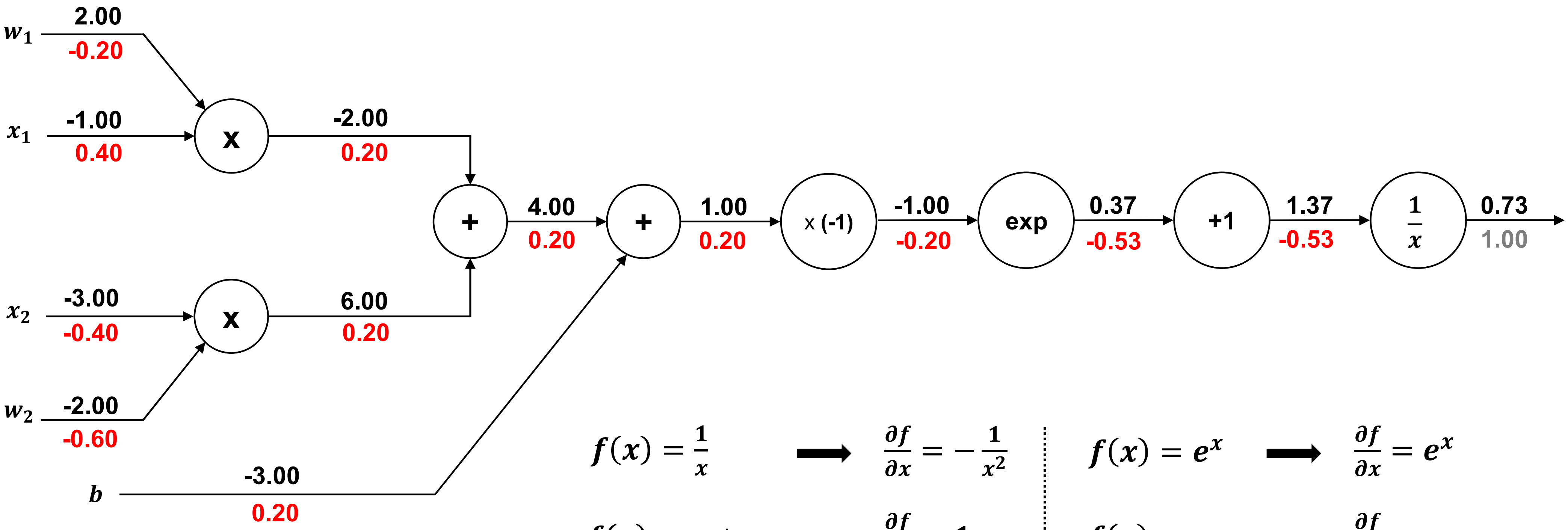

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Backpropagation 동작원리

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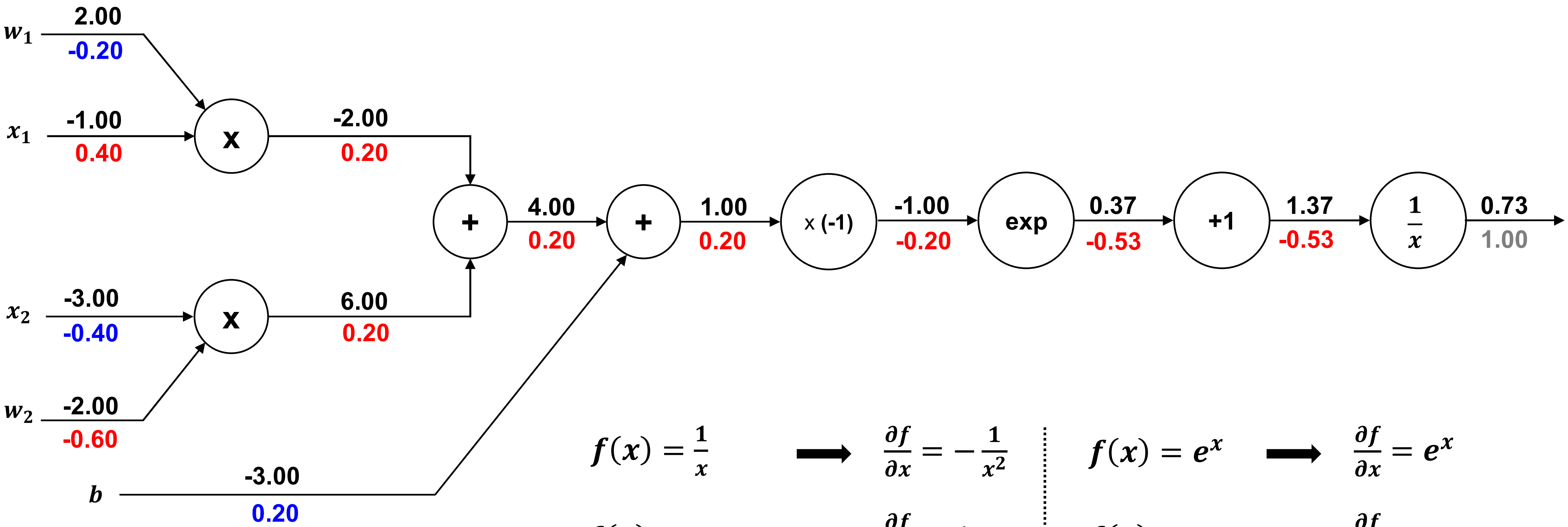

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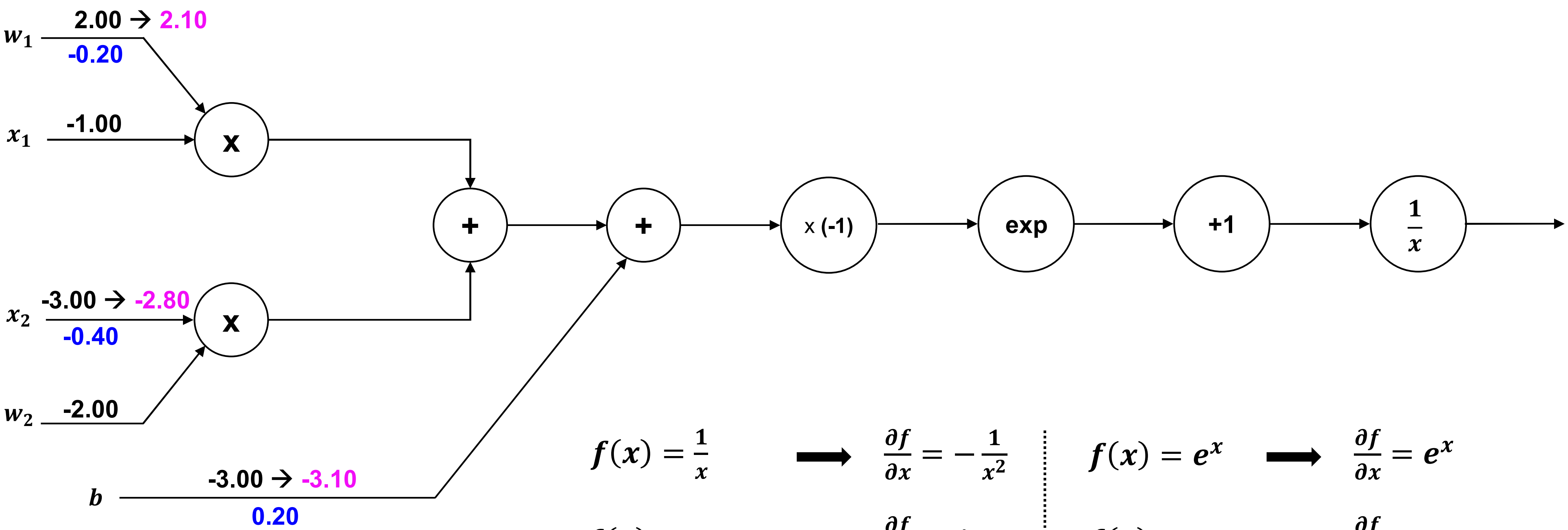
$$f(x) = \alpha x \Rightarrow \frac{\partial f}{\partial x} = \alpha$$

$$w^{t+1} \leftarrow w^t - \mu \nabla f(w^t)$$

Backpropagation 동작원리

- Sigmoid 계층의 backpropagation **(Ex., Learning Rate = 0.5)**



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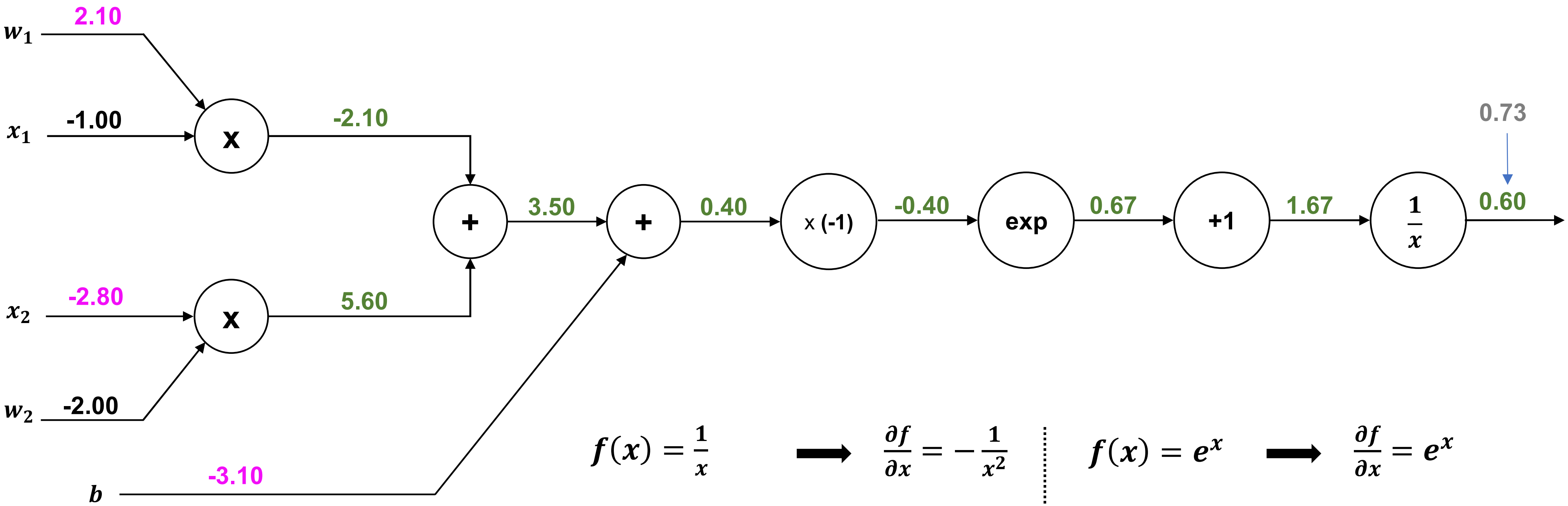

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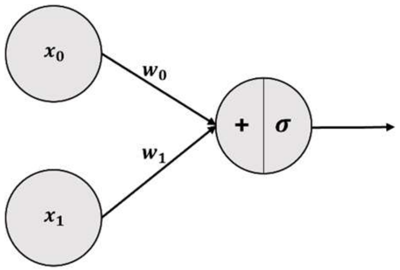
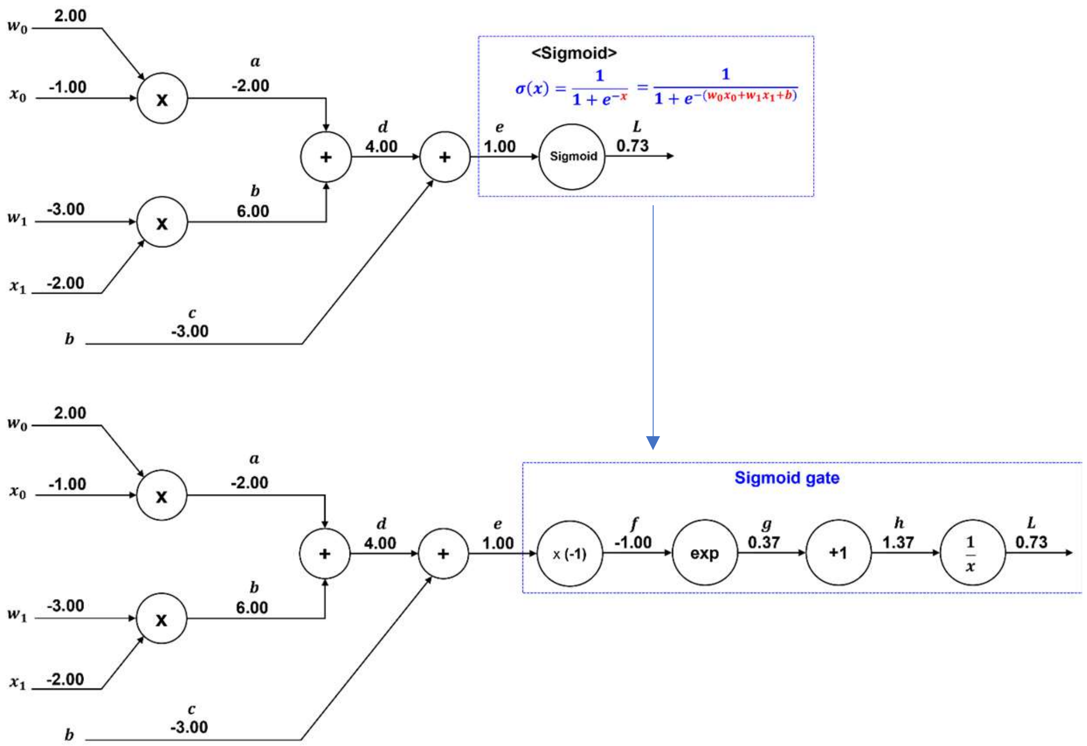
Backpropagation 동작원리

- **Single Layer Perceptron (SLP)**



Forward propagation (pass)

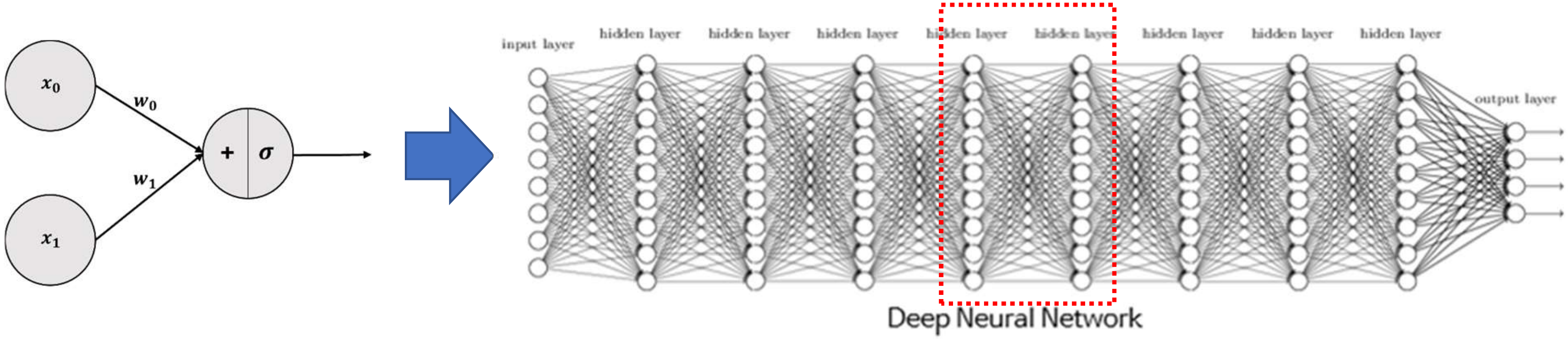


Backpropagation 동작원리

- **SLP → Deep Neural Network (MLP)**

  → **행렬에 대한 Backpropagation**

$\boldsymbol{x} \in \boldsymbol{R}^{m}$ $\boldsymbol{q} \in \boldsymbol{R}^{n}$



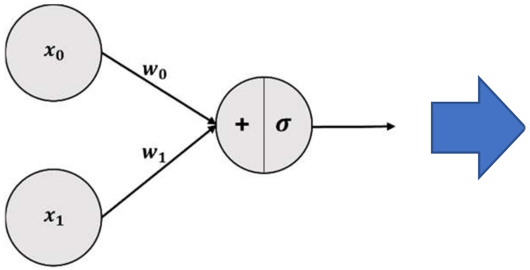
$$\boldsymbol{W} \in \boldsymbol{R}^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}, \quad \boldsymbol{W}^{T} \in \boldsymbol{R}^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}, \quad \boldsymbol{x} \in \boldsymbol{R}^{m} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}, \quad \boldsymbol{b} \in \boldsymbol{R}^{n} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

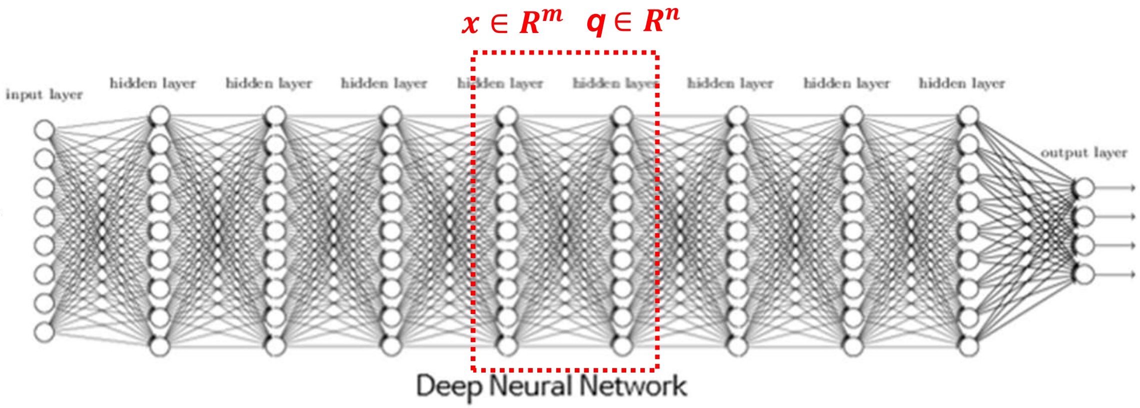
$$\boldsymbol{q} = \boldsymbol{\sigma}(\boldsymbol{W}^{T} \cdot \boldsymbol{x} + \boldsymbol{b}) = \boldsymbol{\sigma}\left\{\begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}\right\} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in \boldsymbol{R}^{n}$$

Backpropagation 동작원리

- **SLP → Deep Neural Network (MLP)**

  → **행렬에 대한 Backpropagation**





$$\boldsymbol{W} \in \boldsymbol{R}^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}, \quad \boldsymbol{W2}^{T} \in \boldsymbol{R}^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}, \quad \boldsymbol{x} \in \boldsymbol{R}^{m} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}, \quad \boldsymbol{b} \in \boldsymbol{R}^{n} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

$$\boldsymbol{y} = \boldsymbol{\sigma}(\boldsymbol{W}^{T} \cdot \boldsymbol{x} + \boldsymbol{b}) = \boldsymbol{\sigma}\left\{\begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}\right\} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

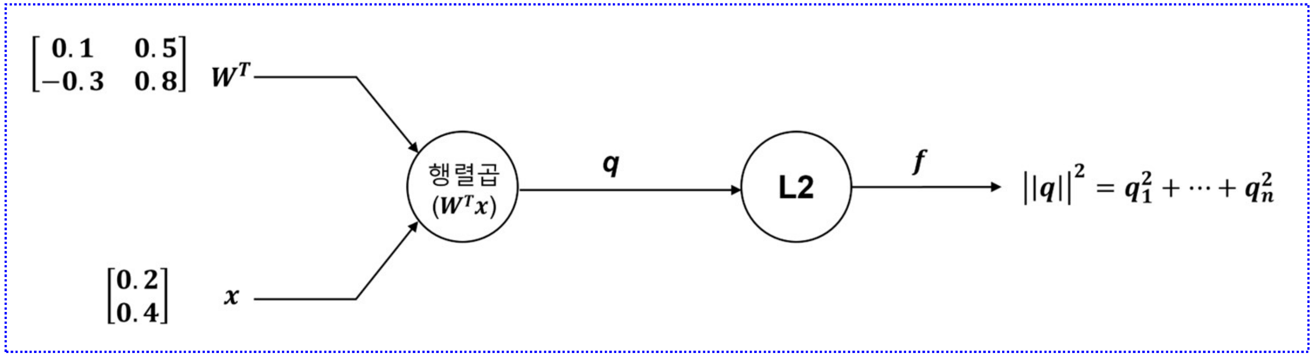
Backpropagation 동작원리

- 행렬의 Backpropagation (Example)

$$\boldsymbol{q} = \boldsymbol{\sigma}(\boldsymbol{W}^T \cdot \boldsymbol{x} + \boldsymbol{b}) = \boldsymbol{\sigma}\left\{\begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}\right\} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix}$$

간소화

$$\boldsymbol{q} = \boldsymbol{W}^T \cdot \boldsymbol{x} = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix}$$



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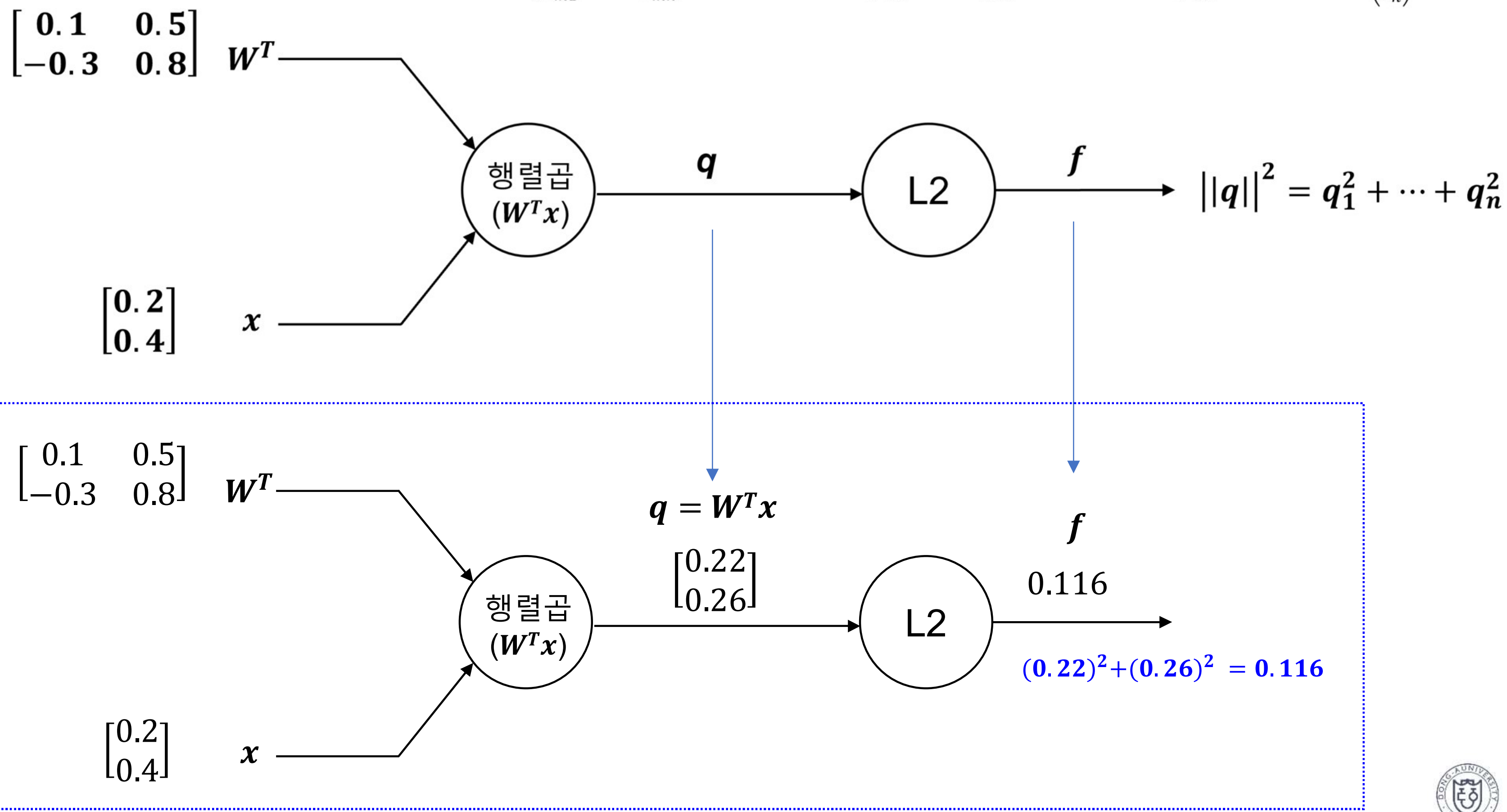
Backpropagation 동작원리

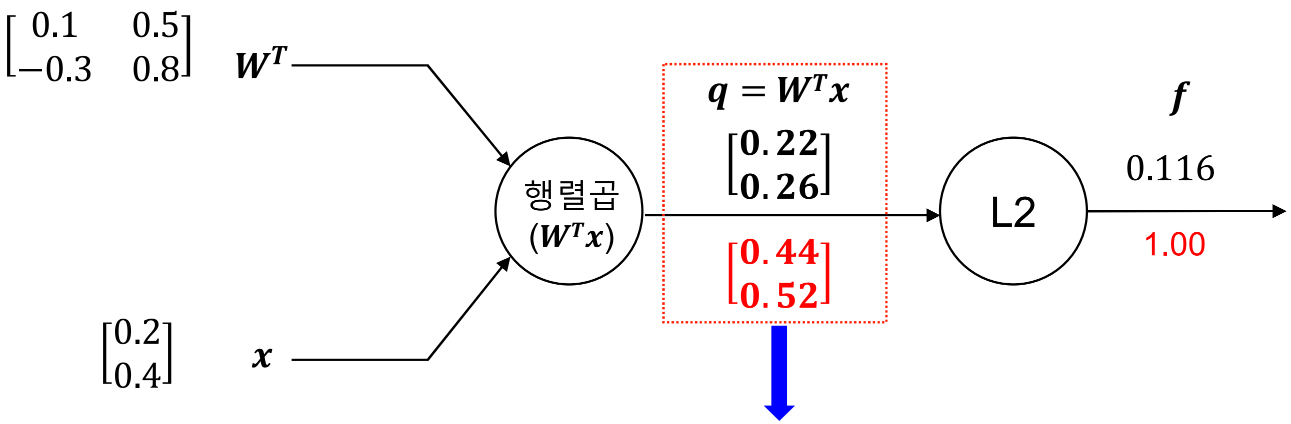
- 행렬의 Backpropagation (Example)

$$W \in R^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}, \quad W^T \in R^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}, \quad x \in R^m = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}, \quad b \in R^n = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \quad q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in R^n$$



Backpropagation 동작원리

- 행렬의 Backpropagation (Example) $W \in R^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}$, $W^T \in R^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}$, $x \in R^m = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$, $b \in R^n = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ $q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in R^n$



$$\frac{\partial f}{\partial q_i} = 2q_i$$

$$\nabla_{q_i} f = 2q_i$$

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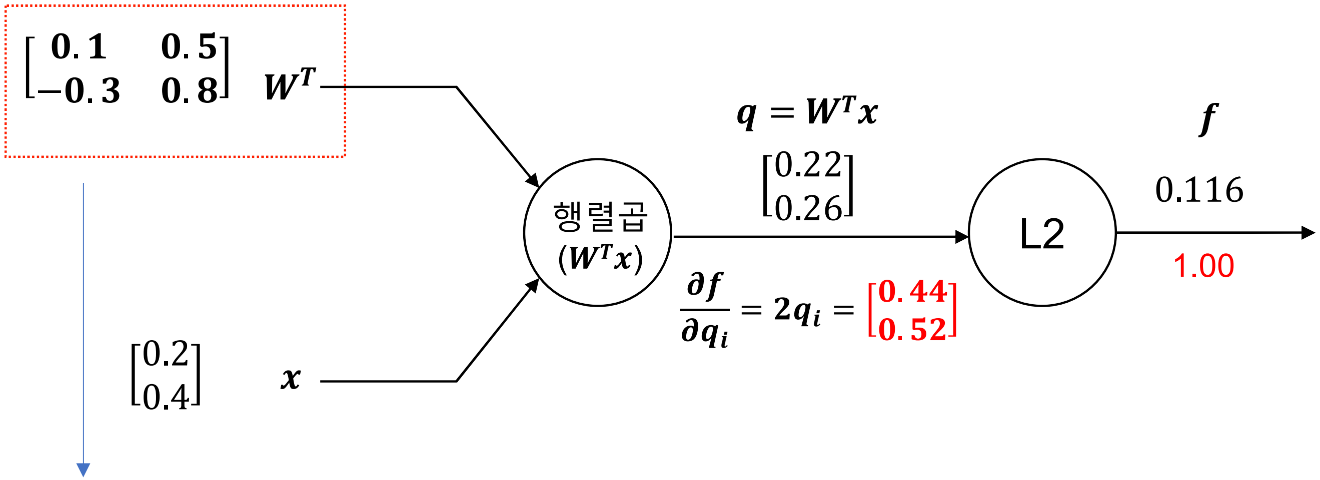
Backpropagation 동작원리

- 행렬의 Backpropagation (Example) $W \in R^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}$, $W^T \in R^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}$, $x \in R^m = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$, $b \in R^n = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ $q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in R^n$



$$\frac{\partial f}{\partial W^T} = \frac{\partial f}{\partial q} \cdot x^T$$

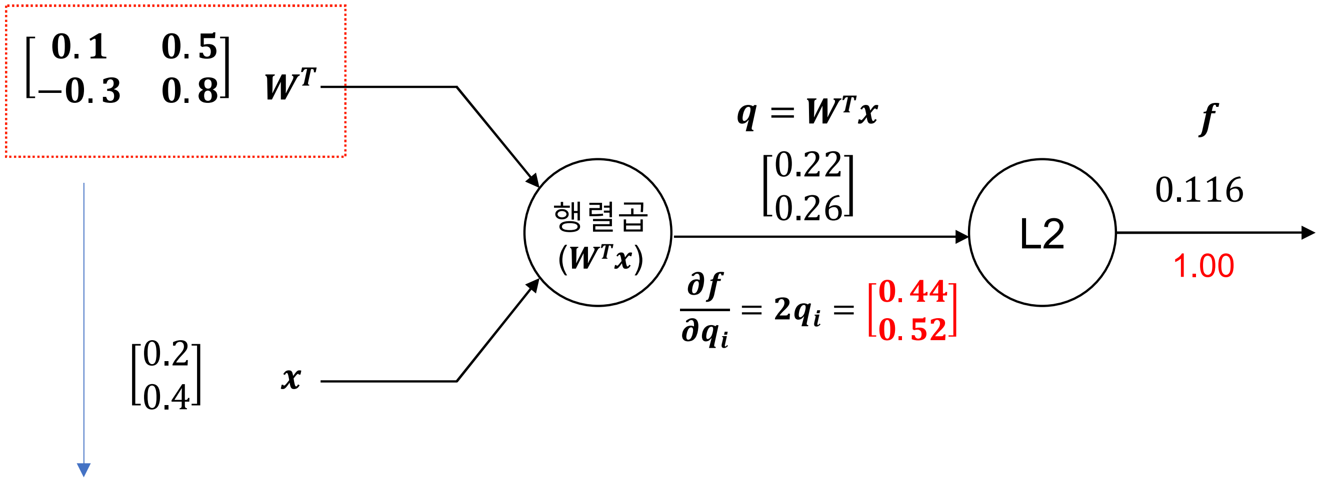
$$\frac{\partial f}{\partial x} = W \cdot \frac{\partial f}{\partial q}$$

$$\frac{\partial f}{\partial W^T} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial W^T} = 2q\boldsymbol{x}^T \in R^{n\times m}$$

$$[n \times m] = [n \times 1] \cdot [m \times 1]^T$$
$$= [n \times 1] \cdot [1 \times m]$$

Backpropagation 동작원리

- 행렬의 Backpropagation (Example) $\boldsymbol{W} \in \boldsymbol{R}^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}$, $\boldsymbol{W}^T \in \boldsymbol{R}^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}$, $\boldsymbol{x} \in \boldsymbol{R}^{m} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$, $\boldsymbol{b} \in \boldsymbol{R}^{n} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ $\boldsymbol{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in \boldsymbol{R}^n$



$$\frac{\boldsymbol{\partial f}}{\boldsymbol{\partial W^T}} = \frac{\boldsymbol{\partial f}}{\boldsymbol{\partial q}} \cdot \boldsymbol{x^T}$$

$$\frac{\partial f}{\partial x} = W \cdot \frac{\partial f}{\partial q}$$

$$\frac{\boldsymbol{\partial f}}{\boldsymbol{\partial W^T}} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial W^T} = 2q\boldsymbol{x^T} = \begin{bmatrix} 0.44 \\ 0.52 \end{bmatrix} [0.2 \quad 0.4] = \begin{bmatrix} \mathbf{0.088} & \mathbf{0.176} \\ \mathbf{0.104} & \mathbf{0.208} \end{bmatrix}$$

$$[n\times m] = [n\times 1]\cdot[m\times 1]^T$$
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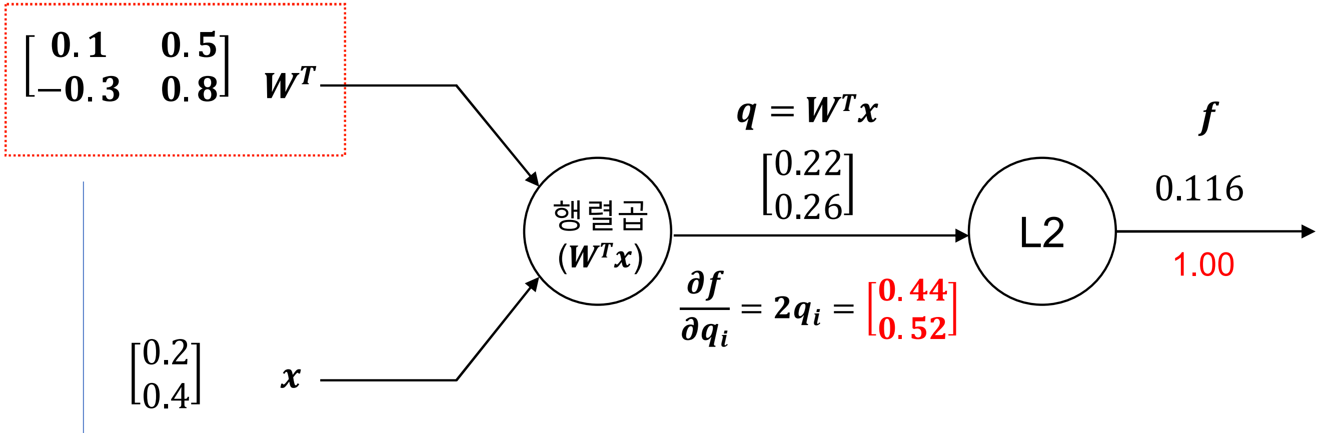
Backpropagation 동작원리

- 행렬의 Backpropagation (Example) $W \in R^{m \times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}$, $W^T \in R^{n \times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}$, $x \in R^m = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$, $b \in R^n = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ $q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in R^n$



$$\frac{\partial f}{\partial W^T} = \frac{\partial f}{\partial q} \cdot x^T$$

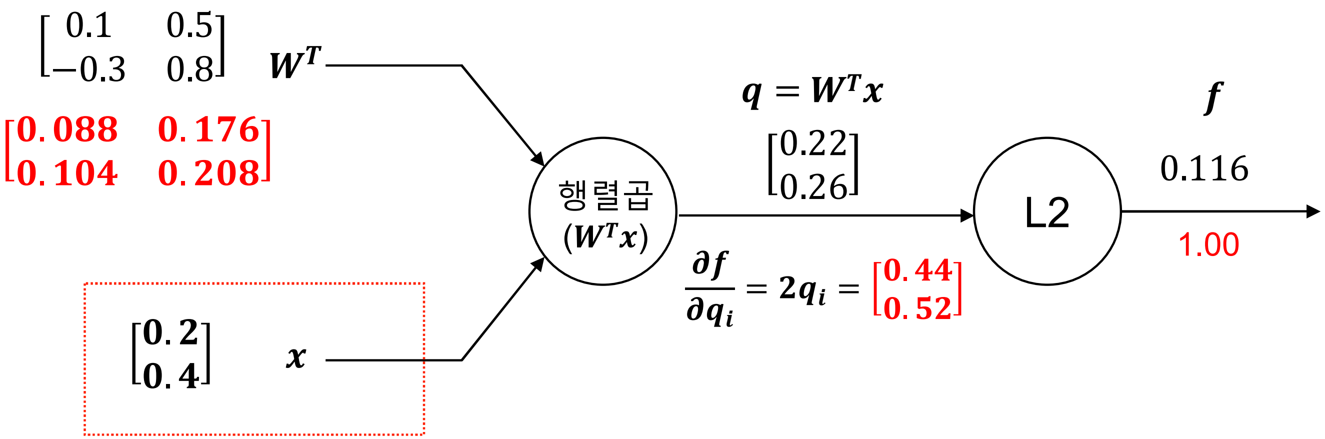
$$\frac{\partial f}{\partial x} = W \cdot \frac{\partial f}{\partial q}$$

$$\frac{\partial f}{\partial W^T} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial W^T} = 2q x^T = \begin{bmatrix} 0.44 \\ 0.52 \end{bmatrix} \begin{bmatrix} 0.2 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.088 & 0.176 \\ 0.104 & 0.208 \end{bmatrix}$$

$$\nabla_{W^T} f = \nabla_q f \cdot \nabla_{W^T} q$$

Backpropagation 동작원리

- 행렬의 Backpropagation (Example) $\boldsymbol{W} \in \boldsymbol{R}^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}$, $\boldsymbol{W}^T \in \boldsymbol{R}^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}$, $\boldsymbol{x} \in \boldsymbol{R}^{m} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$, $\boldsymbol{b} \in \boldsymbol{R}^{n} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ $\boldsymbol{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in \boldsymbol{R}^{n}$



$$\frac{\partial f}{\partial W^T} = \frac{\partial f}{\partial q} \cdot x^T$$

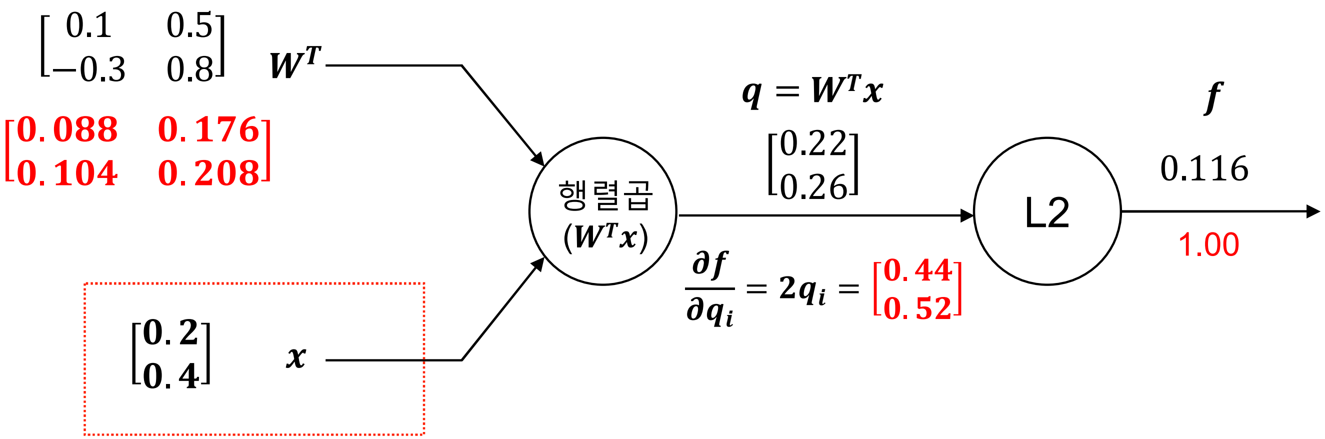
$$\boxed{\frac{\boldsymbol{\partial f}}{\boldsymbol{\partial x}} = \boldsymbol{W} \cdot \frac{\boldsymbol{\partial f}}{\boldsymbol{\partial q}}}$$

$$\frac{\boldsymbol{\partial f}}{\boldsymbol{\partial x}} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial x} = (\boldsymbol{W}^T)^T \cdot 2q \quad \in \boldsymbol{R}^m$$

$$[m \times 1] = [n \times m]^T \cdot [n \times 1]$$
$$= [m \times n] \cdot [n \times 1]$$

Backpropagation 동작원리

- 행렬의 Backpropagation (Example) $\boldsymbol{W} \in \boldsymbol{R}^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}$, $\boldsymbol{W}^T \in \boldsymbol{R}^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}$, $\boldsymbol{x} \in \boldsymbol{R}^{m} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$, $\boldsymbol{b} \in \boldsymbol{R}^{n} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ $\boldsymbol{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in \boldsymbol{R}^{n}$



$$\frac{\partial f}{\partial W^T} = \frac{\partial f}{\partial q} \cdot x^T$$

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \boldsymbol{W} \cdot \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{q}}$$

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial x} = \boldsymbol{W} \cdot 2q \quad = \begin{bmatrix} 0.1 & -0.3 \\ 0.5 & 0.8 \end{bmatrix} \begin{bmatrix} 0.44 \\ 0.52 \end{bmatrix} = \begin{bmatrix} \mathbf{-0.112} \\ \mathbf{0.636} \end{bmatrix}$$

$$[m \times 1] = [n \times m]^T \cdot [n \times 1]$$
$$= [m \times n] \cdot [n \times 1]$$

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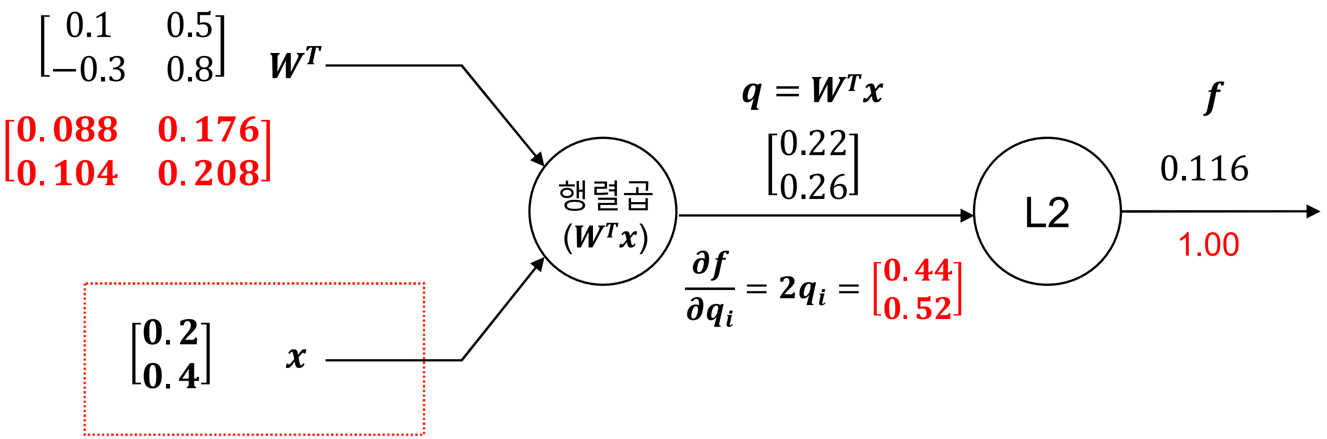
Backpropagation 동작원리

- 행렬의 Backpropagation (Example) $\boldsymbol{W} \in \boldsymbol{R}^{m\times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}$, $\boldsymbol{W}^T \in \boldsymbol{R}^{n\times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}$, $\boldsymbol{x} \in \boldsymbol{R}^{m} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$, $\boldsymbol{b} \in \boldsymbol{R}^{n} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ $\boldsymbol{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in \boldsymbol{R}^n$



$$\frac{\partial f}{\partial W^T} = \frac{\partial f}{\partial q} \cdot x^T$$

$$\boxed{\frac{\boldsymbol{\partial f}}{\boldsymbol{\partial x}} = \boldsymbol{W} \cdot \frac{\boldsymbol{\partial f}}{\boldsymbol{\partial q}}}$$

$$\frac{\boldsymbol{\partial f}}{\boldsymbol{\partial x}} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial x} = \boldsymbol{W} \cdot 2q = \begin{bmatrix} 0.1 & -0.3 \\ 0.5 & 0.8 \end{bmatrix}\begin{bmatrix} 0.44 \\ 0.52 \end{bmatrix} = \begin{bmatrix} \mathbf{-0.112} \\ \mathbf{0.636} \end{bmatrix}$$

$$\boldsymbol{\nabla_x f = \nabla_q f \cdot \nabla_x\, q}$$

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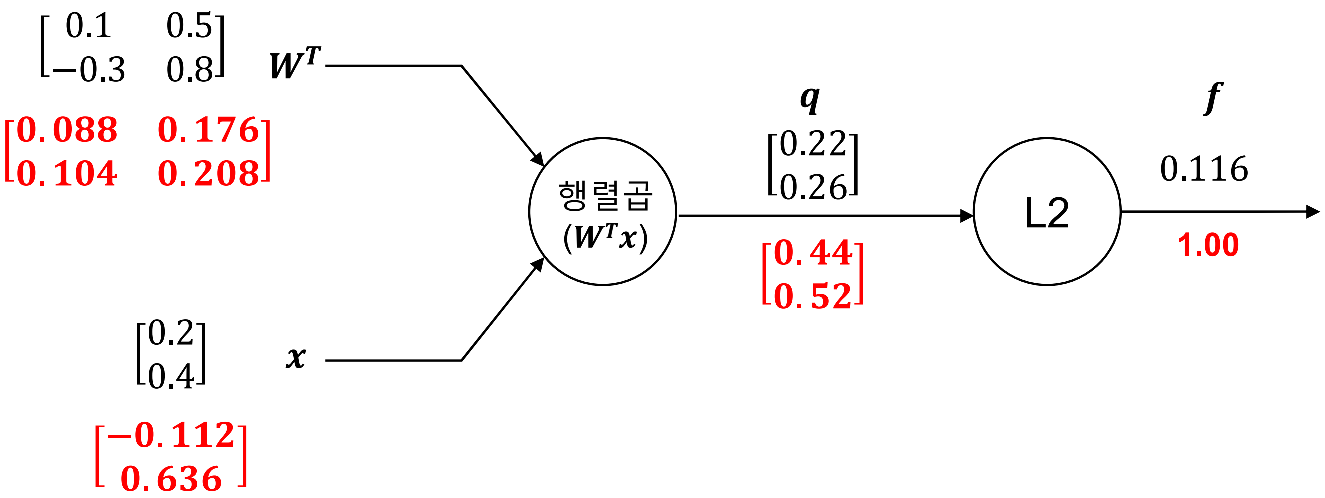

$$\boldsymbol{w}^{t+1} \leftarrow \boldsymbol{w}^{t} - \boldsymbol{\mu}\nabla \boldsymbol{f}(\boldsymbol{w}^{t})$$

Descent Method(하강법)

- 주어진 어떤 지점에서부터 오차가 더 작은 곳으로 이동하려는 방법
  - 탐색 방법 Δw에 따라 하강법이 결정
  - 종류: 경사하강법, 모멘텀 등 → **Optimization(최적화)**

학습률(learning rate, step size)

$$w^{t+1} \leftarrow w^{t} - \mu \nabla L(w^{t}), \quad where\ \nabla L(w^{t})\Delta w < 0$$

탐색 방향

Questions & Answers

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